Engineering 4 Notebook

Elijah LaMothe
Andy Diaz
Vivian Nguyen
Period 6

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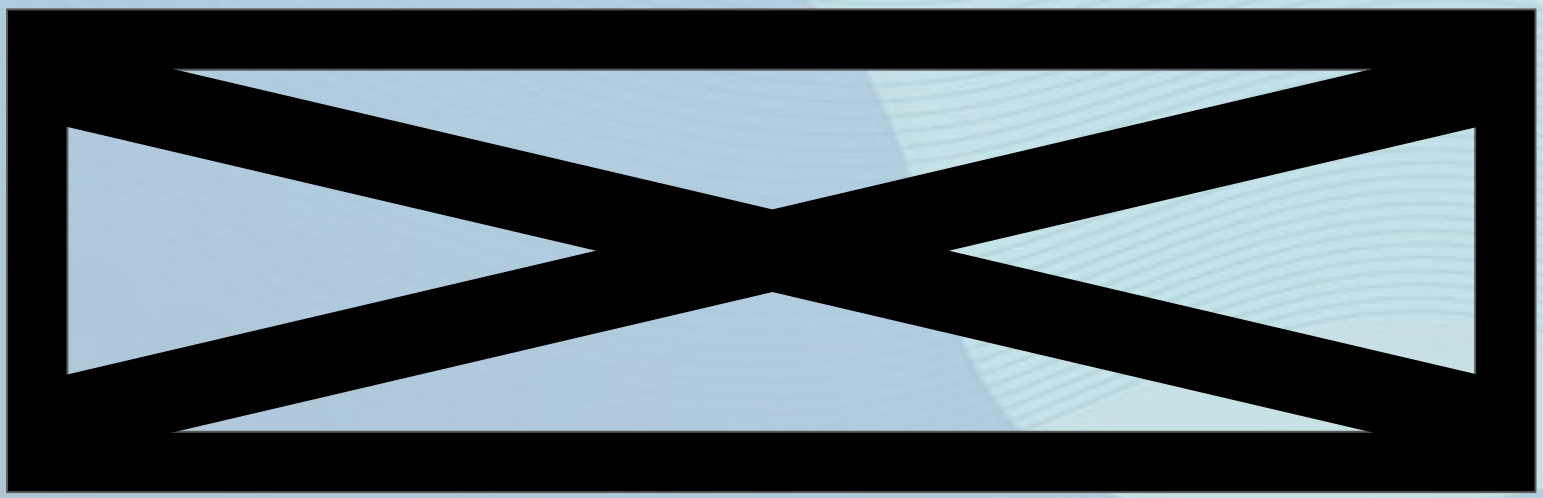
Top 15 Problems: Elijah LaMothe 6

1. Dust is prevalent on planets outside of Earth, and it can pose a problem for probes such as *perseverance.* This is due to the fact that dust can prevent the sun from reaching the solar panels that probes such as *Perseverance* rely on. This causes probes to be less efficient and can hinder the process of observing the surrounding environments or taking samples.
2. Personally, I find keyboards to be pretty inefficient and uncomfortable, the keyboard is so wide that it often causes me to get hand cramps when I type for long periods of time.
3. I hate that electricity bills can be paid for unnecessary things such as charging a battery, turning on the blender, or running lights that connect to outlets.
4. As an engineer, or someone who has engineering-like habits, having to grab a ruler whenever I want to measure something can be annoying, and it often takes a long time to get a really accurate measurement.
5. When running, it is easy to let your running form worsen as you get tired, but this can often lead to injury and cause you to run slower, which is detrimental for professional runners/sprinters.
6. As rockets leave our atmosphere, a process called expendable staging can often take place which is where the thrusters that were used to get it out of our atmosphere are detached from the rocket. These thrusters can often land in the ocean which contributes to ocean pollution and climate change.
7. When sitting down for a long writing assignment, it is easy to run out of lead especially if I forgot to refill it before. Taking the time to find more lead and refill your pencil is annoying and takes time away from your writing, especially if you are doing a timed write of sorts where you need every second.
8. Often, after purchasing fruits such as strawberries, raspberries, blueberries, etc, they will mold after what seems like no time at all. Not knowing when your food will mold can be a huge waste of money and food.
9. After a long day of working, school, or studying, you just want to throw your phone to the side and plop right down in bed, but because we use wired chargers it can be annoying to have to find the wire and plug it into your phone.
10. Cutting fruits or vegetables can be very tedious at times, it can take time away from prepping another part of the meal and even make you susceptible to cutting or injuring yourself in some way.
11. I don't take pictures often, but when I do it can be annoying to have to reach in my pocket or in my bag to find it, open the camera app, and take a picture, it feels like the moment I am trying to capture will be lost by then.
12. When playing basketball (as someone who plays alone because I'm lonely) it can be annoying to have to run and get your rebound after, as well as a waste of effort when you just want to practice your shooting skills.
13. Waiting a super long time for my batteries to recharge can be really annoying, and if I am in the middle of a playing session with my friends, it can cause me to have to wait a long time to play with them again, and by the time they are recharged then nobody is online anymore.
14. During the summer, when it gets insanely hot, sometimes my car engine doesn't work right and I often have to wait until it cools down or the temperature gets a bit lower for the engine to work right. This can be a huge inconvenience and even cause me to be late or completely miss events that I was planning to be at.
15. Getting up to walk to 7 classes a day can be annoying, but what's more annoying is having to pull out my notebook every time I arrive at each class, it doesn't strain me physically but the sheer thought of it can often make me procrastinate pulling it out.

Author:	Date:	Witness:	Date:
Elijah LaMothe	9/5/24	Andy Diaz	9/5/24

Top 15 Problems: Andy Diaz 7

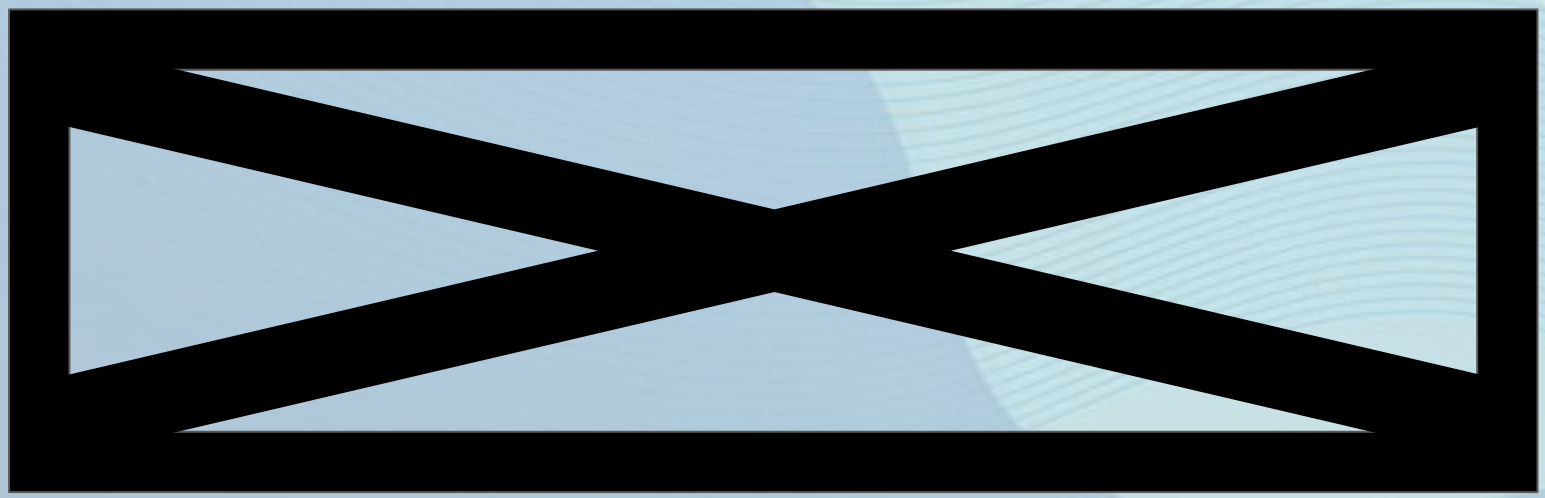
- I hate when I take a shower and my mirror fogs up
- I hate when there is no one to play catch with
- I hate when I have to sort the cans
- I hate when there is a bunch of trash on the side of the road
- I hate when there are ants in my house
- I hate when there is trash in my pool
- I hate when I have to re-rack on squat
- I hate when I have to fold my clothes
- I hate when my dogs leave hair everywhere
- I wish something can give my dogs food and water
- I wish something can shower my dogs
- I hate when I have to cut my grass
- I wish something can fill the pool water when it gets to low
- I hate forgetting to put my clothes to dry when I put them to wash
- I wish something can clean the counter when it gets dirty



Author:	Date:	Witness:	Date:
Andy Diaz	9/5/24	Elijah LaMothe	9/5/24

Top 15 Problems: Vivian Nguyen

- **It is hard to cast a line out far enough creating health risks like joint strains**
- Sleeping in a car or airplane is uncomfortable and can be really bad on muscles and the back
- There are not enough cup holders, and lack of space to put more cups
- Makeup - loose powder gets everywhere and starts to get hard to clean making the area powdery
- Washing paint brushes is hard to wash without residue left in some brussels
- Ironing clothes to unwrinkle takes up too much time
- Forgetting to rinse down the sink after usage builds up toothpaste residue
- Easier way to clean wildlife water tanks efficiently (Extra)
- Managing multiple charging cables in one area gets too messy and unorganized
- Hard to keep track of the food that may expire soon and avoid spoilage (Extra)
- Laptops and technology that tend to overheat on a soft surface (Extra)
- Losing personal items or losing track of where certain things are
- **Keeping plants alive, inconsistent watering, light intake. No knowledge of the environment being grown around**
- **Fans with blades tend to get hazardous in household with animals and children**
- Recyclables and waste tend to get mixed up often and not sorted properly even with categorized trash cans



Author:	Date:	Witness:	Date:
Andy Diaz	9/6/24	Elijah LaMothe	9/6/24

Team Selection Notes

Vivian Nguyen - Presentation Notes Taking

File Edit View Insert Format Data Tools Extensions Help

Criteria	Weight	Andrew Magne	Junior Vasquez	Yaitza Aguilar	Ryleigh	Ryleigh	Elijah LaMothe	Elijah LaMothe	Kayla Franklin	Kayla Franklin
		f there is mail	eeding my cat	ating DDR Mac	tion and Toilet	Fake Money	anet Exploratio	Idea Name	Remote Contr	y Dog Water Bowl
The problem can be accomplished in a given time	2	2	2	1	2	2	1	1	2	2
The problem can be solved with the available resources	2	2	2	2	2	2	2	1	2	2
Initial research indicates that others belive the problem exist	1	1	1	1	1	1	1	1	1	1
The solution is likely to meet a specific need and/or be marketable	1	1	1	1	1	1	1	1	1	1
The problem is interesting enough to keep the team interested.	1	0	1	1	1	1	1	1	1	1
The problem can have positive outcome that can show an impact	1	1	1	1	1	1	1	1	1	1
The problem can be expanded on in the future and maintainable	1	1	1	0	1	1	1	1	1	1
to be manageable in complexity; not so difficult to the point where it	1	1	1	0	0	1	1	1	1	1
	Total	9	10	7	9	10	9	8	10	10

Vivian's Notes

Criteria	Weight	Brynn	Andrew	Ryleigh	Elijah	DJ	Brisa	Kayla	Isaac	Julissa	Anthony	Caleb	Aron	Junior	Yaitza	Vivian	Jaydan	Adriel	Eddie	Sebastion	Rafael	Keon	David
		Mail delivery	Porch Piracy	Hot coffee	anet explorati	ome Overheati	Dirty brushing	Dog Bowl	Alarm	ain water for hom	Dogs water	Dust	Dity solar panels	P.E Clothes	Feeding 7 cats	Plants dying	droom stairs with cl	Heavy objects	Doors being open	Waking up	Slicing up fruits	Sleep Inertia	Self locking gate
The problem can be accomplished in a given time	2	2	2	2	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
The problem can be solved with the available resources	2	2	1	1	1	1	1	2	2	1	2	1	1	1	1	1	1	1	1	1	1	1	1
Initial research indicates that others belive the problem exist	1	0	1	1	1	0	0	1	1	0	0	0	1	1	0	1	0	0	1	1	1	1	0
e solution is likely to meet a specific need and/or be marketable	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
The problem is interesting enough to keep the team interested.	1	1	1	0	1	0	0	0	0	1	0	1	1	0	0	0	1	0	0	0	0	0	1
Is the person reliable	1	1	1	1	1	0	1	1	0	0	1	0	0	1	1	1	1	1	0	1	1	1	1	0
Can I relate to the situation	1	1	0	1	0	1	0	1	1	0	1	1	1	1	1	0	1	0	1	1	1	0	1	0
Can I help in this situation	1	1	1	1	1	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Total	9	8	8	7	6	5	9	8	5	8	7	8	8	6	8	7	6	8	8	7	8	6

Andy's Notes

Criteria	Weight	Windshield Wipers	Mailbox retrieval	Toilet Paper	Fake Money	Phone overheating	Trash floating in pools	Losing TV Remote	Evaporating Gasoline	Running with stuff in pockets	Phone not waking you up	Underoutting Pipes	Pet feeding while on vacay	Fishing lure getting stuck	Dust accumulation	Forgetting to turn off the stove	Wire being unplugged or power being out	Problem of dropping phones	Dirty Solar panels
The problem can be accomplished in a given time	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
The problem can be solved with the available resources	2	2	2	2	2	2	2	2	2	2	2	1.5	2	2	2	2	2	2	2
Initial research indicates that others belive the problem exist	1	1	1	1	1	1	1	0.5	1	0	0.5	1	1	1	0.5	1	1	1	1
The solution is likely to meet a specific need and/or be marketable	1	1	1	1	1	1	1	0	1	0	0.5	1	1	1	0.5	1	0.5	1	1
Theproblem is interesting enough to keep the team interested.	1	0	1	0	1	1	1	0	1	0	0	0.5	0.5	0	0	0	0	0.7	1
I can work with this person, or possible group of people	1	1	1	1	1	0.2	1	1	1	1	1	1	1	1	0.5	0.5	0.5	1	1
The problem can be applicable on a larger scale	1	1	1	0.5	1	1	1	0.5	1	1	1	1	0.5	0.5	0.5	0.8	0.8	0.8	1
The problem can be challenging/engaging enough	2	1	2	0	1	1	2	1	1.5	1.5	1	1	1.5	1	1	1	1	1	2
	Total	9	11	7.5	10	9.2	11	7	10.5	7.5	8	9	9.5	8.5	7	8.3	7.8	9.5	11

Elijah's Notes

[redacted]

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/8/24	Andy Diaz	9/8/24

Top 1 Problem

We hate when trash accumulates in our pools.

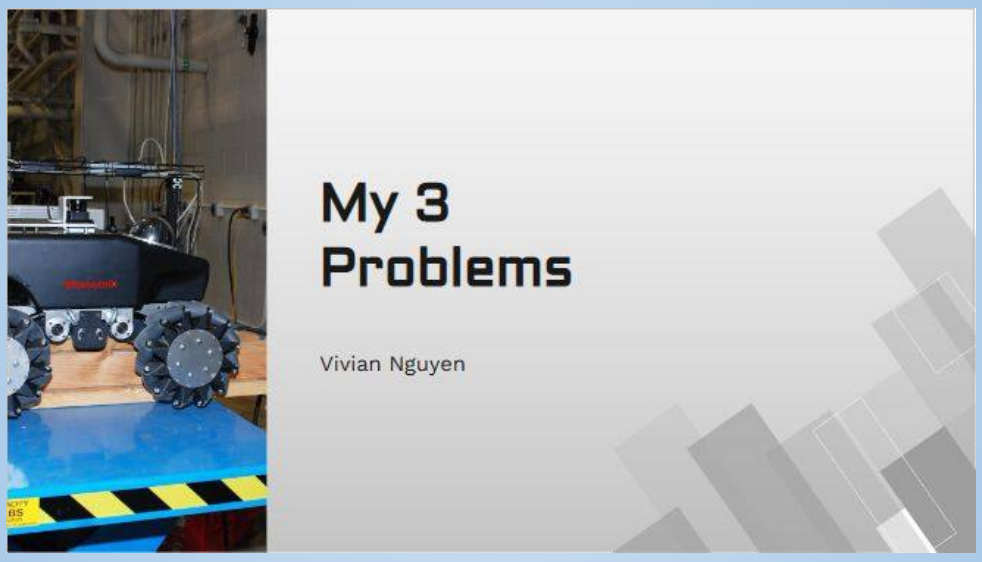
Top 1 Announcement

We are thrilled to announce that our final group members are Andy, Elijah, and Vivian. We chose to work together because of our shared interest in creating a rover mechanism, specifically one for surface pool cleaning. Each of us brings unique strengths to the project: Elijah excels at building, Andy is skilled at enhancing designs, and Vivian contribute to both her building and design skills. Our project will focus on developing a pool surface cleaner, and we're looking forward to solving the problem of surface debris.

Author:	Date:	Witness:	Date:
Elijah LaMothe	9/9/24	Vivian Nguyen	9/9/24

Top 3 Problem Presentations







Author:	Date:	Witness:	Date:
Elijah LaMothe	9/9/24	Vivian Nguyen	9/9/24

Weekly Logs

Week 1: Sep. 9 - Sep. 14

Goals for This Week

Start Research:
- Bacteria
- Percentages of those with pools
- Insects and leaves
- Percentages of owners with either salt water pools or chlorine pools
- What kinds of infections are caused by dirty pools

Tasks	Reflection
Establish Problem statement Research: Mon.-Tue.	- Our initial problem statement: We hate when trash accumulates in our pools. - Ask questions like how, who, what when creating problem statement

Notebook

Notebook Setup	- Cover page, table of contents, included each members initial 15 problems - Plan and design weekly log formatting - Update weekly log

Research

Create document for all research links	Vivian's Research Link (ALL) Elijah's Research: - Research 1 Research 2 Research 3 Andy's Research: - Research 1, Research 2, Research 3, Research 4
Research links	Research Links

Goals for Next Week

- Continue Researching
- Business Statements

Author:	Date:	Witness:	Date:
Elijah LaMothe	9/9/24	Andrew Magner	9/9/24

Swimming Pool Bugs: Types and Risks They Pose

Insects and bugs that can be dangerous:
- Mosquitoes
- Midges
- dobsonflies
- mayflies
- stoneflies

- All may see your pool as place to lay eggs
Most of these bite
- mosquitoes carry diseases

3 swimming pool bug types: water boatmen, backswimmers, and springtails

Waterboeman Bugs
- half an inch
- harmless to humans
- beneficial: known to eat mosquito larvae

feed on

Con: They attract and often confused with backswimmers

Backswimmers Bugs
- They bite; feels like a bee sting
- They countershade meaning they blend in and are hard to see,
- typically live on the bottom

*Difference between water boatmen and backswimmers is backswimmer swim on their back and boatmen swim "right side up"

Springtails
- they are harmless to humans but can get in the pool if there are seedlings or young or decaying plants.
- they like moist areas and eat decaying plants

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/16/24	Elijah LaMothe	9/16/24

How do Water Bugs Get in Your Pool?

- Springtails only end up in pools if they are looking for area with moisture and accidently fall in, but it can be included as a water bug because pool owners don't usually find one or two, but sometimes a whole population

- Water boatmen are naturally attracted to the pool as a habitat. They are either born there are fly in.

- These insects can be attracted to the outside light and the plants, move from nearby pond to the pool and lay their eggs

- If the Boatmen are being removed, both Boatmen and Backswimmers will increase in population

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/16/24	Elijah LaMothe	9/16/24

Weekly Logs

Week 2: Sep. 16 - Sep. 20

Goals for This Week

Continue and finish up research
- Research Summary Sheets

Construct Business Statements
- Rough Draft and Final Draft

Tasks	Reflection
Continue Research Construct Research Statements and Business Problem Statements	Keep Researching - Research Summary Sheets due Friday 9/20 Construct Problem and Business Problem Statements: - No personal opinions - Straight Factual Information - One or two sentences

Notebook

Notebook Setup	- Set goal for the week and reflect on lecture about Business Statements

Research

Create document for all research links	Vivian's Research Link (ALL) Elijah's Research: - Research 1 Research 2 Research 3 Andy's Research: - Research 1, Research 2, Research 3, Research 4
Compile all research into one document	Entire Research Document

Goals for Next Week

Create a problem and business statement

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/16/24	Elijah LaMothe	9/16/24

Research Notes: Dangers of Not Maintaining your Swimming Pool

The Dangers of Not Maintaining Your Swimming Pool

Possibilities of Recreational Water Illnesses
- there are many infections, different types, that can be caused by grow or survive in swimming pools
- these type of illnesses can easily be spread from water that contains germs.
  ↳ getting these illnesses come from contact of contaminated water or breathing the aerosolized mist in the air.
↳ Not just pools but lakes, rivers, oceans, and waterparks

- illnesses can be caused if someone w/ germs comes in contact to the water and contaminates it

~ or ~

- Natural germs that grow to levels that lead to infections

Some recreational water illnesses:
- skin infections
- diarrheal infections
- swimmer's ear
- and respiratory infections from breathing in the water mists of germs

Diarrheal infection is most common reported recreational water illness types and often caused by germs like cryptosporidium, norovirus, and E. coli.

Changes in Chlorine Levels and pH (crucial to killing germs)
- During the summer it is important to maintain the pH and chlorine levels since it is getting used more often.
- pH above 8 less effective to killing germs
- pH below 7 can lead to pipe corrosion

> Recommended: 7.2 – 7.8

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/18/24	Elijah LaMothe	9/18/24

Growth of Algae and other vegetation
- Algae can quickly breed and provide optimum environment for dangerous bacteria like E.coli to increase.
- damages pool
- perfect breeding ground for mosquitoes
- Pink and white slimy mold can also form affecting the filtration of your pool. (NO FILTER = MORE BACTERIA)

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/18/24	Elijah LaMothe	9/18/24

Study Reveal How Much Water Swimmer's Ingest

Study reveal how much water swimmers ingest

- Children ingest 2x more water than adults
- likely to ingest more water in swimming pools than any other body of water.
- Adults swallow 0 to 53mL of water
- children under 18 swallow 0 to 134mL
  ↳ 97% ingested 90mL or less

Importance: knowing the risks for recreational water illness transmissions

- A single mL of water can contain up to a TRILLION oocyles of cryptosporidium → causes infection and sickness w/ one to 100 oocysts.

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/23/24	Elijah LaMothe	9/23/24

Weekly Logs

Week 3: Sep. 23 - Sep. 28

Goals for This Week

- Input remaining research notes and research sheets
- Finish research and business statement. Create a survey with with relevant questions

Tasks	Reflection
Create business/research statement	- Written multiple drafts of research problem statements: make sure factual information is only included
Create survey	- Design and create relevant questions

Notebook

Set up week 3 log, finish entering research notes	- Include links for both the business and research drafts - Include link for survey - Create tasks, reflection, and goals for next week.

Statements

Business/ Statement Draft	https://docs.google.com/document/d/1fRVM-INL8iN48Y1bW4vf9QcO4yZneLLeoZMscsB_Cp4/edit?usp=sharing
Survey Link	https://docs.google.com/forms/d/14qStfGMPnO63JBaasmGRy6LYiBjQHKPG-xoGS4wViys/edit Form: https://forms.gle/MAZfBzPdAuyjVhqK7

Goals for Next Week

- Send out surveys
- FInish up the Business/Problem Statement Drafts (Get it looked over by Santana)

Author:	Date:	Witness:	Date:
Vivian Nguyen	9/27	Elijah LaMothe	9/27

Business/Problem Statement

Problem Statement:

In many areas of Southern California and other rural places, pools have remained a fundamental summer activity individuals look forward to, especially as the temperatures begin to rise in the months of June, July, and August. With both above and in-ground pools becoming increasingly popular, there is a surge in the initiative to ensure these spaces meet the sanitary standards. Based on our sources, there are over 10.7 million swimming pools throughout the U.S. alone. This information becomes even more alarming, as eight out of ten pool inspections find serious health code violations. These pools can carry harmful bacteria such as E. coli, which causes severe infections. The sanitation of pools is necessary as they become increasingly popular so that swimmers' health is not put at risk.

Business Statement:

Studies show that between the 10.7 million in the U.S. alone, eight out of ten pool inspections have found serious health code violations. This is a significant risk to swimmers across the country, risking the chance of infection among them, and jeopardizing the wellbeing of many.

Draft Document:

https://docs.google.com/document/d/1fRVM-INL8iN48Y1bW4vf9Qc04yZneLLeoZMscsB_Cp4/edit?usp=sharing

Author:	Date:	Witness:	Date:
Vivian Nguyen	10/1	Elijah LaMothe	10/1

Weekly Logs

Week 5: Sept. 30 - Oct. 4



Goals for This Week

- Send out surveys
- FInish up the Business/Problem Statement Drafts (Get it looked over by Santana)

Tasks	Reflection
Send out surveys	- Student and Parent Square (Go through Chavez for accessibility)
Final Draft for Business and Problem Statement	- Have Santana look over final draft - Include into team notebook
Survey Editing	Notes: More detailed of what type of pool structure, what time of day pool is cleaned, adjust price ranges, and accessories in the pool (inflatables, toys, etc.)
***ADD NOTES ON PATENT	Many designs seem to use some roller like mechanism so we might be able to do something like that. Every pool cleaner is powered or connected to something externally to power it.
Notebook	
Update Notebook	- Weekly Log: Week 3 - Tasks and add link - Add the Statements in (Business and Problem) Draft Link
Statements	
Survey Link	Engineering 4: Unclean Pools - Google Forms

Goals for Next Week

- Finalize list of patents and gauge what our design could look like.

Author:	Date:	Witness:	Date:
Vivian Nguyen	10/1	Elijah LaMothe	10/1

Weekly Logs

Week 6: Oct. 7 - Oct. 11

Goals for This Week

College Kick-Off Week (short week)
- Finish/Research Patents
- Brainstorm potential designs
- Finish progress of surveys

Tasks	Reflection
Work on Patents	Continue working on patents this week since its short week
College Kickoff: Interviews	Santana went over: - Interviews - Resume - Attire - etc.

Notebook

Update Notebook	- Make Week 6 Log

Research

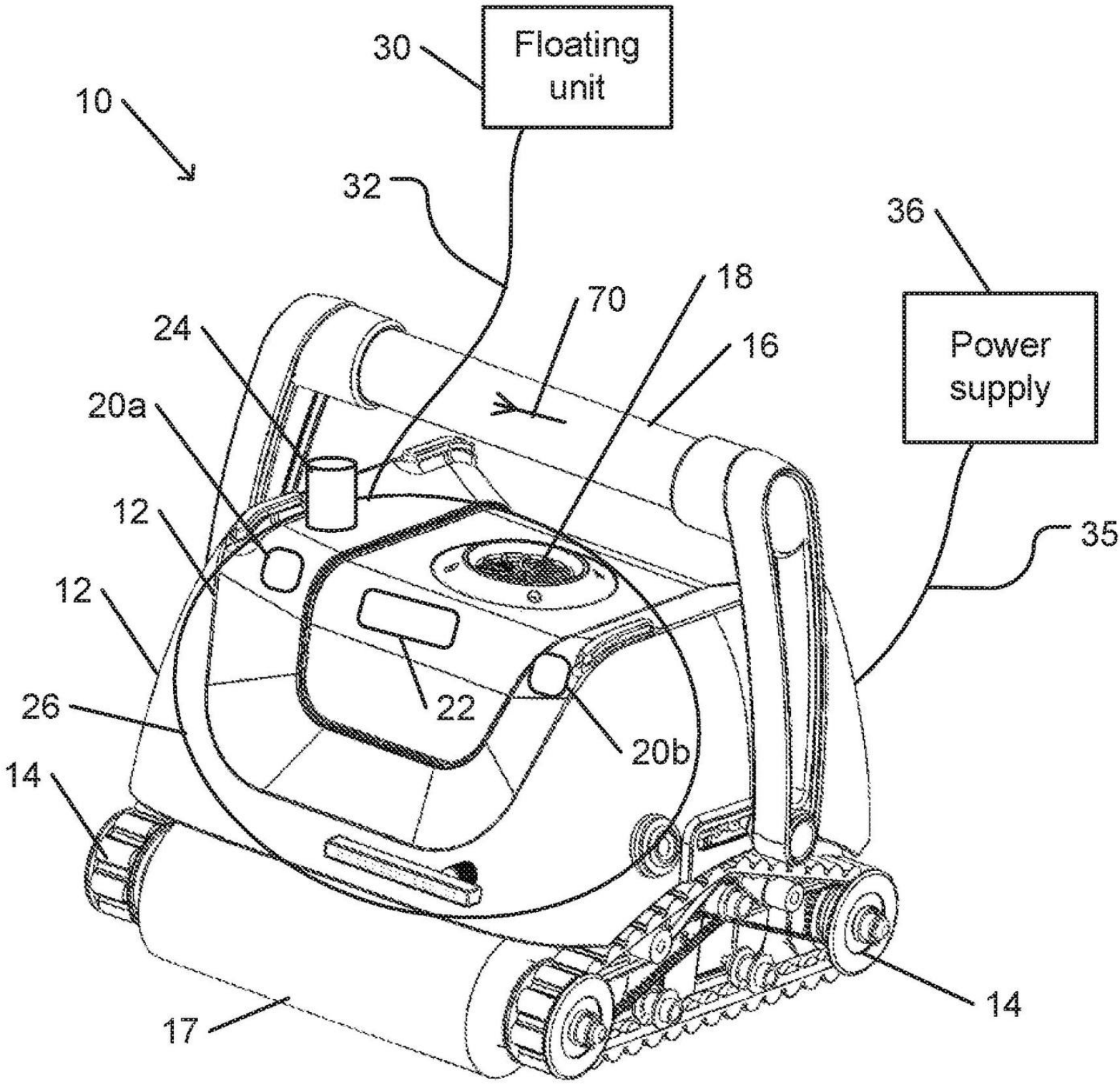
Patent Link	https://docs.google.com/document/d/1iNKoyRyuAQe2FUOTOCnQfakcjT240KPc1RkqgxnNGWk/edit?usp=sharing

Goals for Next Week

Finalize Patents and Surveys

Author:	Date:	Witness:	Date:
Vivian Nguyen	10/7	Elijah LaMothe	10/7

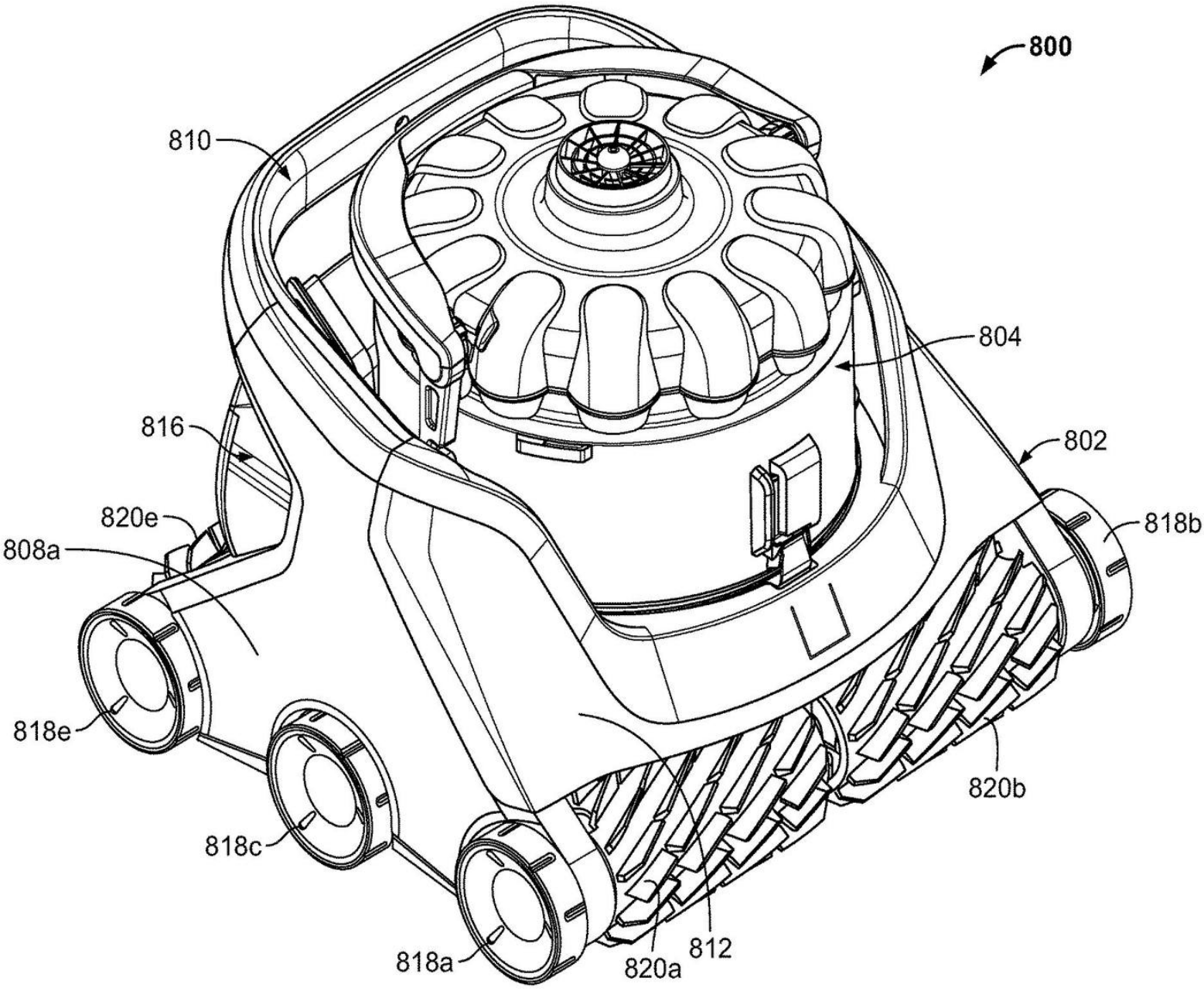
Patented Designs



US10876317B2

Author:	Date:	Witness:	Date:
Elijah LaMothe	10/14	Andy Diaz	10/14

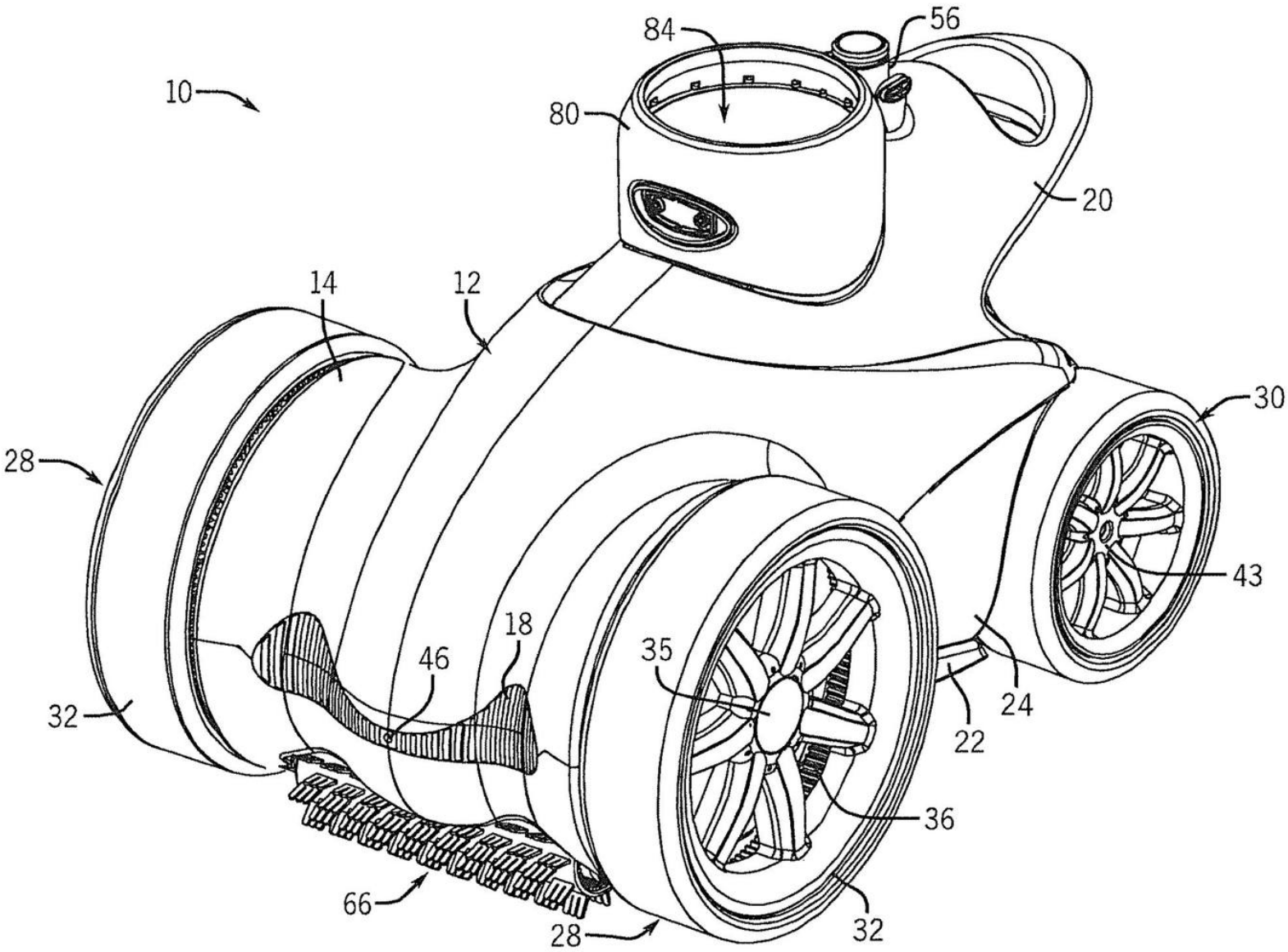
Patented Designs



US10767382B2

Author:	Date:	Witness:	Date:
Elijah LaMothe	10/14	Andy Diaz	10/14

Patented Designs



US10443259B2

Author:	Date:	Witness:	Date:
Elijah LaMothe	10/14	Andy Diaz	10/14

Patented Designs

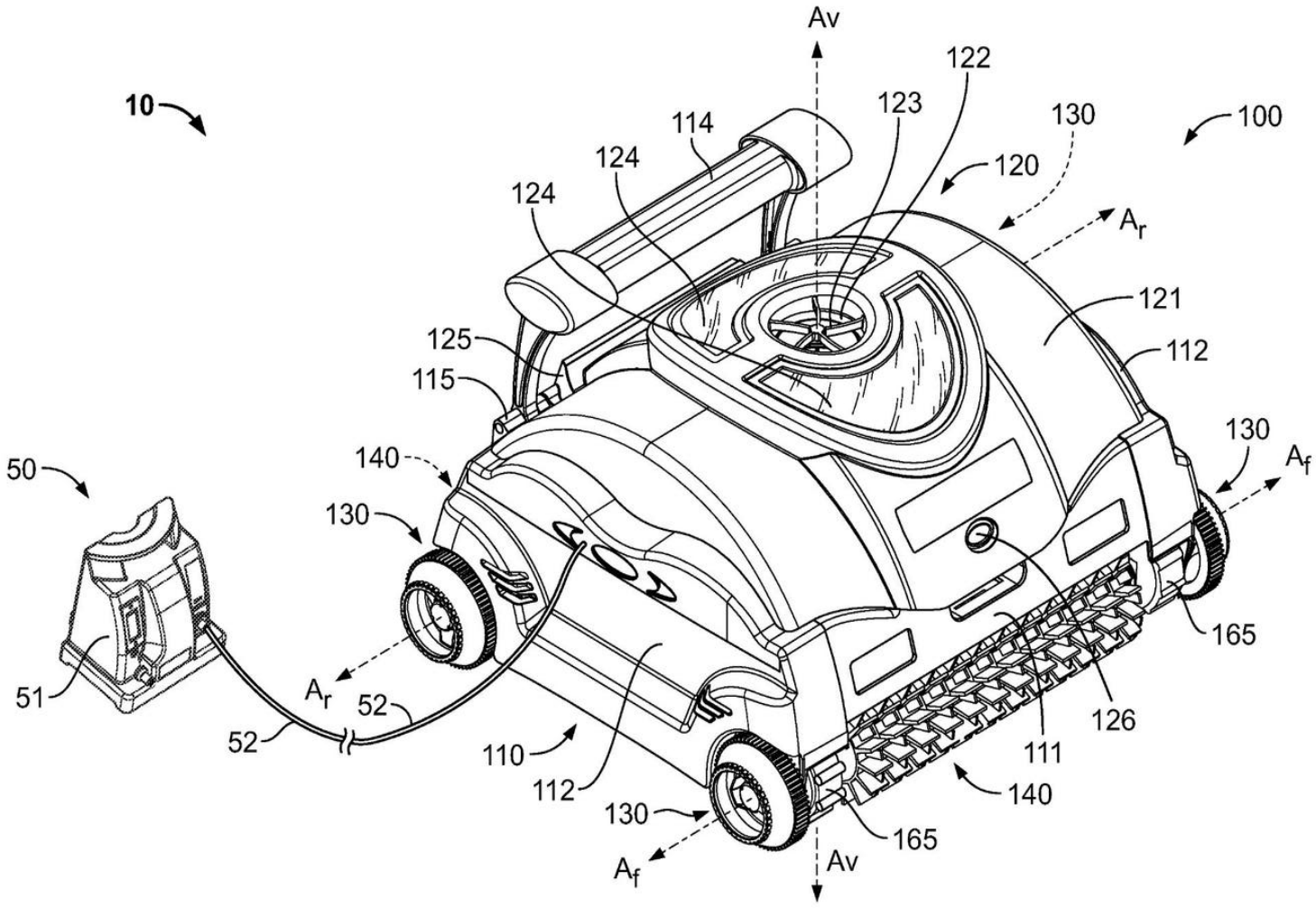


FIG. 1

US9677294B2

Author:	Date:	Witness:	Date:
Elijah LaMothe	10/14	Andy Diaz	10/14

Weekly Logs

Week 7: Oct. 14 - Oct. 18



Goals for This Week

- Finish patent research and compile them into one singular document
- Start the design process if ready
- Take into consideration the survey results and let that govern our design

Tasks	Reflection
Submit Patent research link to Santanna	After submitting the patent research link, we can successfully start our design ideas, but we still need to get the rest of our responses from our survey.
Start designs for the 15 problems submission	We started our design submissions, determining that we want to make devices that skim the top of the pool, travel to the bottom, and a hybrid of the two. So far, Andy's design does not involve any electronics which could play to our benefit.

Notebook

Update Notebook	- Make Week 7 Log - Add all the links that are involved (patent)

Research

Patent Link	Elijah - Patent Research Sheet Andy- Patent Research Sheet Vivian - Patent Research Sheet Compilation of Group Patents - Patents

Goals for Next Week

- Focus on making 15 designs
- Each person do 5 designs
- Submit Patents

Author:	Date:	Witness:	Date:
Vivian Nguyen	10/14	Elijah LaMothe	10/14

Weekly Logs

Week 8: Oct. 21 - Oct. 25

Goals for This Week

- Continue to work on 15 designs and try to get some that use no electronics
- Continue to promote survey on reddit, facebook, instagram, etc.
- Update notebook pages and make sure everything is in check

Tasks	Reflection
15 Designs Submission	Designs without electronics seem to be a very challenging task but can be done,
Continue 15 Designs	Due sometime after 12-week grading period.
Survey	Keep reviewing progress of surveys

Notebook

Update Notebook	- Make Week 7 Log - Add all the links that are involved (patent)

Research

Patent Link	Elijah - Patent Research Sheet Andy- Patent Research Sheet Vivian - Patent Research Sheet Compilation of Group Patents - Patents

Goals for Next Week

- Continue to design 15 drafts
- Check survey progress
- Create presentation for Mrs. Ede's Class (Done by Thursday)

Author:	Date:	Witness:	Date:
Vivian Nguyen	10/21	Elijah LaMothe	10/21

Weekly Logs

Week 9: Oct. 28 - Nov. 1



Goals for This Week

- Create slide presentation to present to Ms. Ede's class
- Keep working on 15 designs
- Continue working on surveys

Tasks	Reflection
Create slides for Ms. Ede's class	Presentation: - Name/Project Title - Problem Statement - Stats - Sources - Market Research - Survey results - Summary *Stakeholders
Notebook	
Update Notebook	Make Log Week 9 - Add goals for this week and tasks
Research	
15 designs	- Continue research and brainstorming potential designs

Goals for Next Week

- Finish up the 15 designs by the end of the week
- Get 250 surveys

Author:	Date:	Witness:	Date:
Vivian Nguyen	10/28	Elijah LaMothe	10/28

Presentation Pitch



Author:	Date:	Witness:	Date:
Vivian Nguyen	10/29	Elijah LaMothe	10/29

Weekly Logs

Week 10: Nov. 4 - Nov. 8

Goals for This Week

- Finish up the 15 designs by the end of the week
- Get 250 surveys

Tasks	Reflection
Santana's Announcement	May 30th - Showcase (Final Deadline) <u>Notes 11/8: Going to Ms. Eid's Thursday 11/14/24</u> - How should they be accessed? - Product Name - 3 words of how the innovations can be described
Notebook	
Update Notebook	Make Log 10 - Add goals for this week and tasks
Research	
Surveys due	- Need to surpass additional 50 in order for it to go in gradebook
Continue 15 Designs	- Finish 15 Designs before grading period

Goals for Next Week

- Ms. Eid's meeting in her classroom on Thursday 11/14
- Finish up designs and notebook
- Check progress of surveys

Author:	Date:	Witness:	Date:
Vivian Nguyen	11/4	Elijah LaMothe	11/4

Weekly Logs



Goals for This Week

- Ms. Eid's meeting in her classroom on Thursday 11/14
- Finish up designs and notebook
- Check progress of surveys

Tasks	Reflection
Santana's Announcement	<u>Notes 11/8: Going to Ms. Eid's Thursday 11/14/24</u> - How should they be accessed? - Product Name - 3 words of how the innovations can be described
Notebook Check	- Due Friday

Notebook

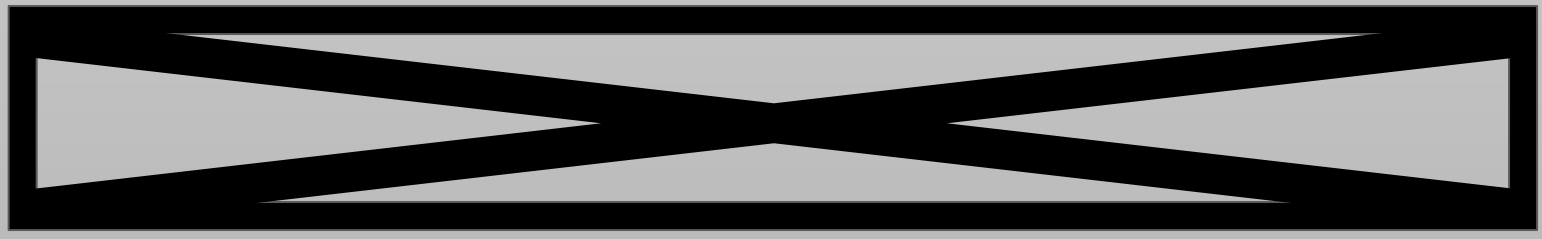
Update Notebook	Make Log 11 - Add goals for this week and tasks Notebook Check on Friday 11/15

Research

Prepare for meeting with Ms. Eid's class	- How should they be accessed? - Product Name - 3 words of how the innovations can be described

Goals for Next Week

Keep working on designs

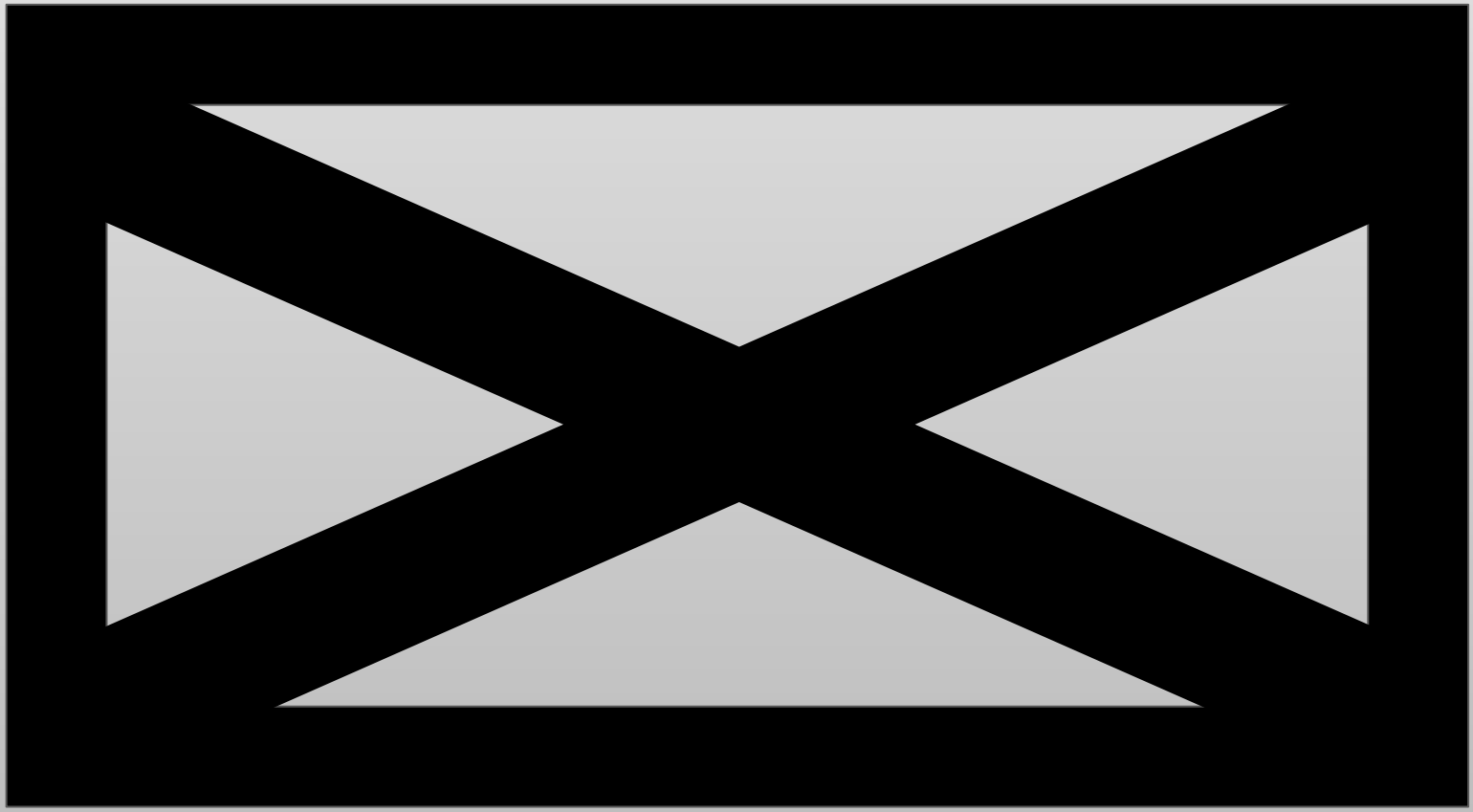


Author:	Date:	Witness:	Date:
Vivian Nguyen	11/13	Elijah LaMothe	11/13

Weekly Logs

Eid's class meeting recap

- Design team partners: Diego, Hector, someone else who wasn't there
- What they asked/What we answered:
- Q: Do we want a futuristic theme? Modern?
- A: Futuristic-ish
- Q: What is the price range of this product?
- A: Between $150-$200
- Q: Would the device include solar panels and if so, how?
- A: We are still going through the design process, and if we were to have solar panels, they would be more of an add-on to our design, not the sole way the device is powered.
- Q: What kind of places would this product be sold at?
- A: Primarily home owner oriented stores such as home-depot or local mom and pop shops that houses products for home maintenance



Author:	Date:	Witness:	Date:
Vivian Nguyen	11/14	Elijah LaMothe	11/14

Weekly Logs

Week 12: Nov.18 - Nov. 22

Goals for This Week

- Ms. Eid's meeting in her classroom on Thursday 11/14
- Finish up designs and notebook
- Check progress of surveys

Tasks	Reflection
Make decision for top three designs	- Top three designs are looking like the skimmer bot, the UFO looking one, and the conveyor type cleaner - Most of us think that the skimmer bot will be the most fun to build
Set up and complete design matrix	- Will most likely just make this on google sheets and decide there - Try to complete this before end of week

Notebook

Update Notebook	Make Log 12 - Add goals for this week and tasks

Research

What kind of batteries/operating systems do we need for this project?	- Research showed that we need around a 7.2 Volt battery with around 7000 Amp Hours - Santana said for this project we shouldn't need anything more than a an arduino even if we decide to use a good amount of sensors and whatnot

Goals for Next Week

THANKSGIVING BREAKKKKK

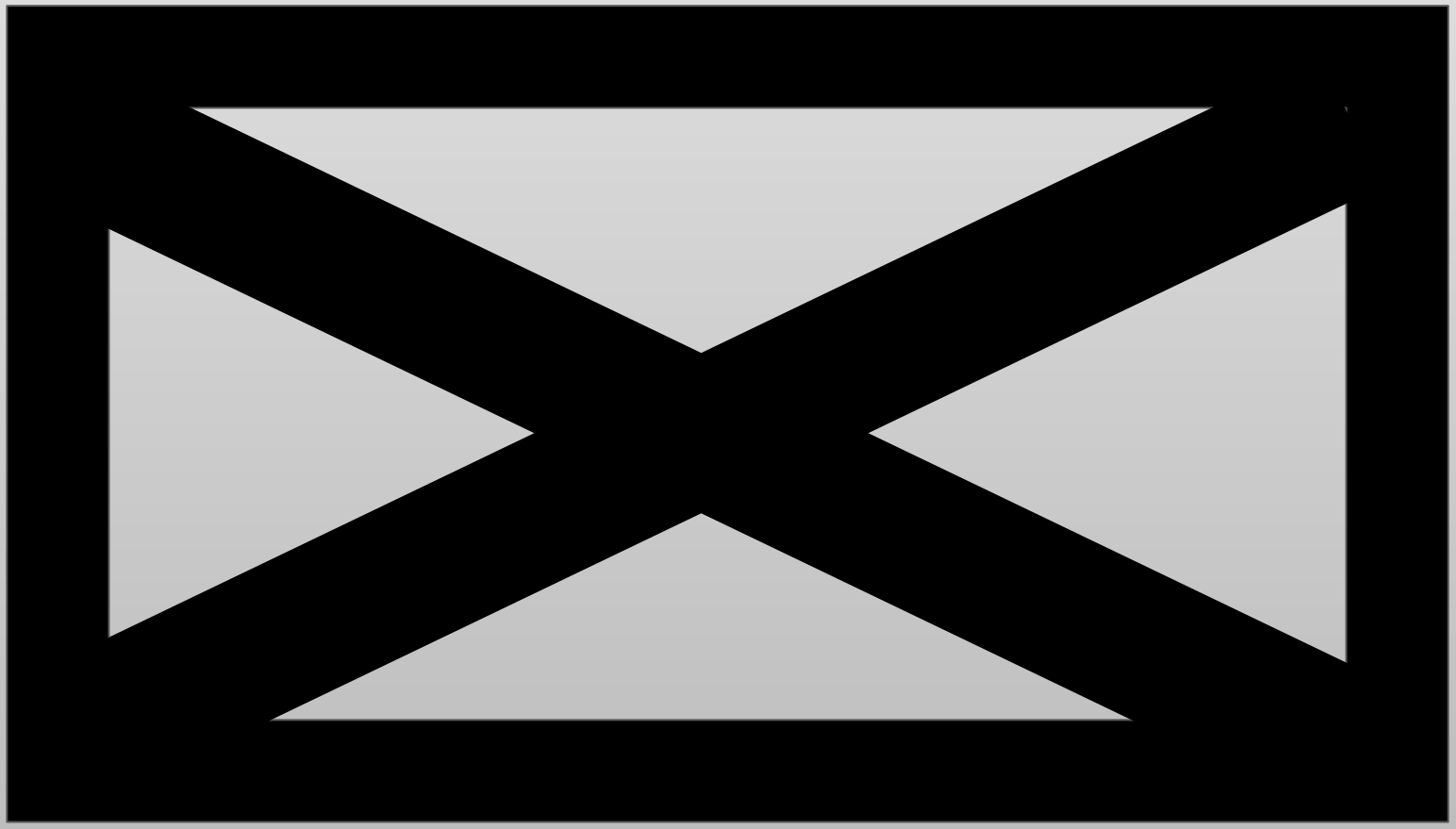
Author:	Date:	Witness:	Date:
Vivian Nguyen	11/22	Elijah LaMothe	11/22

Design Folder

https://drive.google.com/drive/folders/1GPAze-K7pnVzsGcpP0xa3vgkFiyIHiIl?usp=drive_link

Decision Matrix

	Multiplier: x1.5				Multiplier: x1.5					Multiplier: x2	Multiplier: x3			
	5	5	5		5	5	5	5	5	5	5			
	Weight	Efficiency/Speed	Easy Internal Access		Size/buoyancy	Easy Maintenence		Battery Consumption		Complexity	Fun Factor			
												Total	WINNER	
Surface Skimmer	2+1	4	4		3+1.5	4		2		3+3	4+8	42.5		
												Total		
UFO style skimmer	4 +2	2	4		5+2.5	4		4		4+4	2+4	41.5		
												Total		
Conveyor Bot	2+1	5	3		2+1	4		1		2+2	4+8	35		



Author:	Date:	Witness:	Date:
Vivian Nguyen	11/22	Elijah LaMothe	11/22

Weekly Logs

Week 13: Dec. 2 - Dec. 6 36

Goals for This Week

Start collecting and ordering materials
- Place orders into spreadsheet

Tasks	Reflection
Amazon ordering	- Pool Noodle - Poxy Glue - Servos - Bilge Pumps
Research and begin CADDing	- Work and research on sheet metal

Notebook

Update Notebook	Make Log 13 - Add goals for this week and tasks

Research

Look for materials and start building	- Figure out what type of servos to use (preferably waterproof)

Goals for Next Week

- Start building
- Wiring and electrical once parts come in (Solder)
- Coding and get something running

Author:	Date:	Witness:	Date:
Vivian Nguyen	12/3	Elijah LaMothe	12/3

Motor Code

```
const int motorPin1 = 8; // Pin connected to IN1 of the motor driver
const int motorPin2 = 7; // Pin connected to IN2 of the motor driver
const int motorPin3 = 5;
const int motorPin4 = 4;
const int enablePin = 9; // Pin connected to ENA (PWM control) of the motor driver
const int enablePin2 = 3; // Pin connected to ENA (PWM control) of the motor driver


void setup() {
  // Set motor control pins as outputs
  pinMode(motorPin1, OUTPUT);
  pinMode(motorPin2, OUTPUT);
  pinMode(motorPin3, OUTPUT);
  pinMode(motorPin4, OUTPUT);
  pinMode(enablePin, OUTPUT);
    pinMode(enablePin2, OUTPUT);



  // Initialize the motor in the OFF state
  digitalWrite(motorPin1, LOW);
  digitalWrite(motorPin2, LOW);
    digitalWrite(motorPin3, LOW);
  digitalWrite(motorPin4, LOW);
  analogWrite(enablePin, 0); // PWM set to 0 (motor off)
    analogWrite(enablePin2, 0); // PWM set to 0 (motor off)

}

void loop() {
  // Turn the motor on for 3 seconds
  digitalWrite(motorPin1, HIGH); // Set IN1 high
  digitalWrite(motorPin2, LOW);  // Set IN2 low (motor forward direction)
   digitalWrite(motorPin3, HIGH); // Set IN1 high
  digitalWrite(motorPin4, LOW);  // Set IN2 low (motor forward direction)
  analogWrite(enablePin, 255);   // Full speed
    analogWrite(enablePin2, 255);   // Full speed

  delay(3000);                   // Wait 3 seconds

  // Turn the motor off
  digitalWrite(motorPin1, LOW);
  digitalWrite(motorPin2, LOW);
    digitalWrite(motorPin3, LOW);
  digitalWrite(motorPin4, LOW);
  analogWrite(enablePin, 0);     // Motor off
    analogWrite(enablePin2, 0);     // Motor off
```

Author:	Date:	Witness:	Date:
Vivian Nguyen	12/3	Elijah LaMothe	12/3

Weekly Logs

Week 14: Dec. 9 - Dec. 13

Goals for This Week

- Work on code and getting all the motor working
- Get servos up and running

Tasks	Reflection
Work on the code and wiring of the the boards	- Solder up wires and wire management - Get one motor to work in order to get the 2 other pumps working - Work on servos

Notebook

Update Notebook	Make Log 14 - Add goals for this week and tasks - research

Research

Wiring for bilge pumps and implementing motor controller	On the motor controller there are two enable pins and four motor controller ports. You can switch around the motor controller ports to reverse the motors if needed, but that can also be done in the code by switching from HIGH to LOW and vice versa. It is also worth noting that the motor controller needs to be connected to the same power supply as the arduino, and grounded the same as well.

Goals for Next Week

- Set up all three bilge pumps with the servo
- Solder any of the wires that need to be connected, and protect them with shrink wrap

Author:	Date:	Witness:	Date:
Vivian Nguyen	12/9	Elijah LaMothe	12/9

Weekly Logs

Week 15: Dec. 16 - Dec. 19

Goals for This Week

- Set up all three bilge pumps with the servo
- Solder any of the wires that need to be connected, and protect them with shrink wrap

Tasks	Reflection
- Create code for 3 bilge pumps and servo	- Because each motor controller can only take two motors, we had to connect an additional one to the arduino so that a third motor could be powered
- Solder ground and power wires	- It is easier to solder if you hold the tip still and slowly feed the lead wire through at a steady pace. You can use the soldering iron on the shrink wrap as well.

Research

How are the motors and servos powered through code?	Firstly, the power supply should run at around +2 volts than what the motors require, so since our motors are 12V motors, we run the power supply at 14 volts. Additionally, the RC controllers have both enable and motor pins which are connected to the arduino, then declared in the code. This has to be done with each of the motors, and will not run unless each of the pins are labeled directly in the code.

Goals for Next Week

Get CAD design for robot base and cut the polycarbonate.

Author:	Date:	Witness:	Date:
Vivian Nguyen	12/19	Elijah LaMothe	12/19

Motor Code
(3 motors and servo)


```
// Arduino sketch to power a brushed DC motor for a few seconds
#include <Servo.h> //including servo library
Servo huzz;

// Define the motor control pins
const int motorPin1 = 6; // Pin connected to IN1 of the motor driver
const int motorPin2 = 5; // Pin connected to IN2 of the motor driver
const int motorPin3 = 4;
const int motorPin4 = 3;
const int motorPin5 = 12;
const int motorPin6 = 11;
const int enablePin = 7; // Pin connected to ENA (PWM control) of the motor driver
const int enablePin2 = 2; // Pin connected to ENA (PWM control) of the motor driver
const int enablePin3 = 13;




void setup() {
  // Set motor control pins as outputs
  pinMode(motorPin1, OUTPUT);
  pinMode(motorPin2, OUTPUT);
  pinMode(motorPin3, OUTPUT);
  pinMode(motorPin4, OUTPUT);
    pinMode(motorPin5, OUTPUT);
  pinMode(motorPin6, OUTPUT);
  pinMode(enablePin, OUTPUT);
    pinMode(enablePin2, OUTPUT);
        pinMode(enablePin3, OUTPUT);
          huzz.attach(9); //servo attatched to port 9


```

[This is the link to this code on GitHub]

Author:	Date:	Witness:	Date:
Vivian Nguyen	12/19	Elijah LaMothe	12/19

Weekly Logs

Week 16: Jan. 13 - Jan. 17

Goals for This Week

- Update notebook from over the week and past few weeks
- Create CAD model of robot base and
- Implement potentiometer or switch for motor controller

Tasks	Reflection
Create CAD design of robot	- The sheet model design of the robot in onshape has edges, or rather places where the polycarbonate is meant to be folded, which can be done with our laser cutter.
Add potentiometer or switch	- The potentiometer did not work as well as we thought, we had issues with the wiring and managing of power and ground, as well as how to program it, so we went with a switch instead.

Research

How does the potentiometer or switch work? How will we cut/design the robot base?	The potentiometer has a range of values from 0-1023, which we would use to limit the amount of current surging through the system. So, when the potentiometer value is 0, there is no power, and when it is 1023, there is max power. However, this system is overcomplicated and could easily be replaced with a switch. After realizing this, we put a switch on the power wire for the system and it now works smoothly.

Goals for Next Week

- Build and construct the storage unit for the material

Author:	Date:	Witness:	Date:
Vivian Nguyen	1/14	Elijah LaMothe	1/14

Potentiometer and Servo code

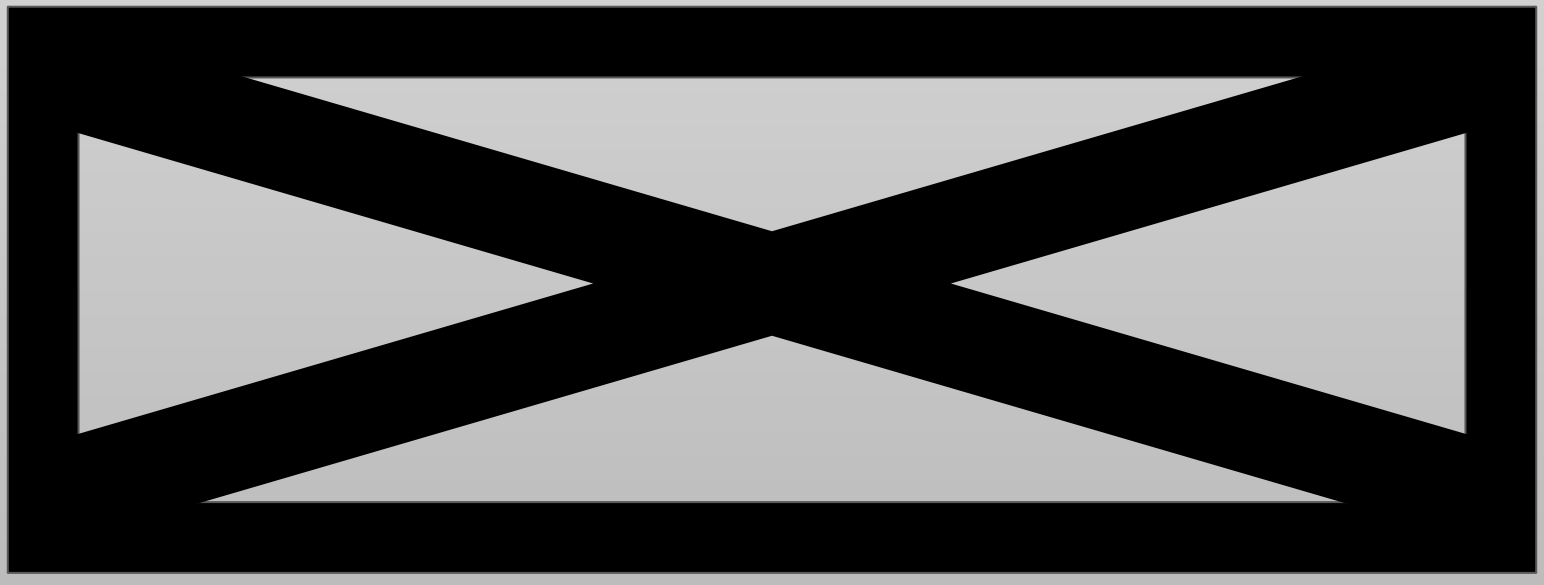
```
#include <Servo.h>

Servo myServo;  // Create a Servo object to control the servo

int potPin = A0;  // Pin where the potentiometer is connected
int val = 0;       // Variable to store the potentiometer value

void setup() {
  myServo.attach(9);  // Pin where the servo is connected
}

void loop() {
  val = analogRead(potPin);            // Read the potentiometer
  val = map(val, 0, 1023, 0, 180);     // Map the value to servo angle range (0-180 degrees)
  myServo.write(val);                  // Set the servo position
  delay(15);                           // Wait for the servo to reach the position
}
```



Author:	Date:	Witness:	Date:
Vivian Nguyen	1/14	Elijah LaMothe	1/14

Weekly Logs

Week 17-18: Jan. 21 - Jan. 31

Goals for This Week

- Create prototypes
- Begin CAD

Tasks	Reflection
CAD	- Elbow adapter for the outlet of bilge pump Purpose: Redirect the running water away in another direction Status: Printed - Bilge pump Adapter Purpose: Connect the bilge pump in another direction to ultimately attach to the base Status: Printed - Filtration Basket Purpose: The inlet of the debris, it will have servos connected to the basket, filters the debris out of the water as it passes through the water Status: CAD Completed
Build and Assembly	- Decided to begin creating a prototype so we can start getting something up and floating - Used a plastic container as our base and attached 3 bottle to the bottom of the container for an open air space for basket Plan: - Container will contain all our power and wiring so the water does not reach - Drill the bilge pump adapters into the bottom of the bucket, continuing to waterproof the inside of the container

Goals for Next Week

- Begin printing out CADS
- Complete wire management and get the robot up and running again
- Update Notebook
- Add parts to prototype exterior

Author:	Date:	Witness:	Date:
Vivian Nguyen	1/21	Elijah LaMothe	1/21

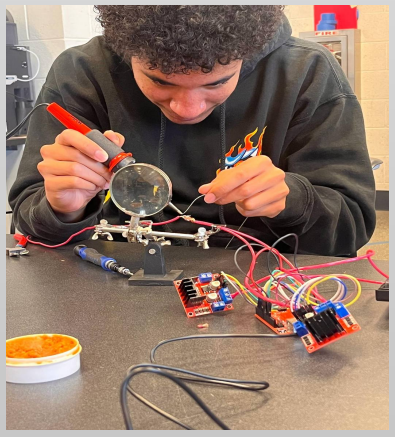
Weekly Logs

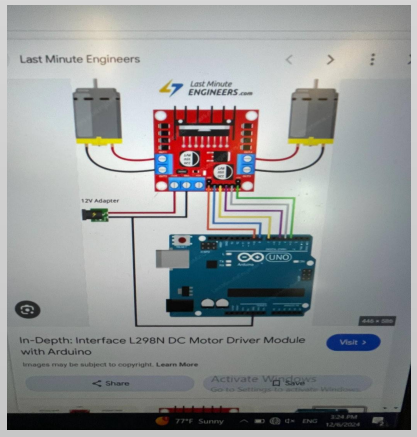
Week 19: Feb. 3 - Feb. 7

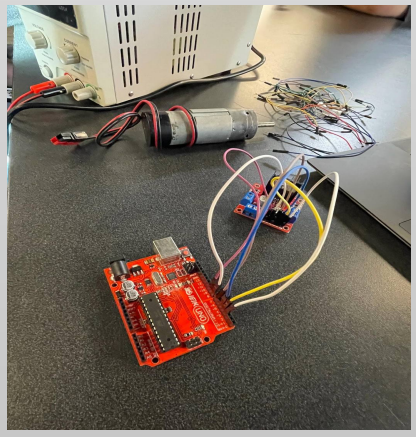
Goals for This Week

- Begin printing out CAD
- Complete wire management and get the robot up and running again
- Update Notebook
- Add parts to prototype exterior

Tasks	Reflection
CAD	- Completion of bilge pump and filtration basket BAD - All we need to do is print - Alter our prototype design
Wiring Management	- Connect battery adapter to arduino and motor control - Solder lower gauge wire with battery adapter wire
Adjustments	- Took out servos to test the bilge pumps





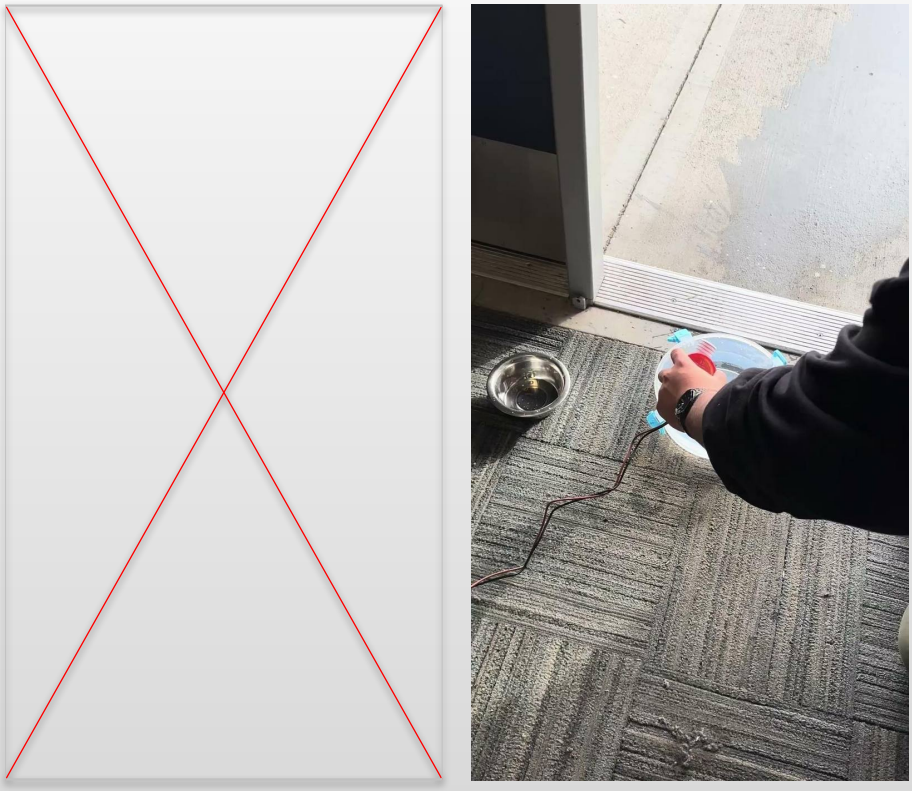


Goals for Next Week

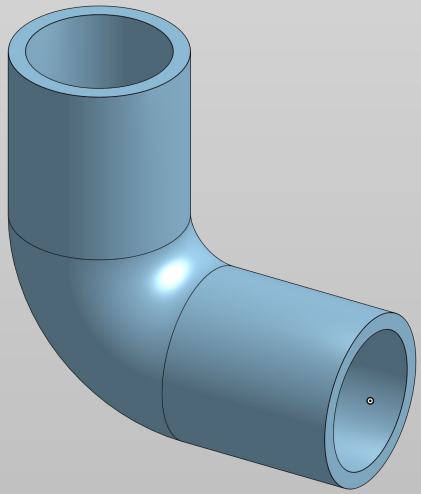
- Complete the bilge pump adapter and add into onto out base
- Get the robot up and running, floating specifically
- Get all the batteries and power supply running
- If anything, put it into the water

Author:	Date:	Witness:	Date:
Vivian Nguyen	2/4	Elijah LaMothe	2/4

Wiring, Build, CAD Results, and Assembly Prototype Photos



Stock bilge pump test



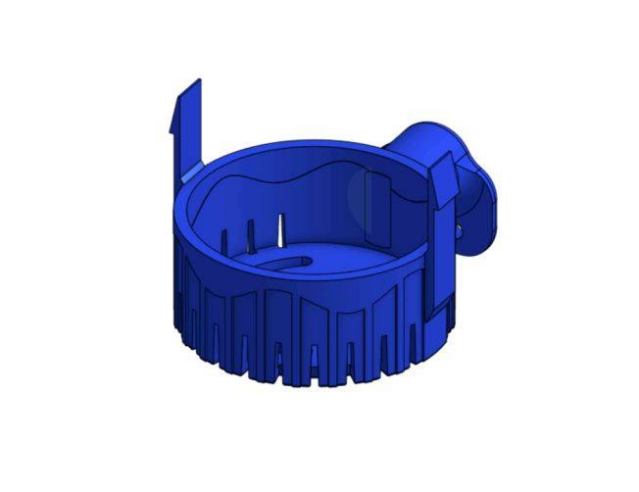


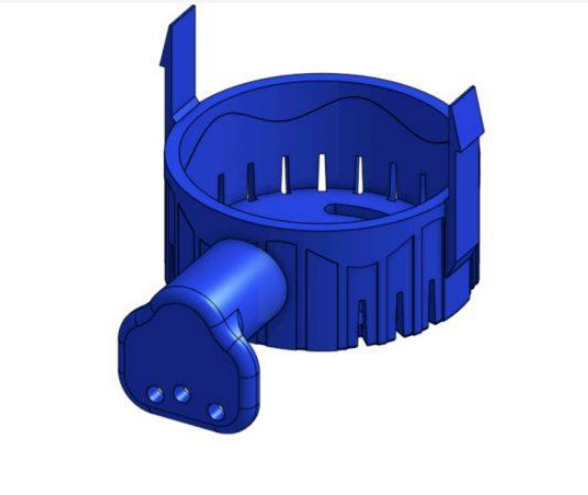


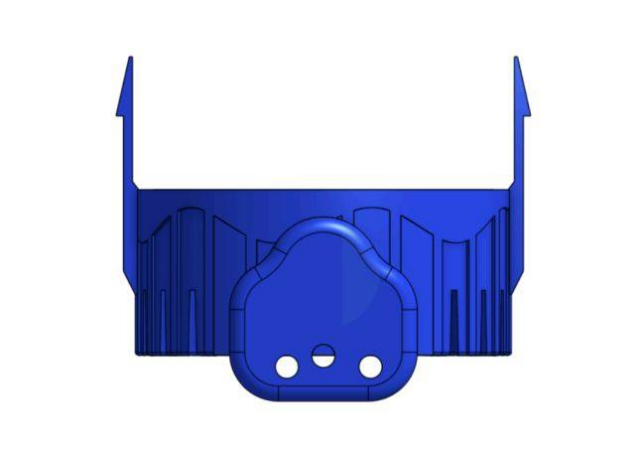
Bilge pump elbow

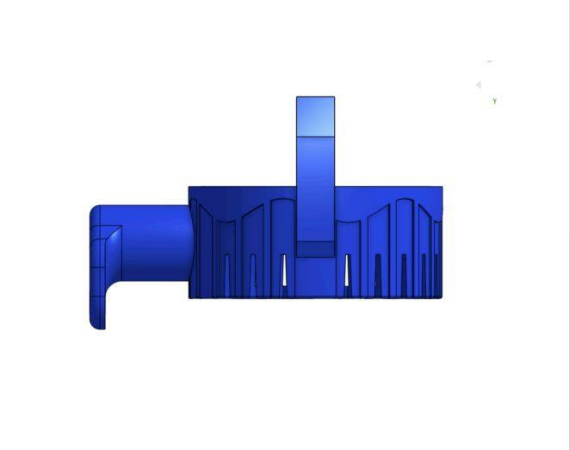
Author:	Date:	Witness:	Date:
Vivian Nguyen	2/4	Elijah LaMothe	2/4

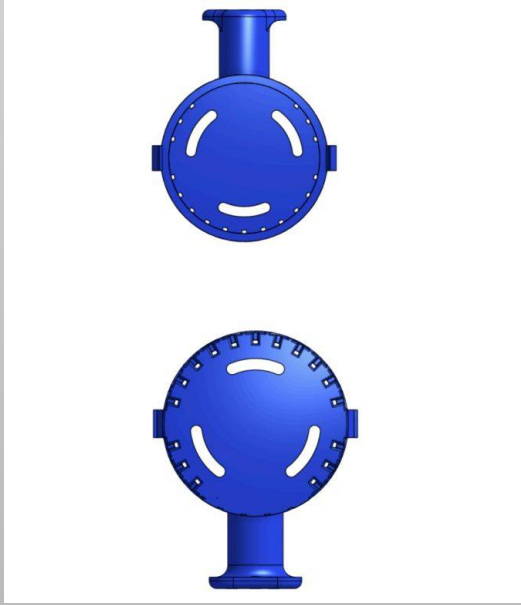
Bilge Pump Base Adapter CAD







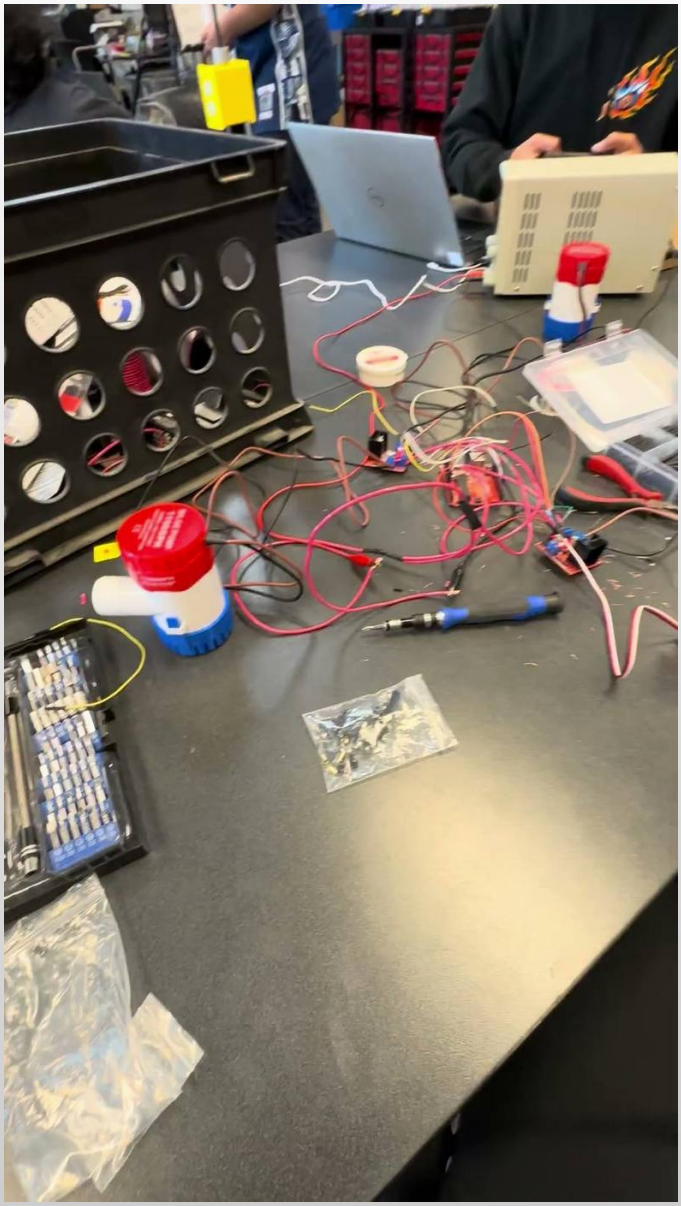


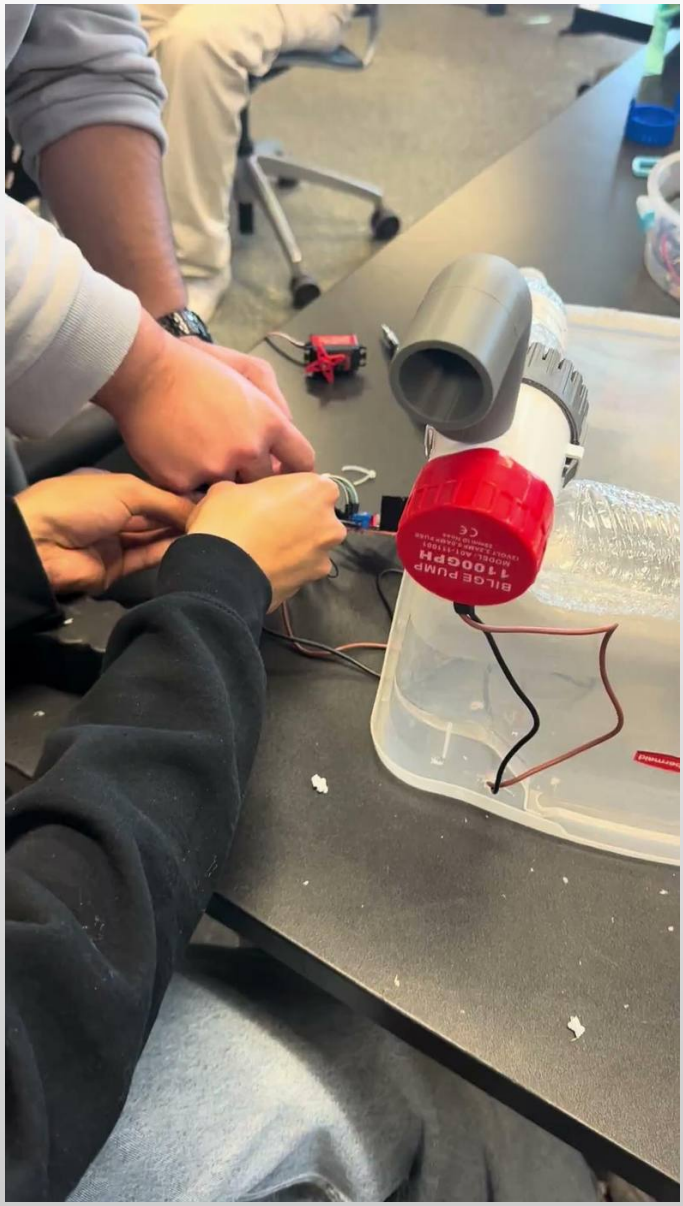




Author:	Date:	Witness:	Date:
Vivian Nguyen	2/11	Elijah LaMothe	2/11

Assembly Prototype Photos/Video





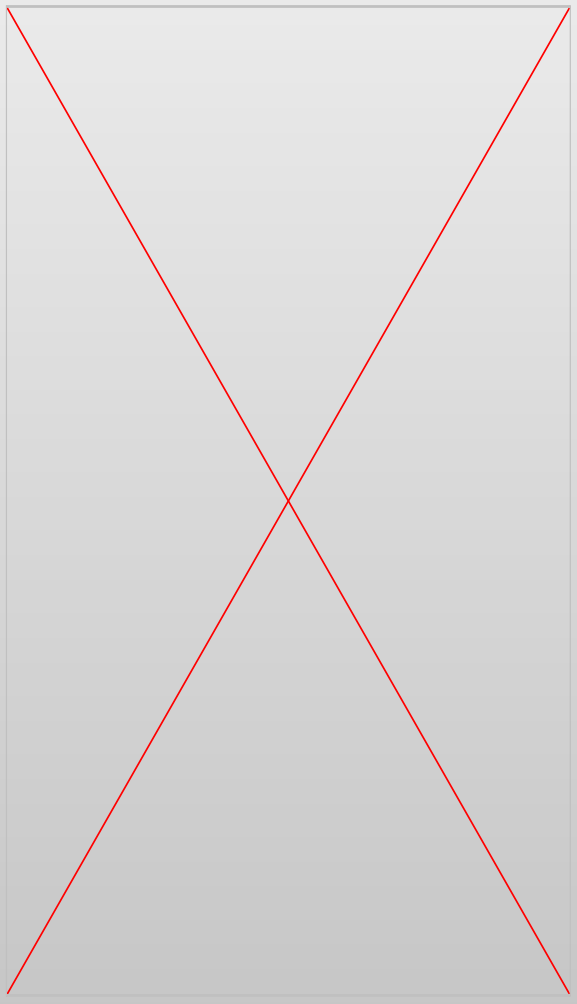
Author:	Date:	Witness:	Date:
Vivian Nguyen	2/11	Elijah LaMothe	2/11

Weekly Logs

Week 20: Feb. 11 - Feb. 15

Goals for This Week

- Completed all CAD except the filtration basket - plan to finish by this week
- Wiring complete - waiting for assembly of prototype
- Assemble



Goals for Next Week

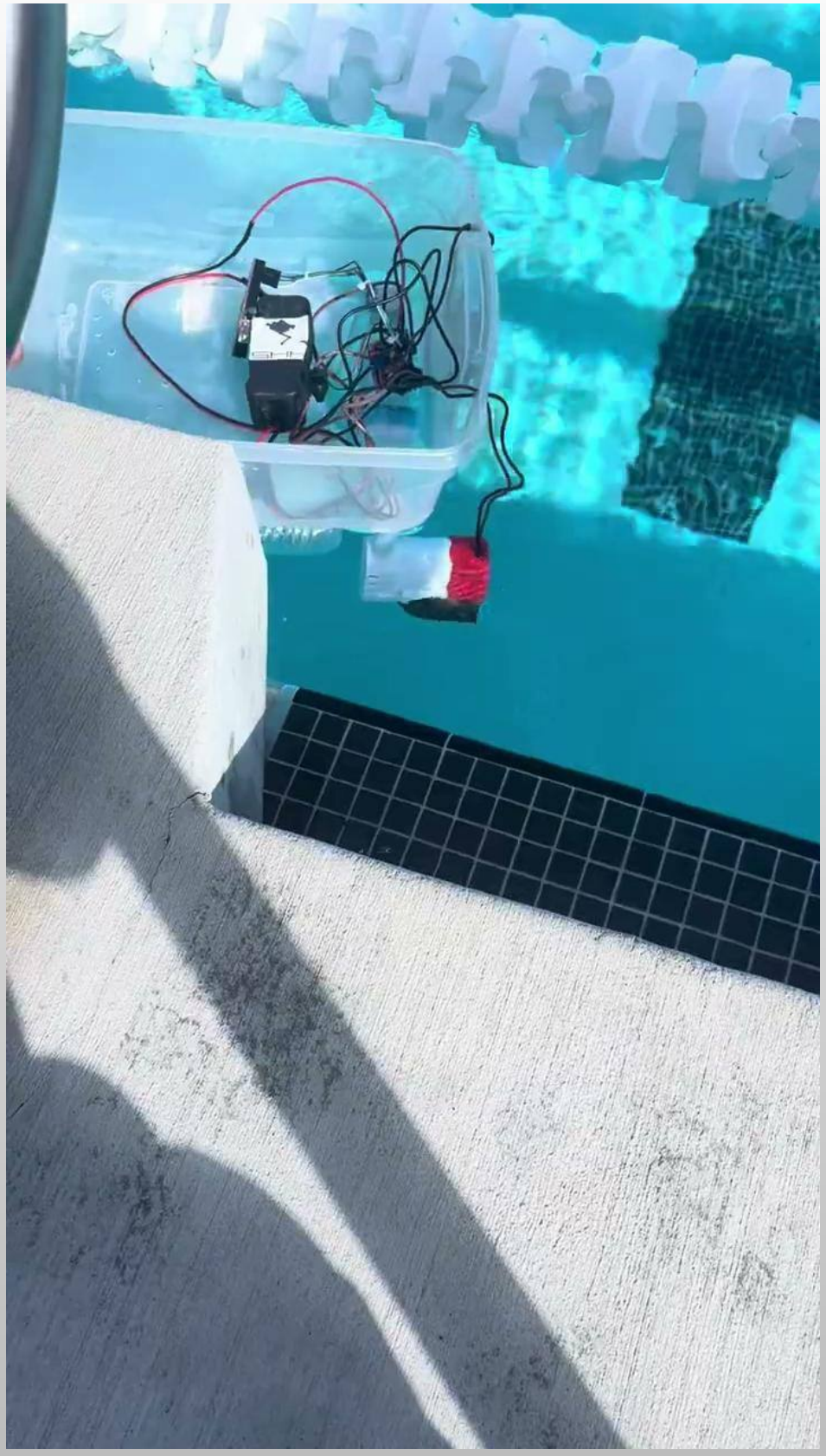
We plan to add the filtration basket to the apparatus

- Continue testing the device and mechanism

Tasks	Reflection
CAD	- Completion of bilge pump base adapter, servo adapter, elbows
Wiring Management	- Solder new wiring for the batteries - Loop wiring through the side of the base container
Assembly	- Drill base adapter onto the plastic container and add hydrepoxy for extra support
Testing	- We took the apparatus out to the pool to see if it functions how we hoped. - Only thing we ran into was our code was not pushing one of our bilge pump in the right direction - The apparatus floats on the water but does has an imbalance depending on where the battery is the placed.

Author:	Date:	Witness:	Date:
Vivian Nguyen	2/11	Elijah LaMothe	2/11

Testing Video at the Pool



Author:	Date:	Witness:	Date:
Vivian Nguyen	2/11	Elijah LaMothe	2/11

Sonar Code

```
#include <Wire.h>
#include <Adafruit_SSD1306.h>


Adafruit_SSD1306 display(128, 64, &Wire, -1);


void setup() {
  // put your setup code here, to run once:

  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.display();

}

void loop() {
  // put your main code here, to run repeatedly:

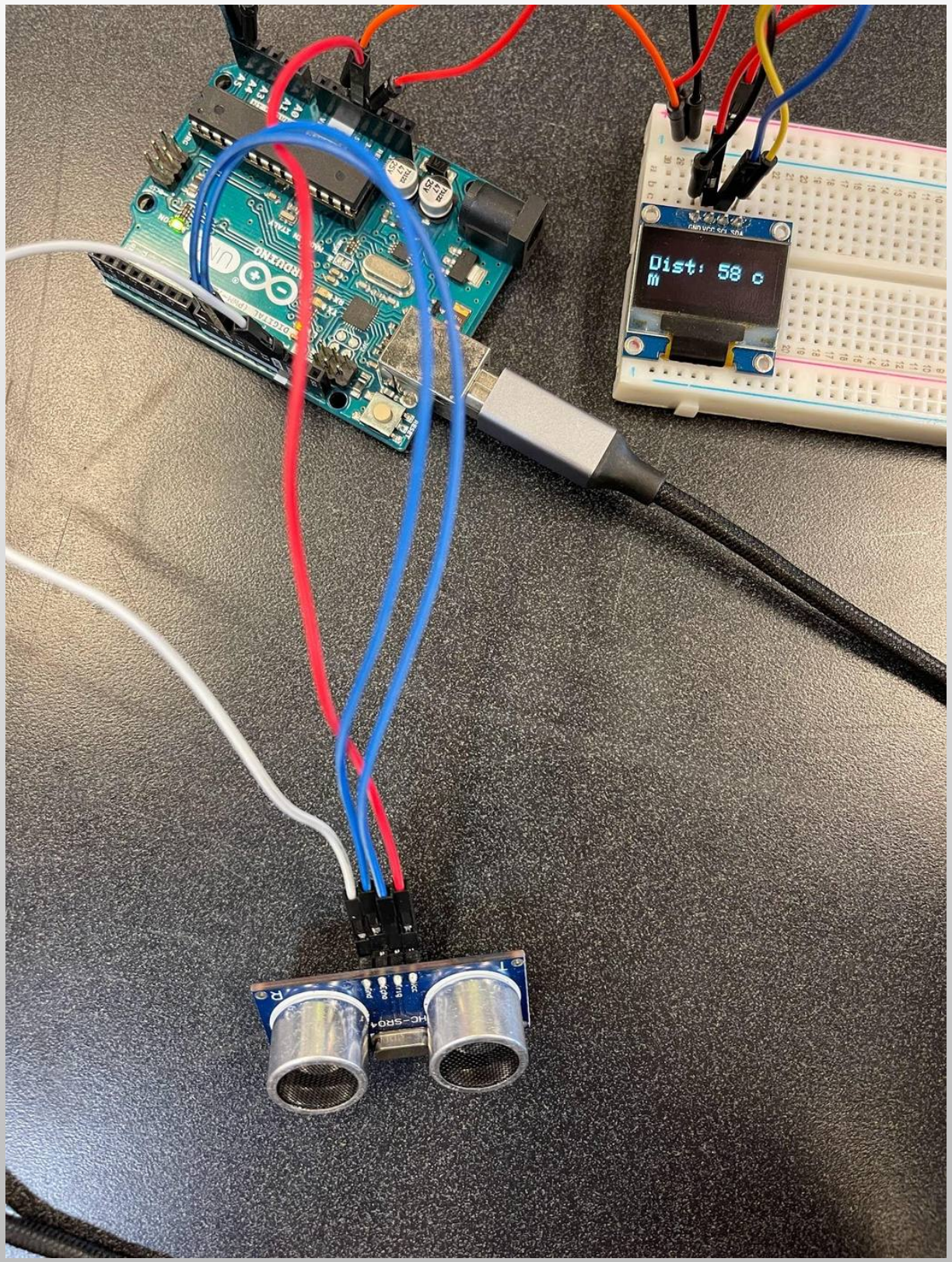
  display.setTextSize(1.5);
  display.setTextColor(WHITE);

  display.clearDisplay();
  display.setCursor(0,0);
  display.println("Hello Stewie");
  display.display();

}
```

Author:	Date:	Witness:	Date:
Vivian Nguyen	2/12	Elijah LaMothe	2/12

Sonar Setup



Author:	Date:	Witness:	Date:
Vivian Nguyen	2/12	Elijah LaMothe	2/12

Weekly Logs

Week 21: Feb. 24 - Feb. 28

Goals for This Week

- Continue testing the robot
- Work on the filtration baskets and adding it onto the base
- Update notebook and code

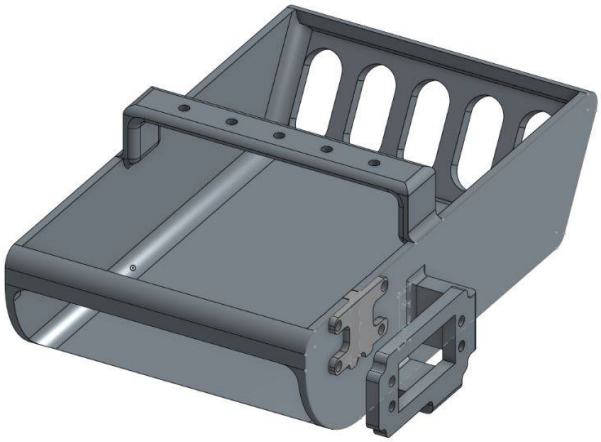
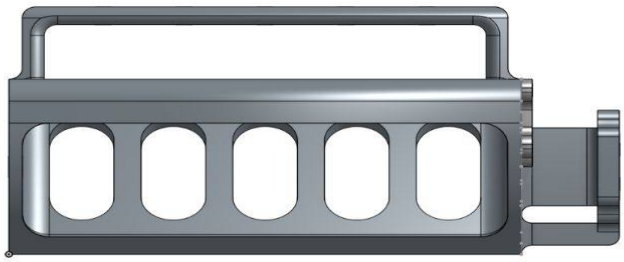
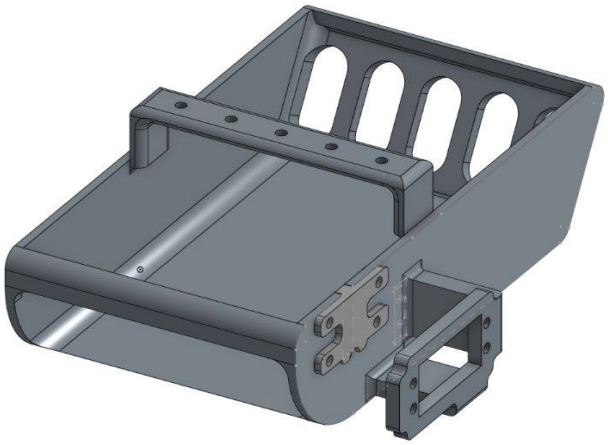
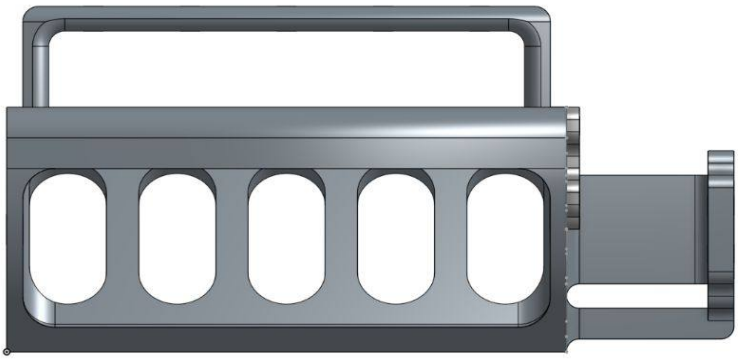
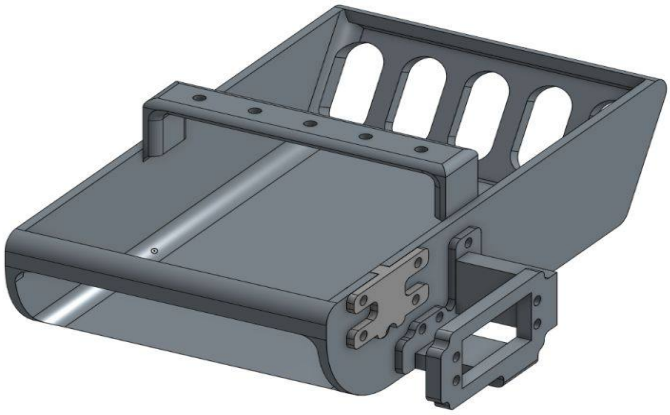
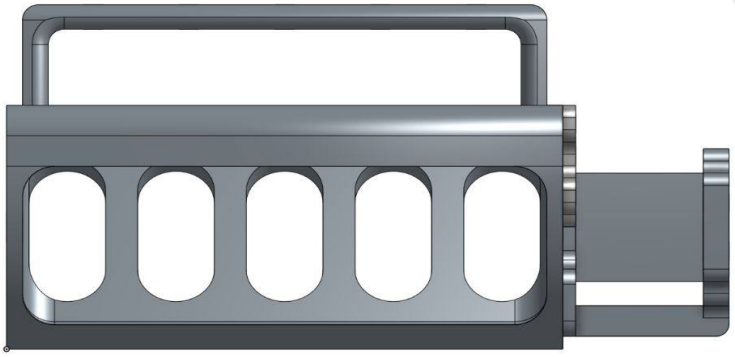
Tasks	Reflection
Assembly	- Cut out the foam pieces to attach onto the corners of the base
CAD	- Test out half the filtration basket - Modify the CAD and print final result - We plan to add onto bottom of the base tomorrow
Code and Wiring	- Work on the sonar, still progressing. Working the display of the sonar. Make sure that the distance that the sonar is reading shows on the display. Also start developing algorithm for motor redirection when sonar detects wall.

Goals for Next Week

- Test robot in pool again and see if the redirection algorithm and filtration basket both work alongside the bilge pumps.
- Identify the servo positions that allow debris to come in and stop it from coming out
- Possibly cut new base design from polycarbonate

Author:	Date:	Witness:	Date:
Vivian Nguyen	2/24	Elijah LaMothe	2/24

Filtration Basket Iterations

Isometric View	Front View
First Iteration	
	
Second Iteration	
	
Third and Current Iteration	
	

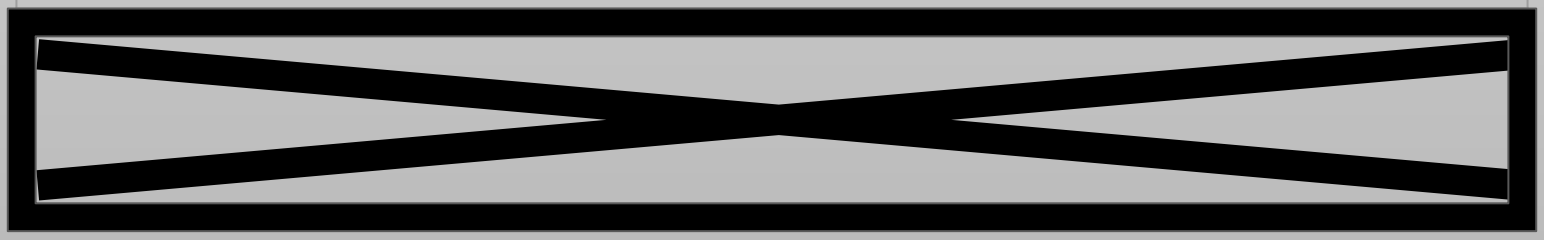
Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Basket Iterations Explanations

Iteration	Explanation
First Iteration	While the first iteration of the basket worked and did everything we needed, we had decided to change the servo hub which didn't fit within this current design and we also had notice that it was slightly to large to fit on the bottom of our prototype. Another design problem that needed to be fixed was the servo mount as it was slightly to small to fit the servo in, causing us to sand it down until it did.
Second Iteration	Our second iteration fixed all the problem of the first basket as it was smaller, the distance of the servo mount to the basket was fixed which allowed the new servo hub to fit, and slightly made the hole for the servo to go through more bigger. Feedback from Mr.Santana and Kilyer had pointed out that instead of having the servo mount as one piece with the basket we should have it as it own piece that's abled to be screwed onto the basket.
Third and Current Iteration	In our most recent iteration of the filtration basket we had added holes to be able to screw in with the newly designed servo mount.

Side Note

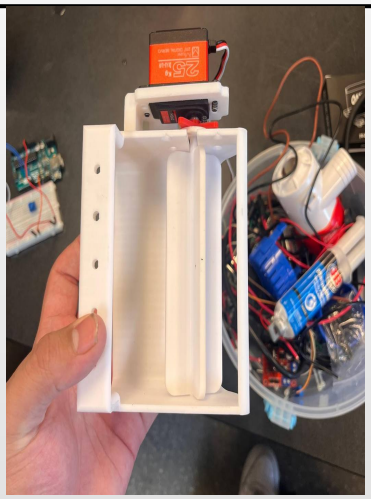
When printing the first two iterations of the filtration basket we had decided to print only the front half as to save filament as we only needed to test the front half of the basket to test the distances and size when assembling the whole filtration system.

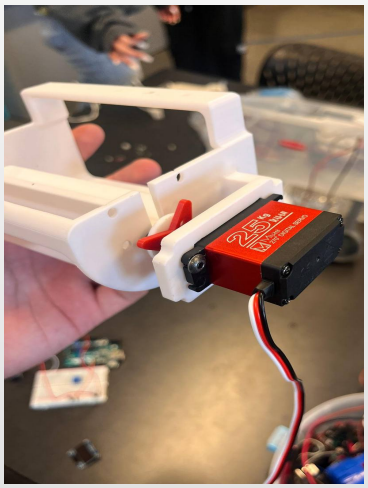


Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Fully Printed and Assembled Filtration Systems for All Iterations

1st



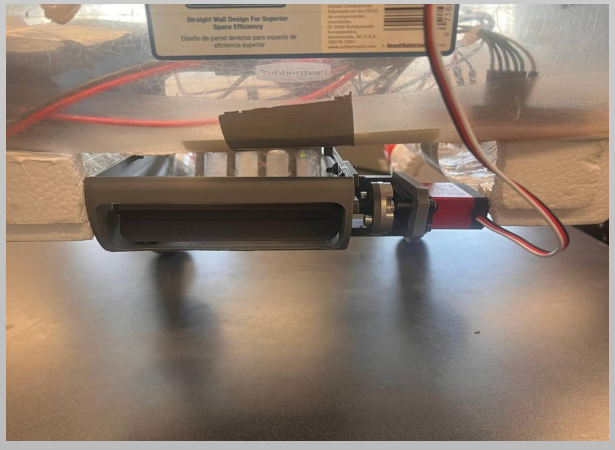


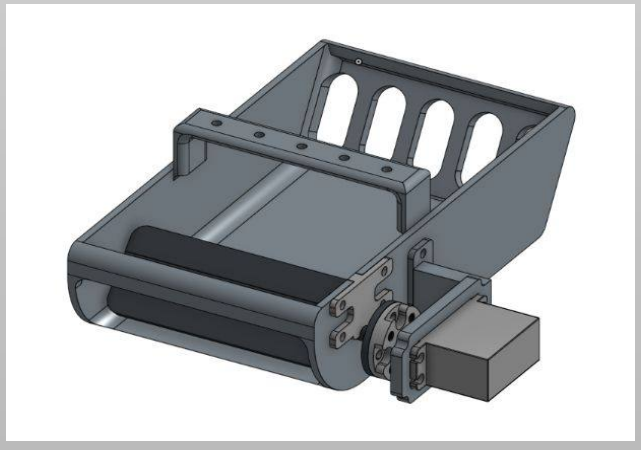
2nd





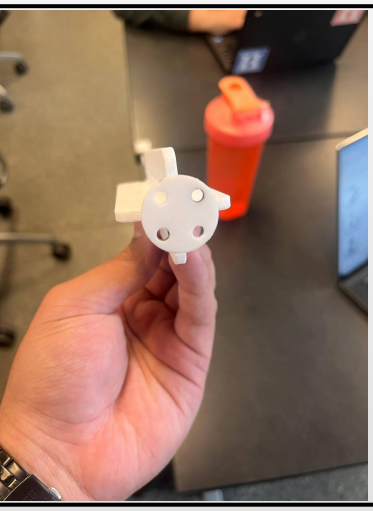
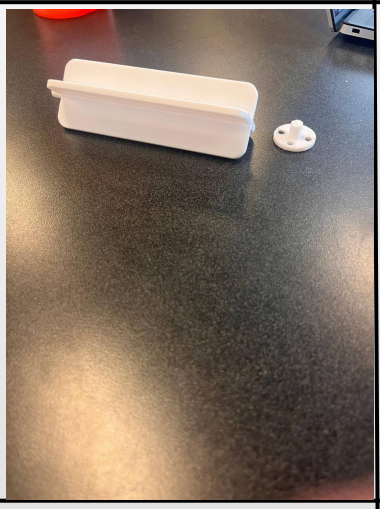
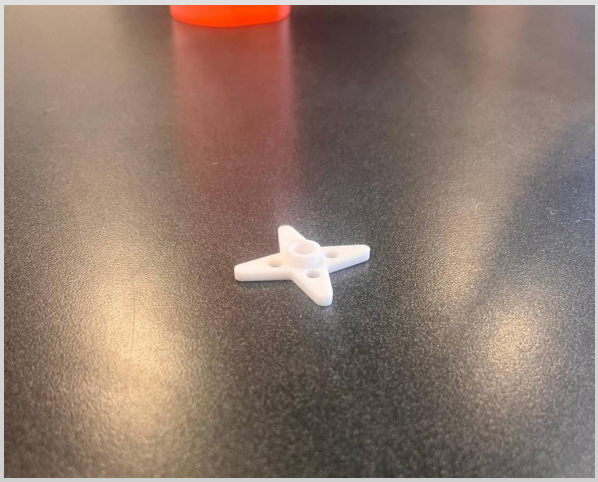
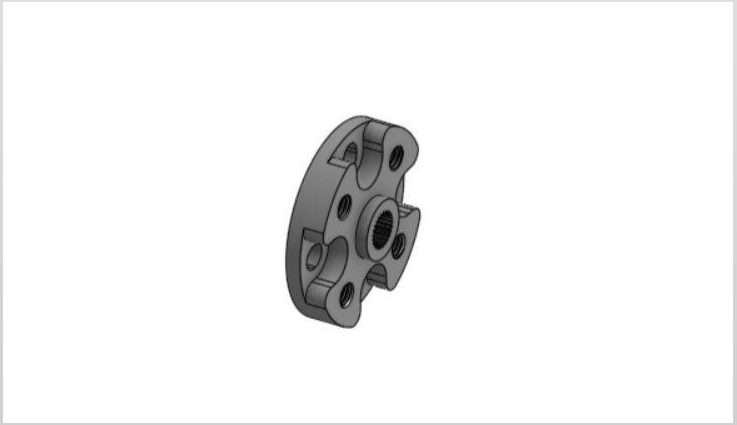
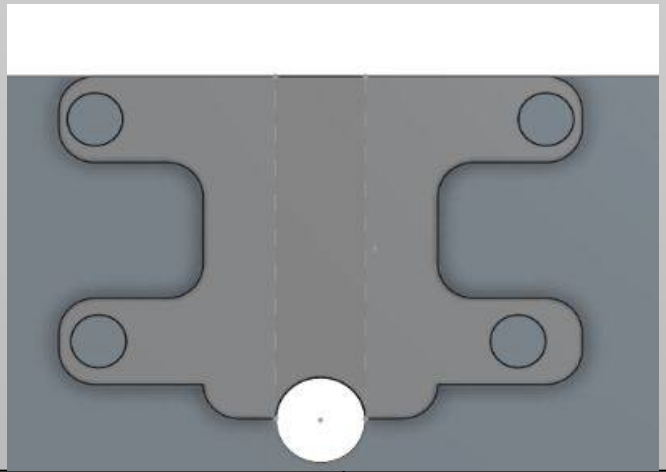
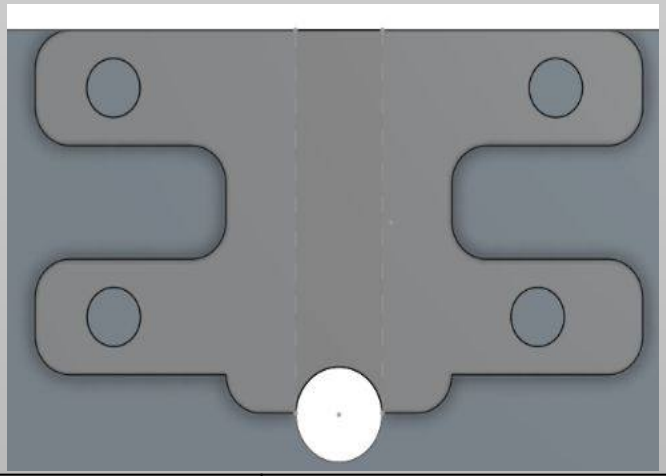
3rd and Current





Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

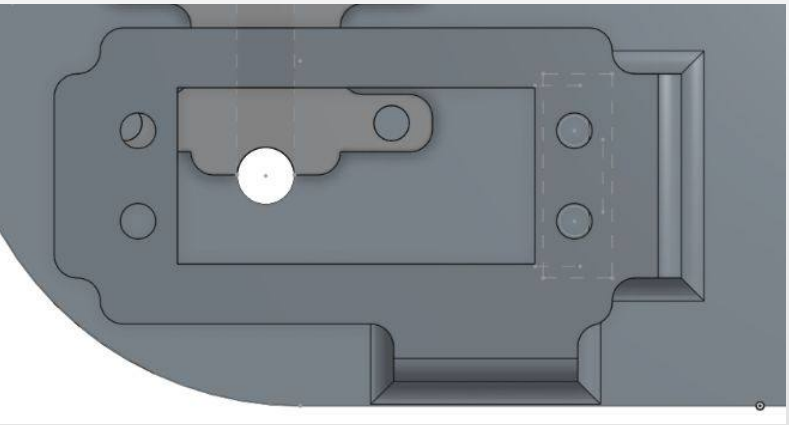
Changes in Parts

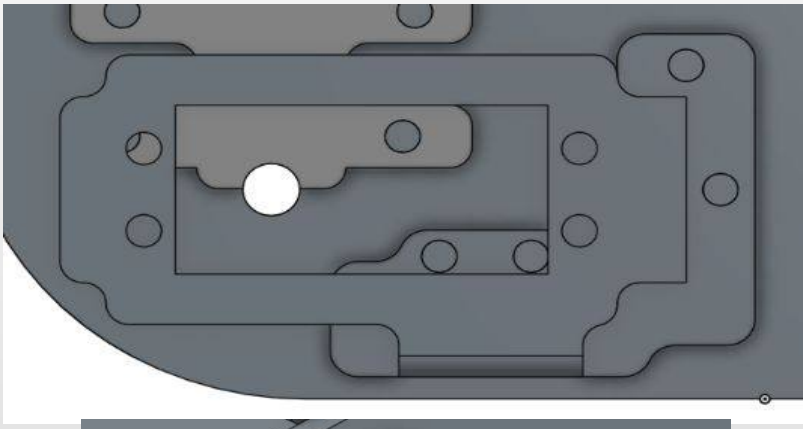
Original	New
Wall	
 	
Servo Hub	
	
Detachable Basket Part	
	

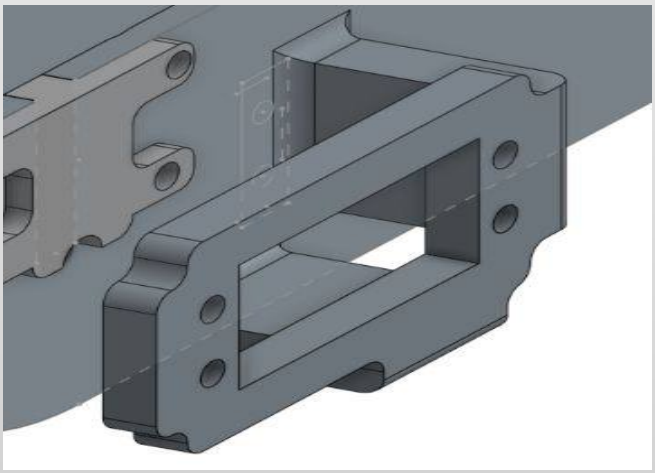
Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

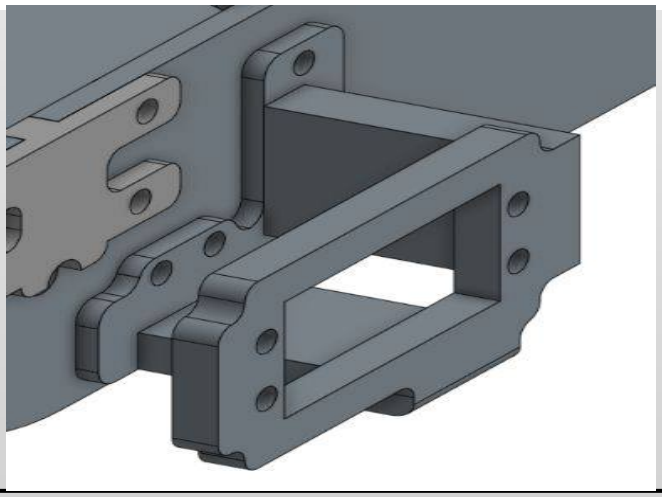
Changes in Parts

Original	New
Servo Mount	





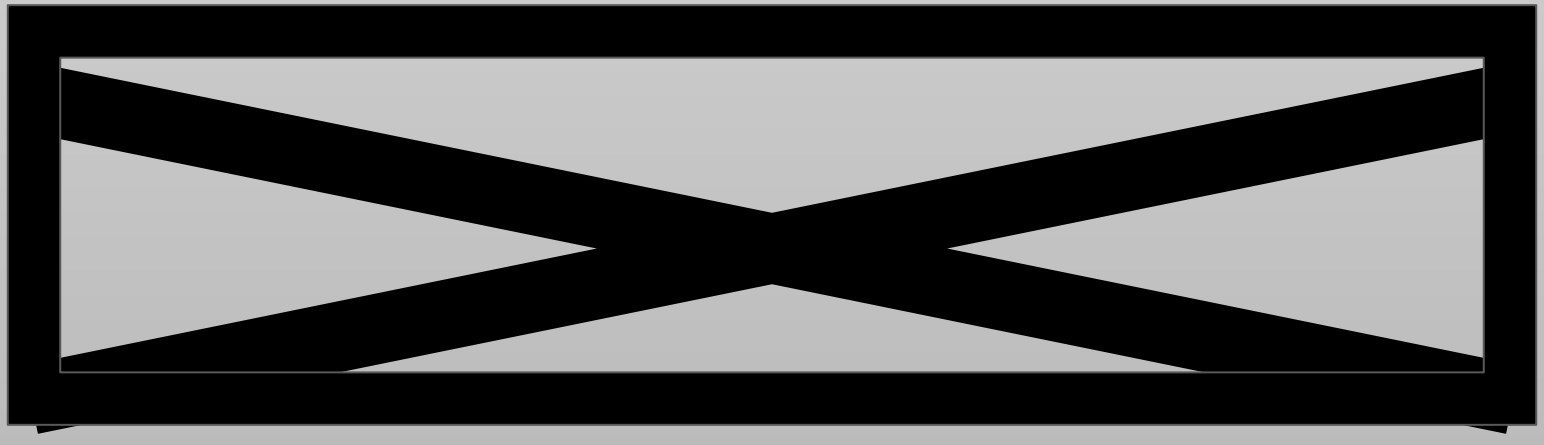




Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

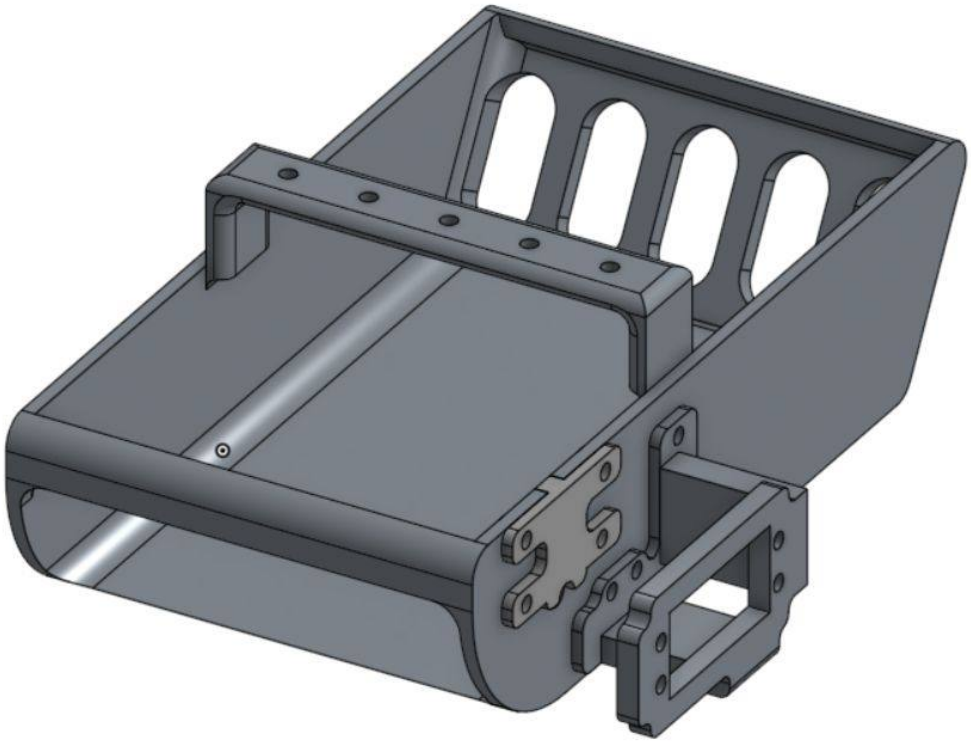
Changes in Parts Explanations

Part	Explanation
Wall	We had made the round piece at the front more bigger to match the hole distances on the new servo hub. When testing the wall caded part we accidently broke it causing us to add more fillets and slightly increasing the infill density when printing a new one.
Servo Hub	When printing and placing the caded servo hub on the servo it had fit well, but after a few spins the printed part became loose causing us to switch to a gobilda servo hub.
Detachable Basket Part	Based on the wall breaking we decided to slightly adjust this part by making it a bit more thicker to prevent it from breaking.
Servo Mount	We decided to change the servo mount from being connected to the basket as one piece to being its own piece because if the servo mount needs to be modified again we don't need to reprint the whole basket, which saves us filament. This change was due to the feedback we got from Mr.Santana and Kilyer.



Author:	**Date:**	**Witness:**	**Date:**
Andy Diaz	2/28	Elijah LaMothe	2/28

Filtration Basket

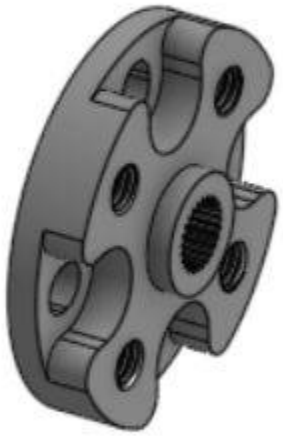


Wall

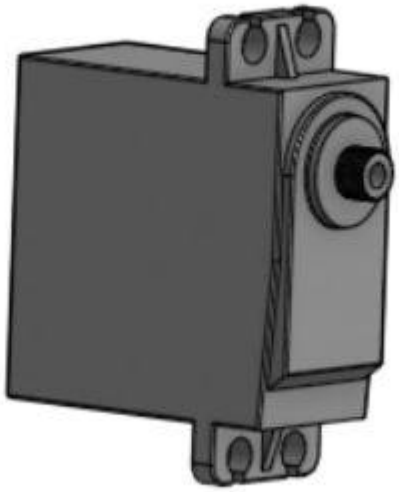


Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Servo Hub



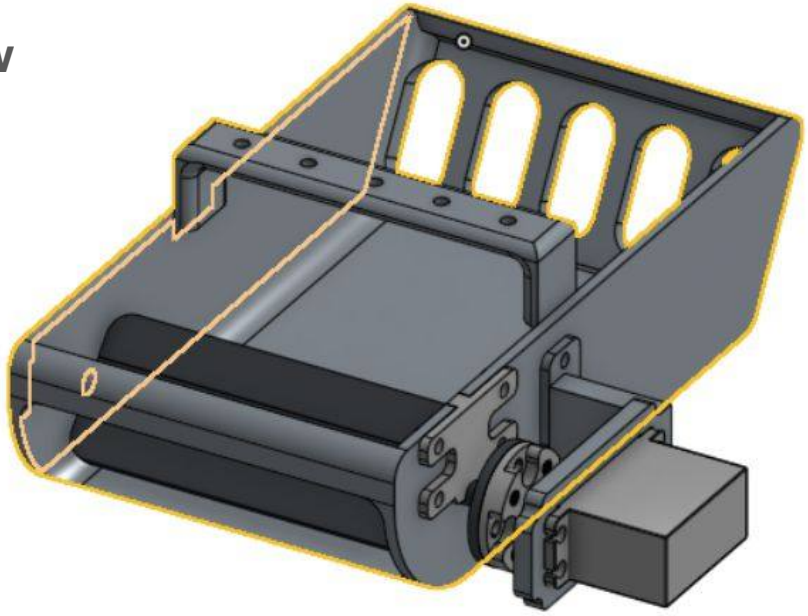
Servo



Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Filtration Basket Assembly CAD

Assembly Isometric View

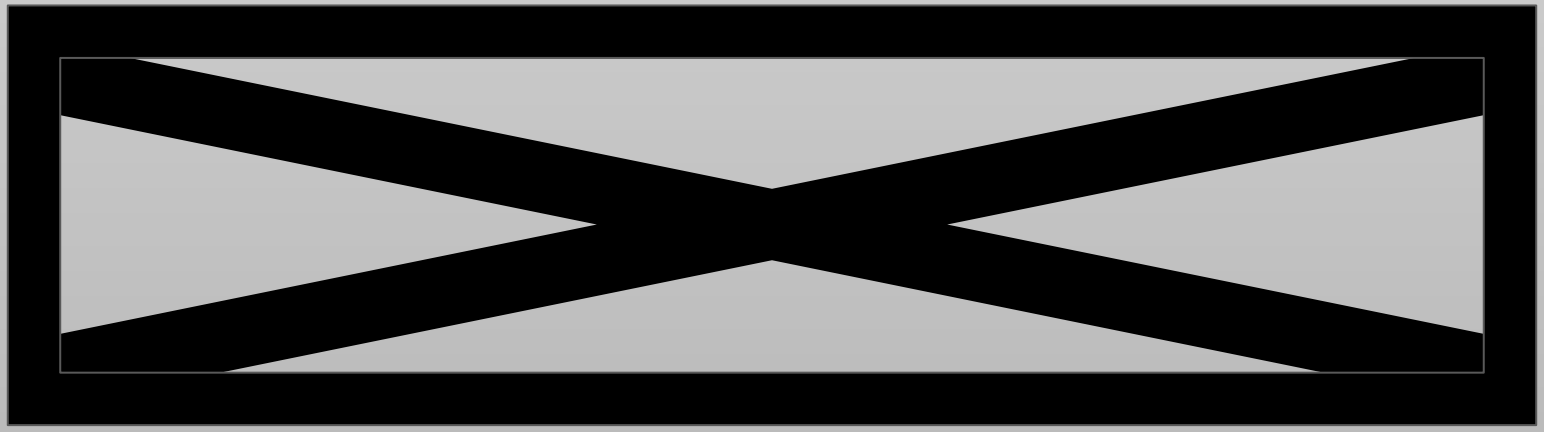


Assembly Top View



Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Part	Explanation
Filtration Basket	The base that holds everything together. It holds all the debris and lets the water flow out of the basket through the holes in the back. It connects to the base of the prototype by the holes on the top of the basket handle which is made to be screwed in.
Wall	The wall allows the debri and water to go inside the basket by being spun. It's called the wall because once the debri goes into the basket it cannot come out as the wall is blocking it.
Servo Hub	The servo hub connects the wall and the servo together.
Servo	The servo is the device that is spinning the wall and is screwed into the servo mount.
Filtration Basket Assembly	Located at the bottom of the prototype pool skimmer the filtration basket assembly allows water and debri to flow into a basket. The debri and water is allowed to go into the basket by a part called the wall which is being constantly spun by a servo. The debri is stored into the basket as it cannot escape as the wall is blocking its path and also the mesh covering the holes which only allows water to flow through it.

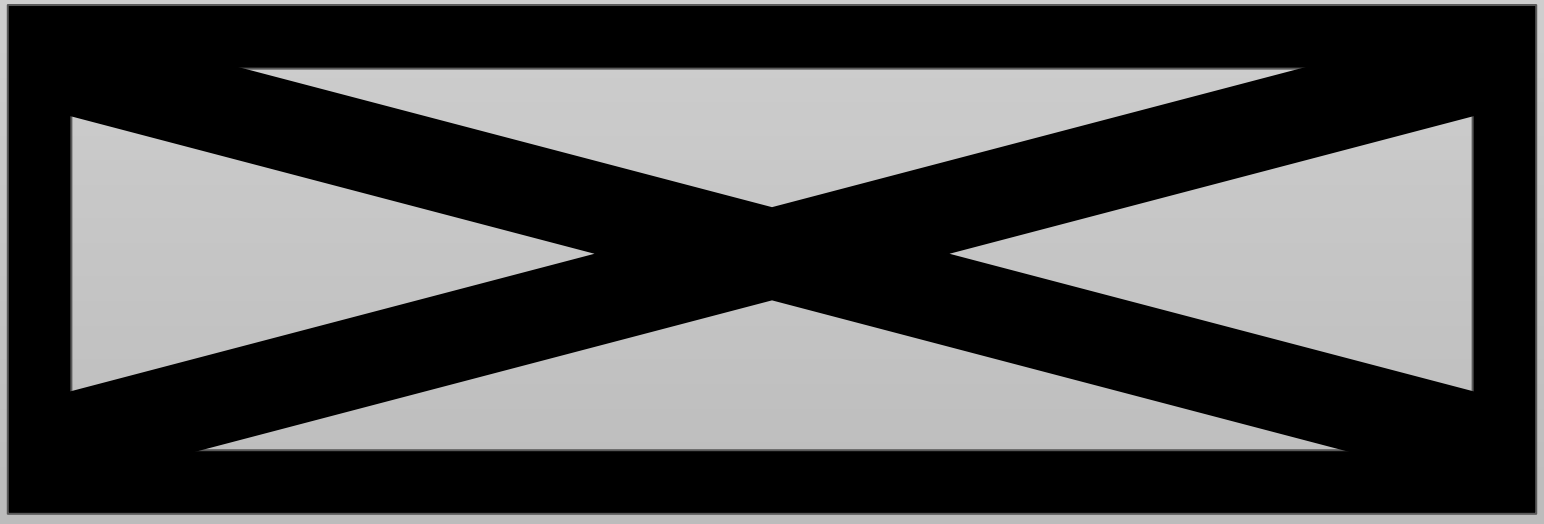


Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Parts Order Sheet

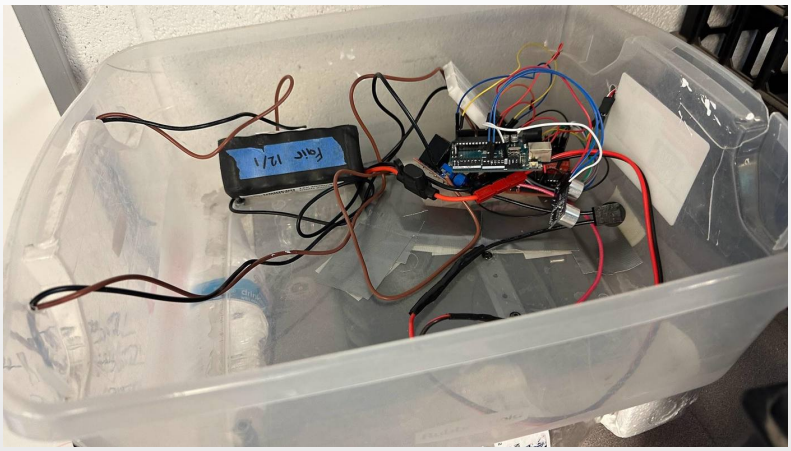
Amazon Parts Order Requisition

Team (Just 1 Person)	Part Name	Description	Amazon Link	Price	Qty	Shipping	Tax	Total
Santana (EXAMPLE)	IMU (EXAMPLE)	Sensor to detect orientation (EXAMPLE)	LINK	$0.00	0	$0.00	8.75%	$0.00
Anthony A	Barrington Hardwoods Maple Lumber Square Turning Blanks (4pc)	Wood for the box	LINK		1	$0.00	8.75%	$0.00
Anthony A	18 inch Ground Anchors Screw (4pc)	To prevent the chicken gain to get struck	LINK		1	$0.00	8.75%	$0.00
Anthony A	Fast Drying Polyurethane Spray	To protect our wood from moist	LINK		2	$0.00	8.75%	$0.00
Anthony A	rfid kit for arduino	Sensors that will open an door	LINK	$9.99	1	$0.00	8.75%	$10.86
Anthony A	Dog Bowl	A bowl to place the food inside	LINK	$1.99	1	$0.00	8.75%	$2.16
Ryleigh :D	Peltier cooler module kits	3 sets of parts needed to cool a surface using electricity	LINK	$14.99	1	$0.00	8.75%	$16.30
Ryleigh :3	10 peltier modules	Multiple parts that can be used to cool a surfce	LINK	$29.99	1	$0.00	8.75%	$32.61
Ryleigh :>	Sweatbands	Band to reduce sweating during physical activity	LINK	$21.85	1	$5.75	8.75%	$29.51
Ryleigh :)	5 waterproof push button switches	Part to turn the device on and off	LINK	$9.99	1	$0.00	8.75%	$10.86
Ryleigh :P	10 DS18B20 Temperature Sensor Digital Thermometer	Sensors made to detect changes in temperature	LINK	$8.99	1	$0.00	8.75%	$9.78
Ryleigh :O	Copper foil tape	Conductive tape made of copper	LINK	$30.98	2	$0.00	8.75%	$36.40
Ryleigh B)	NEW Copper grill mats	Thin sheets of copper that can conduct heat	LINK	$59.95	5	$0.00	8.75%	$86.18
Vivian N	MarineWeld Syringe	Binding component for polycarbon	LINK	$6.98	2	$0.00	8.75%	$8.20
Vivian N.	Pool Foam Rope	Used to keep our apparatus afloat	LINK	$14.99	1	$0.00	8.75%	$16.30
VIvian N.	Bilge Pump	Thruster for robot navigation	LINK	$14.99	3	$0.00	8.75%	$18.92
Vivian N.	Brushless Motor with ESC	Extra component for robot navigation, if we decide to reti	LINK		2	$0.00	8.75%	$0.00
Vivian N.	Waterproof Servo	Used for wall closing mechanism to catch debris	LINK	$16.99	4	$0.00	8.75%	$22.94
Vivian N.	1/2" PVC Pipe	Apparatus framework; hold our buoy mechanism in place	LINK		0	$0.00	8.75%	$0.00
Vivian N.	Fabric Mesh	Water flows through restricting debris from exiting	LINK	$7.48	0	$0.00	8.75%	$7.48
Andrew M	HiLetgo RFID Kit	ID senor to unlock door slip	LINK	$5.69	1	$0.00	8.75%	$6.19
Andrew M	6 pack 12x12 aluminum sheet metal	used to build the insert that goes onto the door	LINK		1	$0.00	8.75%	$0.00
Ryleigh :>	IRLZ44N MOSFETs	Used to make arduino send more amps	LINK	$9.99	1	$0.00	8.75%	$10.86
Ryleigh :D	IRLB8721 MOSFETs	Used to make arduino send more amps	LINK	$9.99	1	$0.00	8.75%	$10.86
Ryleigh :(	Heatsinks for the MOSFETs	Heatsinks to take out the heat	LINK	$7.99	1	$0.00	8.75%	$8.69
Ryleigh :(	1N4007 Rectifier Diode	Used for flyback protection	LINK	$5.99	0	$0.00	8.75%	$5.99
Vivian N	MarineWeld Syringe	Binding component for polycarbon	LINK	$6.98	1	$0.00	8.75%	$7.59
Vivian N.	PVC Elbows	Redirect water current from the initial bilge pump outlet	LINK		1	$0.00	8.75%	$0.00
Soleil P.	100 pack Pet Training Pads	Pee pads to absorb cat waste	LINK	$20.99	1	$0.00	8.75%	$22.83
Vivian N.	PVC Pipes	Framework of the sheet metal to attach pool noodles	LINK		1	$0.00	8.75%	$0.00
Julissa V	Bilge Pump	Pump the water	LINK	$14.99	0	$0.00	8.75%	$14.99
Aron Ruiz	Circular Fingerprint Module,	will allow us to be able to use finger tech to unlock doors	LINK	$27.69	0	$0.00	8.75%	$27.69
Aron Ruiz	Wifi motherboard	will allow us to comunicate via bluethooth or wifi	LINK	$16.00	0	$0.00	8.75%	$16.00
Aron Ruiz	Motor	will allow the deadbolt to be movved	LINK	$15.00	0	$0.00	8.75%	$15.00
Aron Ruiz	Electro plate	allows us to be able to lock the handle	LINK		0	$0.00	8.75%	$0.00
Julissa V	VEVOR Beverage Dispenser	this will be our water dispensor	LINK	$45.77	1	$0.00	8.75%	$49.77



Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Drilling Accident



In the process of drilling holes to attach the filtration basket, the drill was pushed down to hard and cracked the plastic base. To temporarily fix the base, we used duct tape in order to keep the base from leaking water through.

Complete Apparatus Assembly

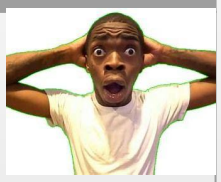
Author:	Date:	Witness:	Date:
Andy Diaz	2/28	Elijah LaMothe	2/28

Weekly Logs

Week 22: Mar. 3 - Mar. 7

Goals for This Week

- Test apparatus at the pool



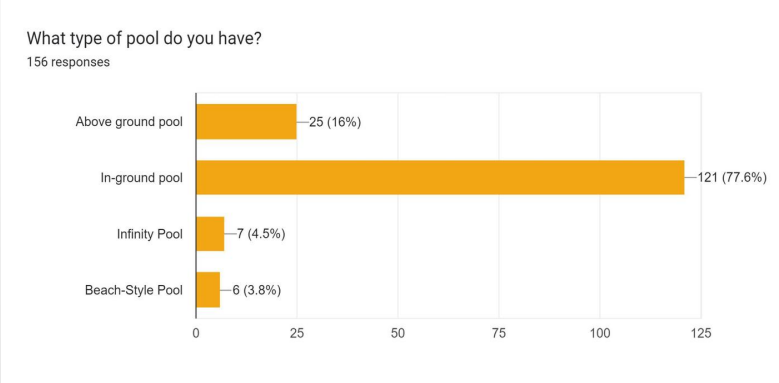
Tasks	Reflection
Testing	- As we tested the device in the pool, we needed to place the battery to even out how the apparatus floated on the water - The issue we ran into was that our left motor was not pushing out water at its maximum effort and we need to seal the screws that hold the two motors in the front to avoid any leakage in our base. Improvements: - Adjust the thrust of the motors, so change the adapters to have a smaller outlet in order to generate enough thrust out of the motors and push our apparatus quicker.
CAD	- Working on the CAD for the sheet metal base framing, the bilge pump outlets, and adjust the placement of the bilge pumps so that they are centered and have balanced weight on the water.

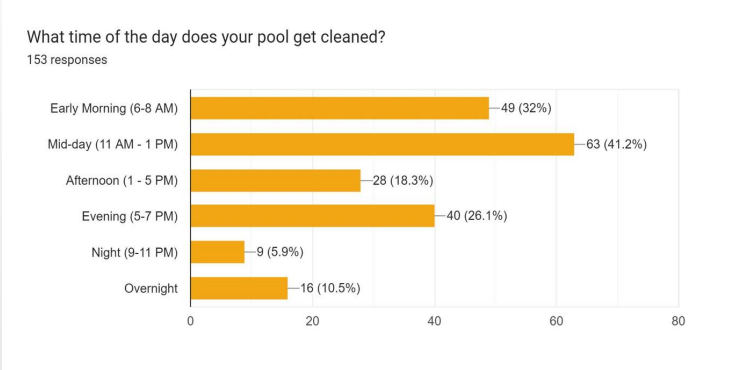
Goals for Next Week

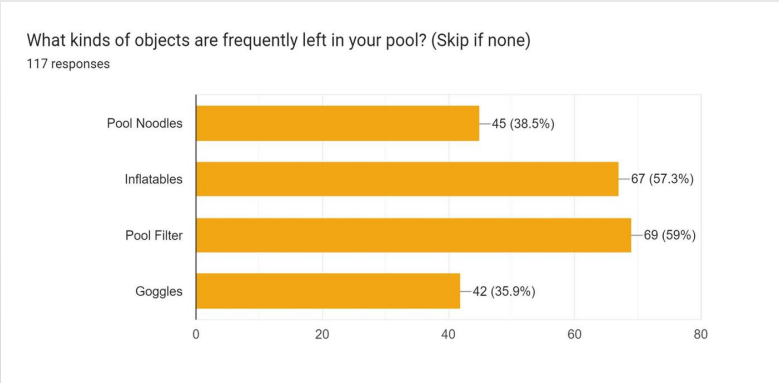
Find a new method that will help the robot propel forward faster. This could be anything from redesigning the propulsion adapter or just finding fast bilge pumps. However, Mr. Santana did mention that when he competed in RoboSub, an underwater robotics competition, he used slower motors than our current ones and still worked fine, so therefore the speed of our pumps are likely not the issue.

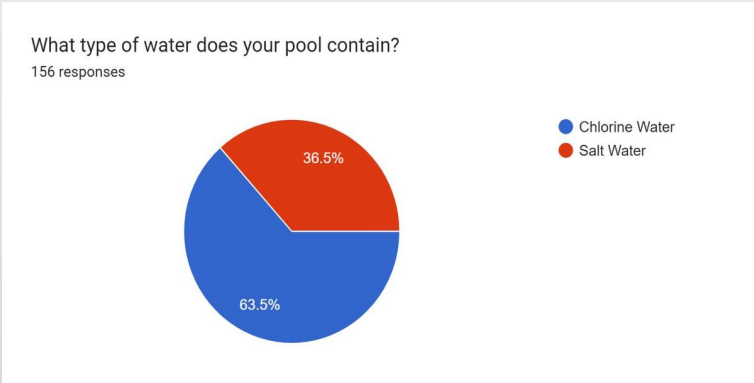
Author:	Date:	Witness:	Date:
Vivian Nguyen	3/3	Elijah LaMothe	3/3

Survey Results







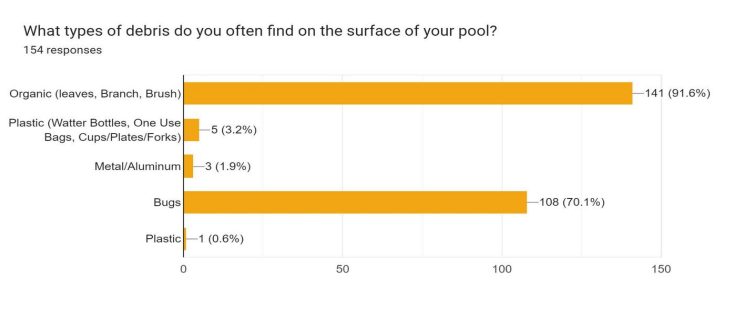


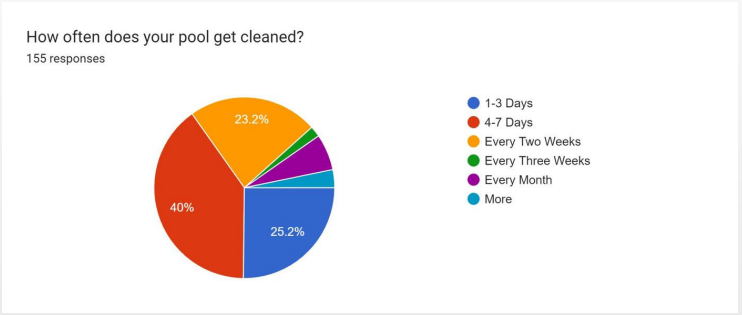
Summary/Analysis

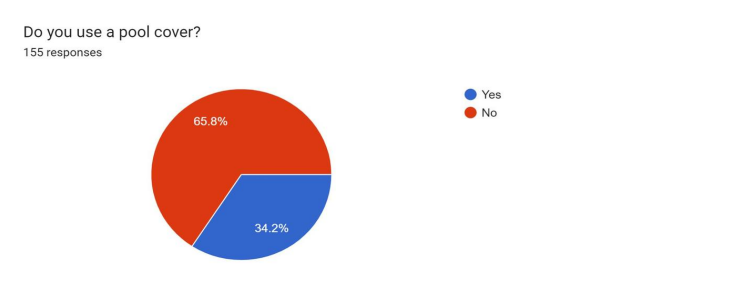
Many more respondents said that they clean their pools during the day and not during the night, which makes us interested in incorporating a solar charging mechanism to take advantage of natural resources rather than recharging a battery. Additionally, many people leave objects in their pool, meaning that we might have to adopt a navigation system for our robot to avoid running into obstacles. To do this, we brainstormed possible solutions such as a sonar or distance sensors.

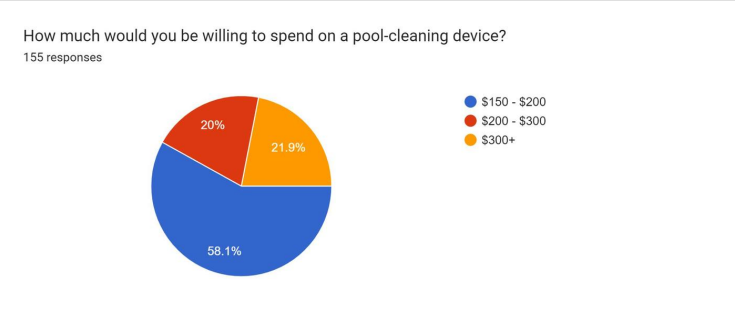
Author:	Date:	Witness:	Date:
Vivian Nguyen	3/5	Elijah LaMothe	3/5

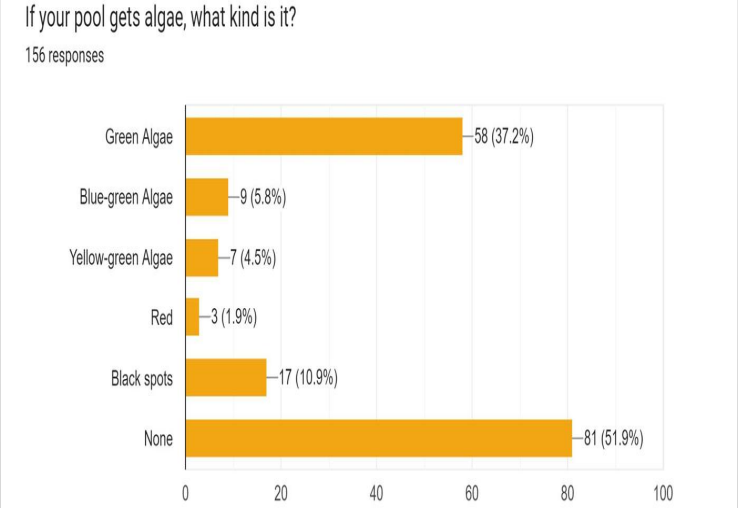
Survey Results











Summary/Analysis

Organic material was the primary type of debris that is found in pools, meaning that our robot design should be focused on clearing leaves, branches, sticks, and other similar material from the surface of the pool. Additionally, a majority of respondents said that they clean their pools every 4-7 days, meaning that our robot's battery doesn't have to last very long, and can clean the pool faster. However, this also implies people will have to recharge the battery of the robot if utilizing solar energy is not an option.

Author:	Date:	Witness:	Date:
Vivian Nguyen	3/5	Elijah LaMothe	3/5

Weekly Logs

Week 23: Mar. 10 - Mar. 14

Goals for This Week

- Continue CAD for filter basket
- Work on code for sonar and thrust

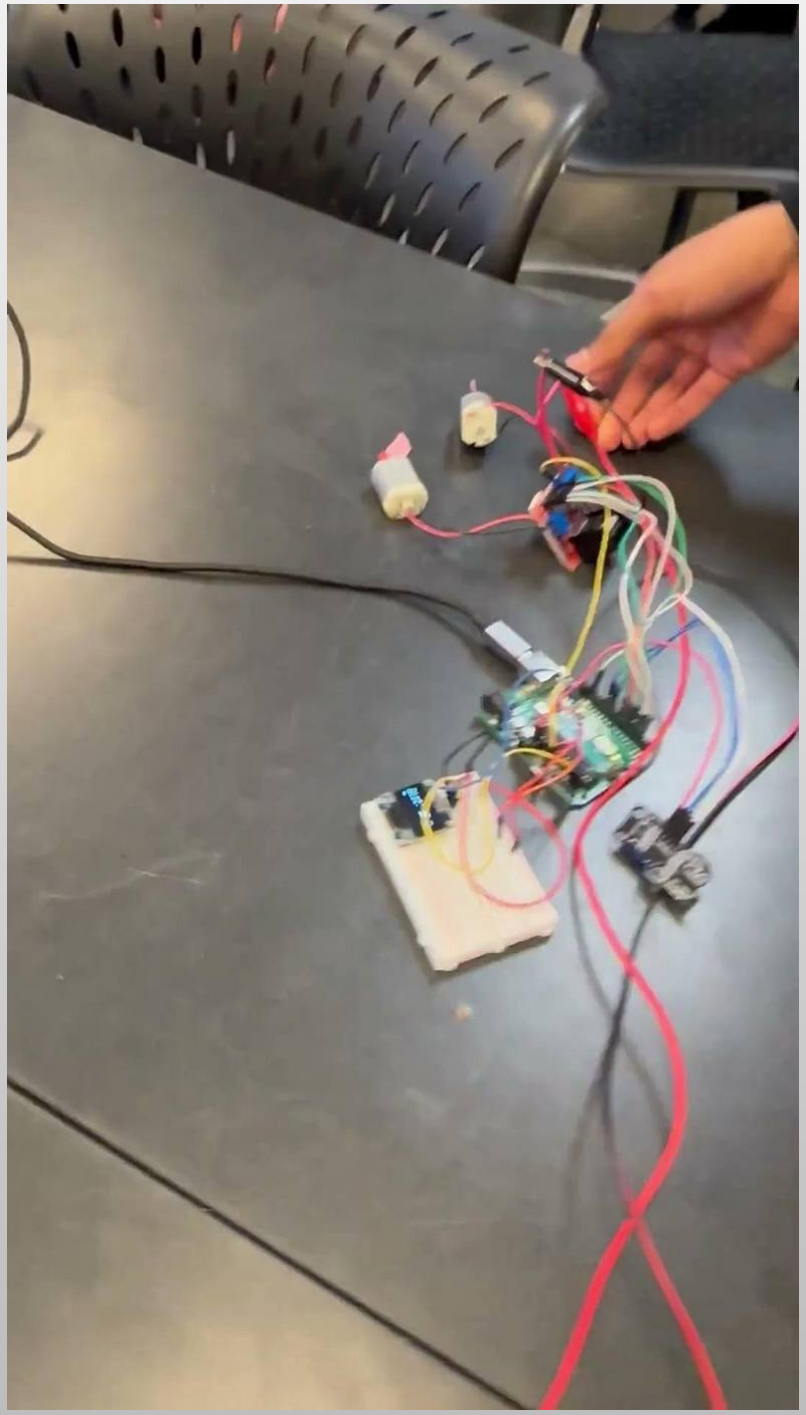
Tasks	Reflection
CAD	- CAD a custom propeller for vex flaps to slide into the stud - Continue working on the sheet metal model of the base to cut out of the polycarbonate, make sure that the model is able to fit the current version of the filter basket
Build	- Uninstalled the filter basket to replace the propeller - Need to install/attach the custom bilge pump thrusters - Install the sonar sensor attachment onto the front of the robot. Make sure that it is horizontal and facing a wall
Program	- Continue working on the code for the sonar sensor, and incorporate a turning function when it detects anything less than 10 centimeters in front of it

Goals for Next Week

Implement the sonar redirection algorithm on the robot and test how it works in the pool. Find alternative methods for bilge pump propulsion because the current method isn't generating nearly enough force. This can be done by purchasing stronger bilge pumps, or finding another method to propel the robot forward, possibly just using a motor and propeller altogether.

Author:	**Date:**	**Witness:**	**Date:**
Vivian Nguyen	3/10	Elijah LaMothe	3/10

Sonar Redirection Algorithm vid



Author:	Date:	Witness:	Date:
Vivian Nguyen	3/10	Elijah LaMothe	3/10

Sonar Redirection Algorithm Code


```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128   // OLED width in pixels
#define SCREEN_HEIGHT 64   // OLED height in pixels
#define OLED_RESET    -1   // Reset pin (not used)
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

const int trigPin = 9;  // Trigger pin for ultrasonic sensor
const int echoPin = 10; // Echo pin for ultrasonic sensor
long duration;
int distance;

const int motorPin1 = 6; // Pin connected to IN1 of the motor driver
const int motorPin2 = 5; // Pin connected to IN2 of the motor driver
const int motorPin3 = 4;
const int motorPin4 = 3;

const int enablePin = 7;  // Pin connected to ENA (PWM control) of the motor driver
const int enablePin2 = 2; // Pin connected to ENA (PWM control) of the motor driver

void moveForward() {
  digitalWrite(motorPin1, HIGH);
  digitalWrite(motorPin2, LOW);
  digitalWrite(motorPin3, HIGH);
  digitalWrite(motorPin4, LOW);

  analogWrite(enablePin, 200);  // Adjust speed
  analogWrite(enablePin2, 200);
}

void stopMotors() {
  digitalWrite(motorPin1, LOW);
  digitalWrite(motorPin2, LOW);
  digitalWrite(motorPin3, LOW);
  digitalWrite(motorPin4, LOW);
analogWrite(enablePin, 0);  // Full speed
analogWrite(enablePin2, 0);

}
```

[GitHub Link]

Author:	Date:	Witness:	Date:
Vivian Nguyen	3/10	Elijah LaMothe	3/10

Weekly Logs

Week 24: Mar. 17 - Mar. 20

Goals for This Week

- Continue CAD for filter basket
- Work on code for sonar and thrust

Tasks	Reflection
CAD	- CAD a custom propeller for vex flaps to slide into the stud - The length of the filter propeller does not match to an equal amount of the vex flaps so in order to ensure symmetry across the propeller, we had to cut one of the flaps in half
Build	- Uninstalled the filter basket to replace the propeller - Use a dremel to cut an axle for the new cad model of the filter basket propeller - This would prevent the plastic from breaking and allow the piece to last longer - Use a dremel to cut the bilge pumps so that all that is exposed is the motor axle and the propeller
Program	- Fine tune the code for the sonar sensor implementation, make sure that nothing is covering the circuit board section of the sensor or else it will throw off the readings

Goals for Next Week

FInish cutting the rest of the bilge pumps and take off the current mounts that keep the old bilge pumps in place. Attach a better propeller onto the new cut bilge pumps and test how it works in the water (this can only happen if the propeller part that was placed on the parts list can be ordered.

Author:	Date:	Witness:	Date:
Vivian Nguyen	3/17	Elijah LaMothe	3/17

Bilge pumps (cut version)







We cut the bilge pumps into this version to be able to run a propeller directly on the axle of the bilge pump rather than have it run through a channel (suction system) like it came when we bought it. Doing this would allow us to maximize our speed and make easy changes to the propeller of the bilge pump should we encounter any issues.

Author:	Date:	Witness:	Date:
Vivian Nguyen	3/21	Elijah LaMothe	3/21

Broken Filter Basket Propeller



Author:	Date:	Witness:	Date:
Vivian Nguyen	3/21	Elijah LaMothe	3/21

Weekly Logs

Week 25: Mar. 24 - Mar. 28

Goals for This Week

- CAD a bilge pump holder for the new (cut) versions of the bilge pumps
- Test the new holders and see if any adjustments need to be made

Tasks	Reflection
CAD	- Use caliper to get the dimensions of the bilge pump and CAD the holder accordingly - Make sure the bilge pump holder has holes on the top so it can be screwed in. Also make sure these holes are accessible with a pipe wrench for easy maintenance
Build	- Assemble the new bilge pump mounts - The holder should have a flap/flat surface in the front which prevents the bilge pump from moving forward/backward (this is basically just a countermeasure)
Program	Nothing got done for programming this week, but in the following weeks we should make sure that both bilge pumps are getting the same amount of power when running, and the sonar redirection algorithm is working with the robot inside the pool

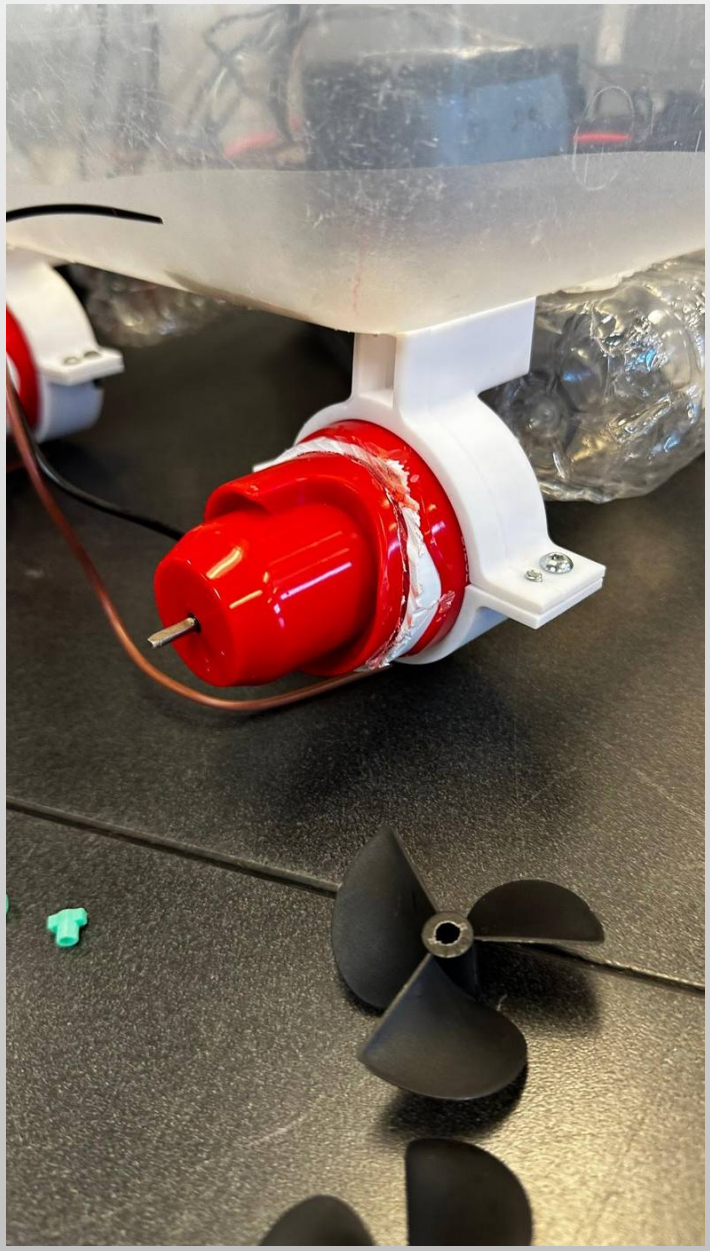
Goals for Next Week

Attach the bilge pump and their holders on the base of the robot. Possible test how this works in the pool if time permits, but only one of the three members of the team will be present for nearly the entirety of the week. If any adjustments need to be made to the CAD for the holders, make them.

Author:	Date:	Witness:	Date:
Vivian Nguyen	3/24	Elijah LaMothe	3/24

Bilge Pump Holder (cut version)





Author:	Date:	Witness:	Date:
Vivian Nguyen	3/31	Elijah LaMothe	3/31

Weekly Logs

Week 26: Mar. 31 - Apr. 4

Goals for This Week

Attach the bilge pumps and their holders on the base of the robot. Perform any maintenance that might need to be done to the robot so that when all the members are back, we can test its performance in the pool.

Tasks	Reflection
Build	- Attach the bilge pumps to the base of the robot - Make sure the holders have two points of contact with the base (IE: make sure they are screwed in in two holes at least)
Maintenance	- Tighten the screws for the filter basket and the servo that turns its propeller - Add styrofoam pieces to the front of the robot to make sure it floats more evenly in the pool
Program	Nothing got done for programming this week, but in the following weeks we should make sure that both bilge pumps are getting the same amount of power when running, and the sonar redirection algorithm is working with the robot inside the pool

Goals for Next Week

Make custom propellers for the bilge pumps (since the ones we ordered are taking forever to get here). Test out how the custom propellers work inside the pool. All findings should be recorded and if any changes need to be made to the shape of the propeller make sure to document them in the notebook.

Author:	Date:	Witness:	Date:
Vivian Nguyen	3/31	Elijah LaMothe	3/31

Weekly Logs

Goals for This Week

Make a custom 3D printed propeller
Test how he propellers work in the pool (did not happen)
Glue water bottles onto the bottom
Test the sonar detection

Tasks	Reflection
Finish CAD for custom propellers	We made two versions of the propeller, only one of which worked. This is because one of them was mirrored, when it didn't need to be. We can use the same propeller for both the left and right side of the robot. The direction of airflow was also vastly different between the two. See the model on page 79
Clean up wiring inside robot body	Puncture holes in the base of the robot for the bilge pump and servo wires to travel through. Excess wire can either be zip tied or managed somewhere on the inside of the base. In the future it would service us to have a dedicated compartment for each electronic piece and the wires that connect to them
Update notebook with videos/photos of iterations and current versions	Upload a video of the sonar algorithm working in real time and include the code that goes with it. Show pictures of the broken filter basket propeller and possibly a stress test simulation alongside it. Show the new bilge pump models and explain why we decided to cut them.

Goals for Next Week

Test how the robot maneuvers the pool with the new bilge pumps and custom propellers and adjust accordingly. Get all of the excess material out of the base and make sure it is ready to test. Make sure there are no holes on the base where water could leak through.

Author:	Date:	Witness:	Date:
Vivian Nguyen	4/7	Elijah LaMothe	4/7

Weekly Logs

Week 28: Apr. 14 - Apr. 17

Goals for This Week

Test how the custom propellers work in the pool
Adjust the base of the robot in case there is an imbalance/leak
Clean up wiring on the base/get rid of all excess material

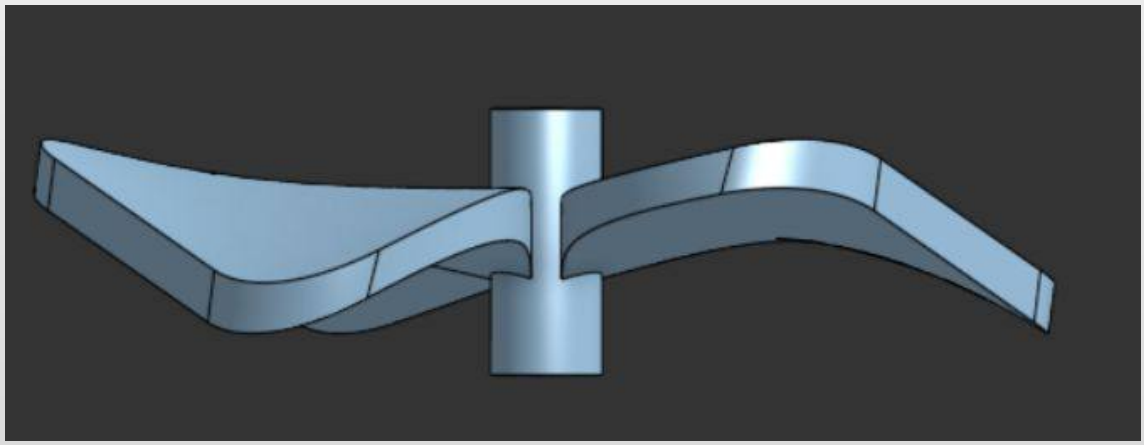
Tasks	Reflection
Test robot in pool with new propellers	The robot moved very well with the custom propellers, significantly faster than before. We observed that the propellers were not low enough in the water to maximize the robots speed so we will need to adjust the bilge pump mounts.
CAD/print adjusted propellers and test	We found that our current propellers had more air directed on the outside of the propeller rather than a concentrated area toward the center. We can get more concentrated air flow by making the angle of the propeller more severe. *see page 80 for reference
Find a way to have the robot move straight (getting rid of drift)	Power is not reliable enough to ensure the robot is moving straight, so we decided to explore the possibility of implementing a BNO055 IMU sensor. This will allow us to track the robots heading as it moves throughout the pool and correct itself it moves off-track.

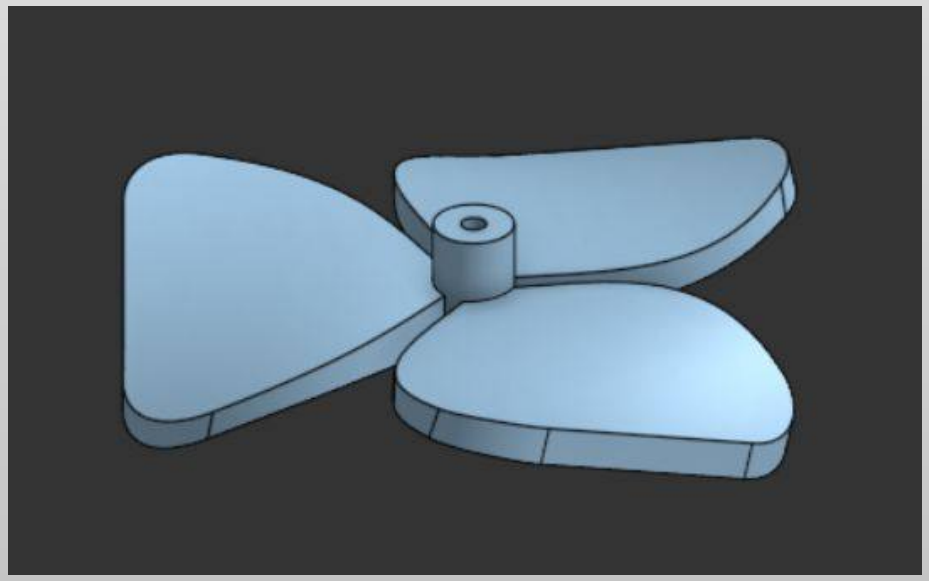
Goals for Next Week

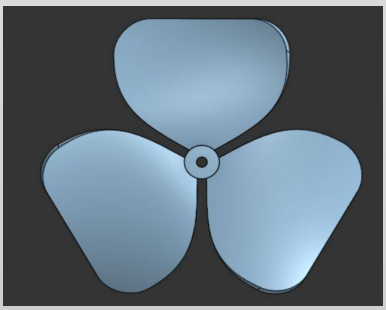
Work on soldering the IMU to conductive wire so it can connect to the arduino. Also find a way to fix the IMU so that it is stable, this can be done using CAD or something simpler like screws/zip ties. Test the adjusted propeller to see if a more concentrated air flow allows the robot to move faster. Add/adjust the supports for the base so the robot is more stable when navigating the water. Adjust the basket filter so that it is flush with the water.

Author:	Date:	Witness:	Date:
Vivian Nguyen	4/14	Elijah LaMothe	4/14

Custom Propeller V.1 and performance





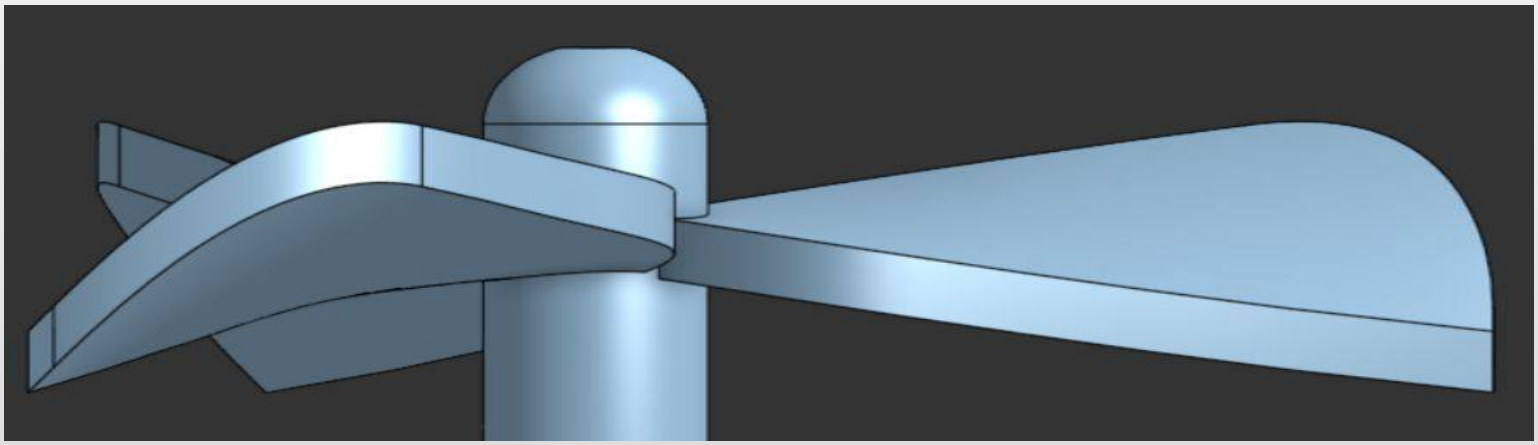


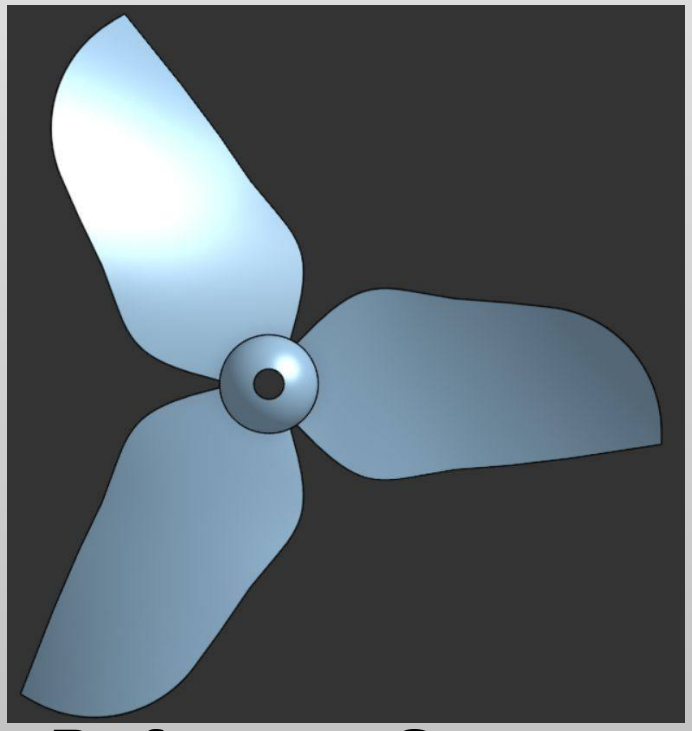
Performance Summary

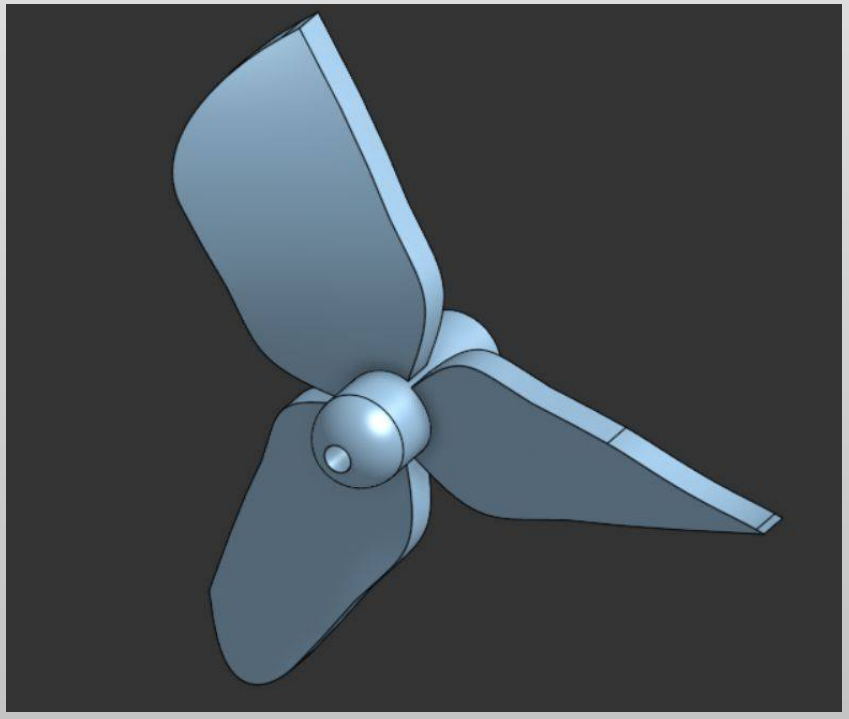
These propellers have a recorded wind reading of 2.1 meters per second, the flow of air was also toward the outer edge of the propeller rather than the center. With these propellers, we were not able to move very quickly in the pool compared to our old bilge pump model.

Author:	Date:	Witness:	Date:
Vivian Nguyen	4/15	Elijah LaMothe	4/15

Custom Propeller V.2 and performance







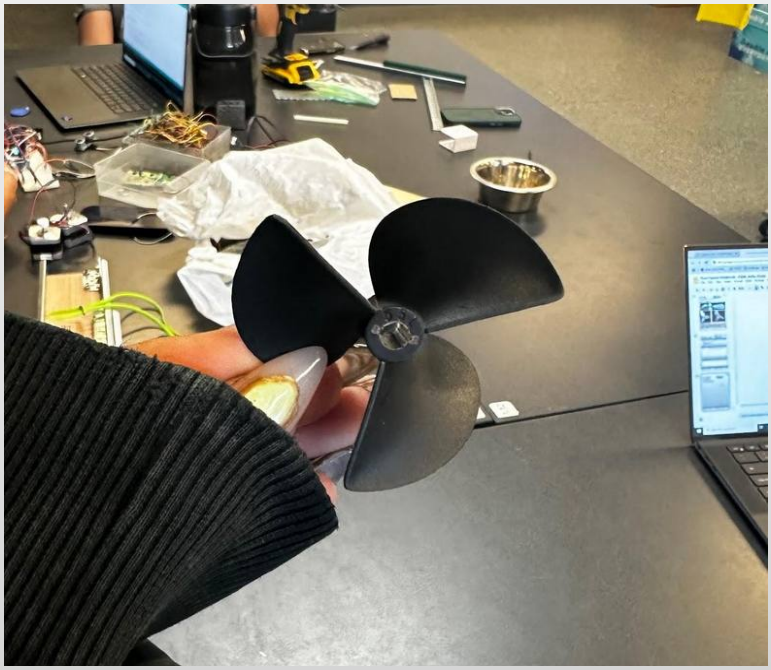
Performance Summary

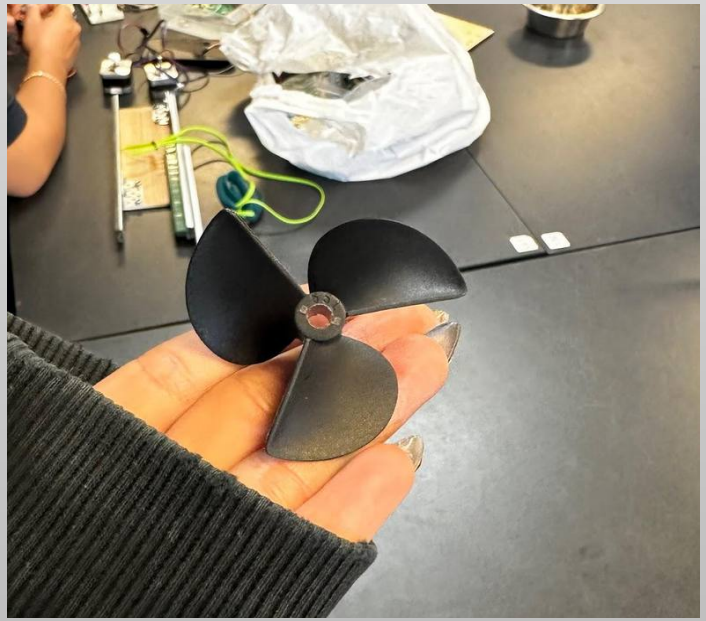
These propellers have a recorded wind reading of 2.7 meters per second, surpassing the speed of our version 1 model by a huge margin. While these propellers have a high recorded wind reading, the flow of air is toward the outer wing of the propeller rather than the middle, making them inefficient for underwater navigation.

Author:	Date:	Witness:	Date:
Vivian Nguyen	4/15	Elijah LaMothe	4/15

Pre-made Propeller







Author:	Date:	Witness:	Date:
Vivian Nguyen	4/15	Elijah LaMothe	4/15

Weekly Logs

Week 29: Apr. 28 - May. 2

Goals for This Week

- Test the new propellers, create the adapter for the D-shaft on bilge pump
- Begin cutting out the sheet metal
- Try and get the IMU working
- Keep updating notebook
- Test the apparatus this week (Update: Test next week; Santana is gone)

Tasks	Reflection
New Propellers	- New fan has concentrated current in the center of the propellers; plan to test in the water next week - The angle of the propellers is also much more severe than the old propellers making its ability to cut through water much better
IMU implementation	- After discussion, we have decided that implementing the IMU might be too complicated and unnecessary considering the amount of time we have left for this project - Alternatively, we could reduce the amount of power one bilge pump receives, and continue to perfect the sonar algorithm
Sheet metal model	- This week we could not cut the sheet metal model out of polycarb as it was suggested that we test it on cardboard first

Goals for Next Week

- Test the speed of the robot in the pool with the new propellers
- See how well the sonar redirection algorithm works and if any changes need to be made

Author:	Date:	Witness:	Date:
Vivian Nguyen	4/28	Elijah LaMothe	4/28

Weekly Logs

Week 30: May. 5 - May. 9

Goals for This Week

- Test the robot in the pool with the new propellers
- Also test out the sonar redirection algorithm and change the logic if it does not work well
- Attach the sonar holder to the robot (make sure it is level with the pool wall)

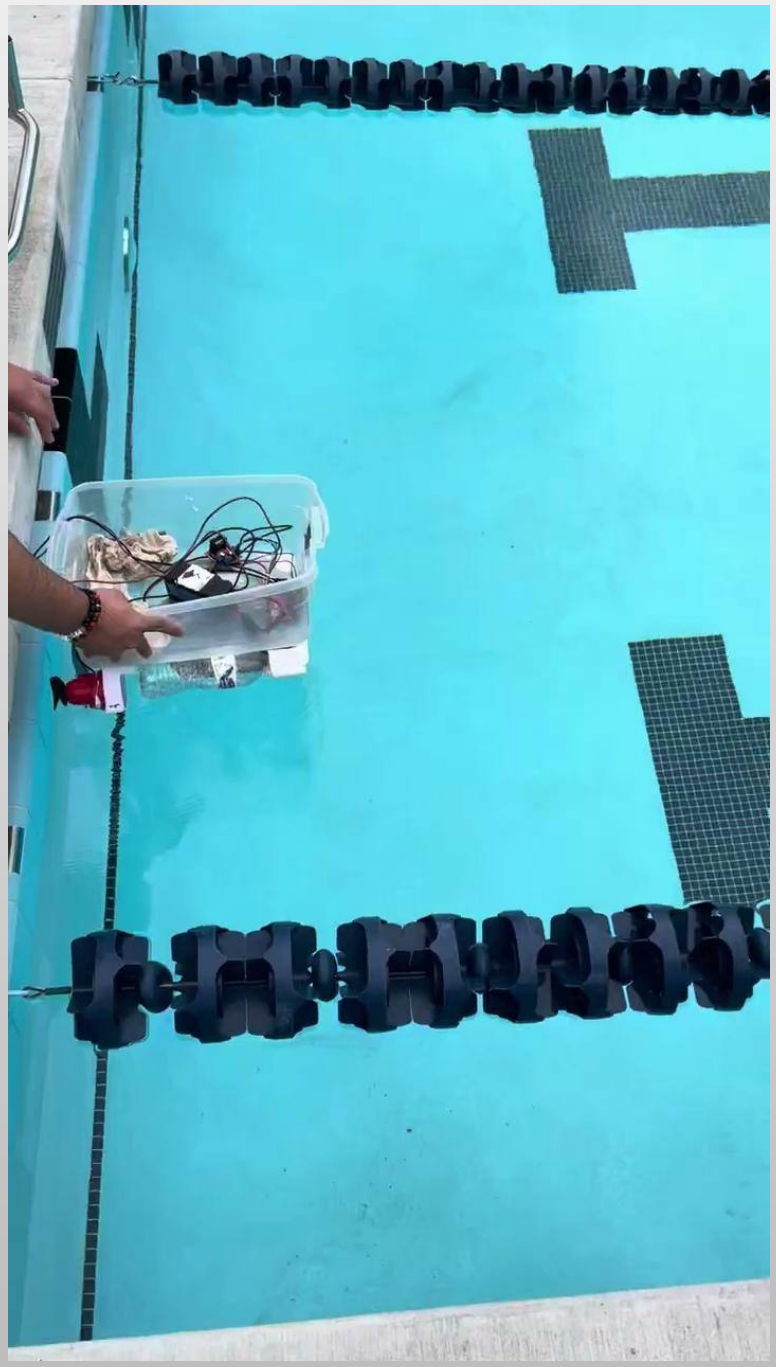
Tasks	Reflection
New propeller performance	- The robot moves significantly faster in the water with the new propellers - As predicted, the angle of the propeller was the primary factor in the performance of the propeller and ultimately led to us being so much faster
Base Force Test	- The base of the robot is much heavier in the back where the bilge pumps are mounted. It is nearly 2 times heavier than the front - To balance these forces, we shifted the battery toward the front of the apparatus (the battery weights approx 6 newtons)
Sheet metal model	- We identified discrepancies with the sheet metal model, not only was it visually not buoyant enough, but the pocket in which water is flowing through is too cluttered with walls and area in which the water would not flow smoothly (see demonstration on slide 85)

Goals for Next Week

- Come to a conclusion about the base of the robot
- CAD, build, and assemble base model so that we can test the sonar redirection algorithm and find the optimal placement for the sensor

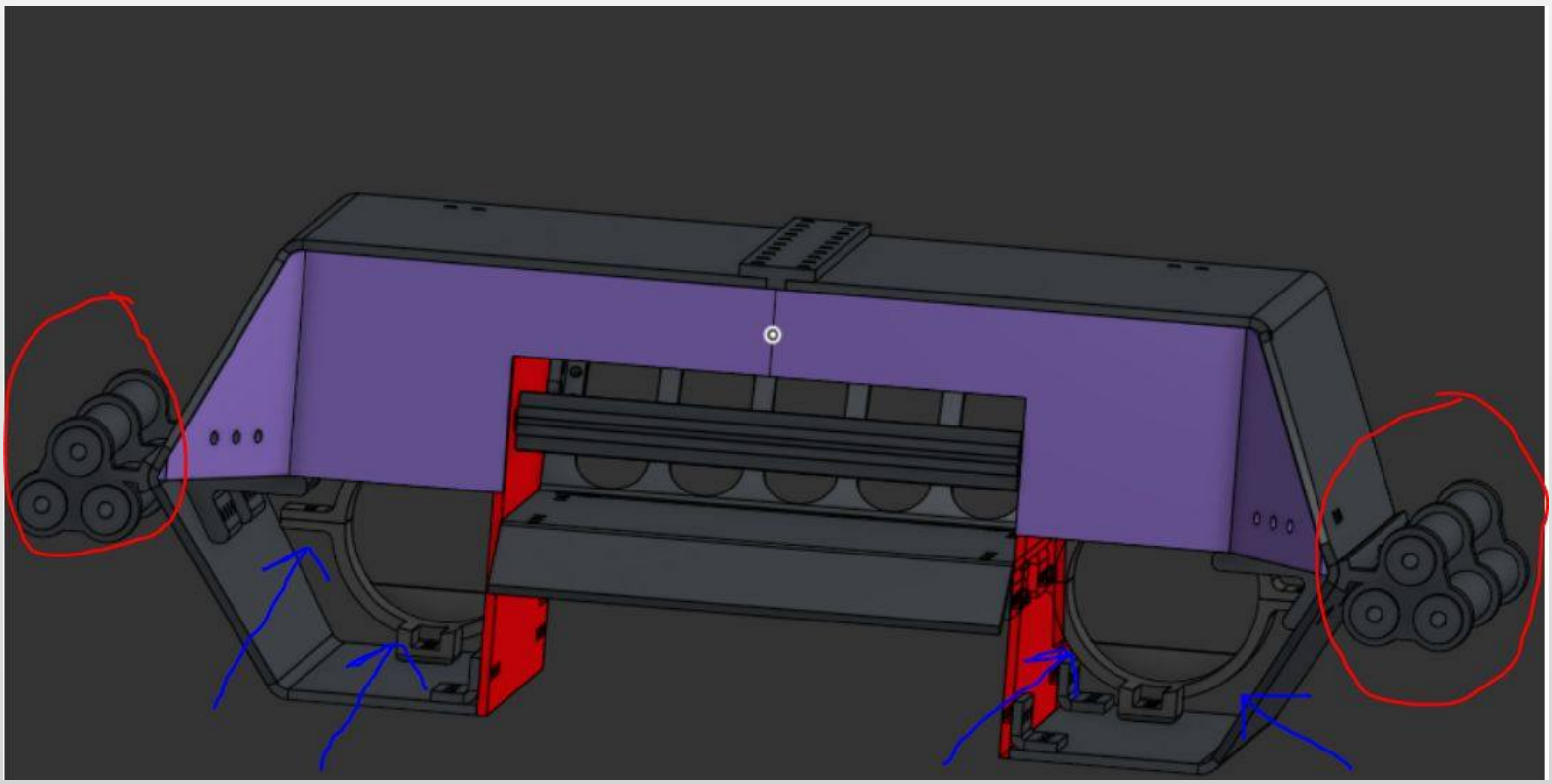
Author:	Date:	Witness:	Date:
Vivian Nguyen	5/5	Elijah LaMothe	5/5

Pre-made propeller performance in water



Author:	Date:	Witness:	Date:
Vivian Nguyen	5/6	Elijah LaMothe	5/6

Polycarb base design conflicts

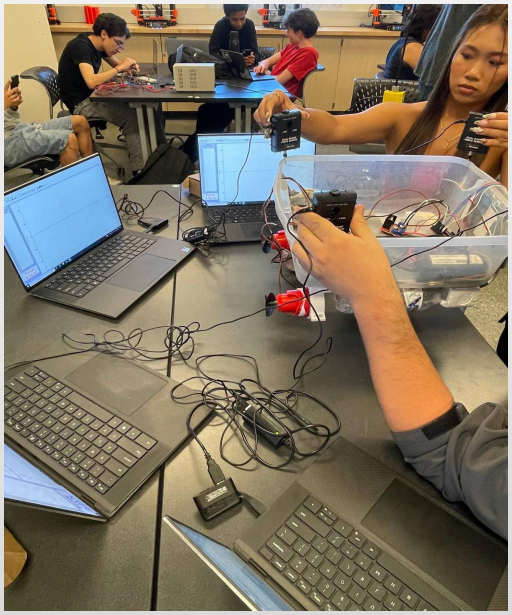


The pool noodle mounts for the robot were too little and not buoyant enough to support the weight of the robot (as santana commented through a discussion)

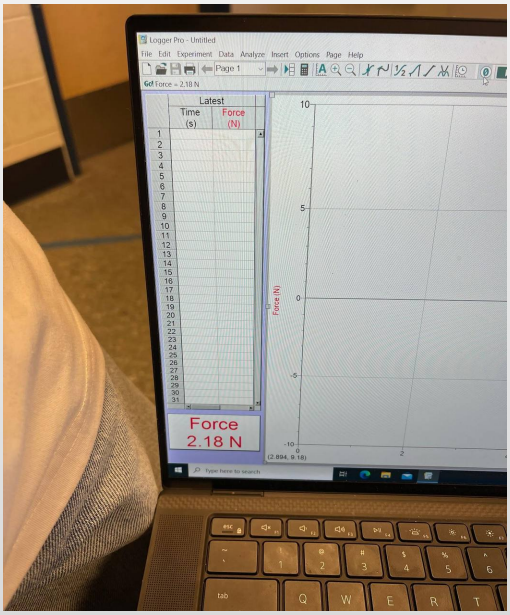
The water flowing through the bottom portion of the robot hits too many walls and obstructions which would ultimately slow down the robot and cause inconsistencies with its pathing.

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/6	Elijah LaMothe	5/6

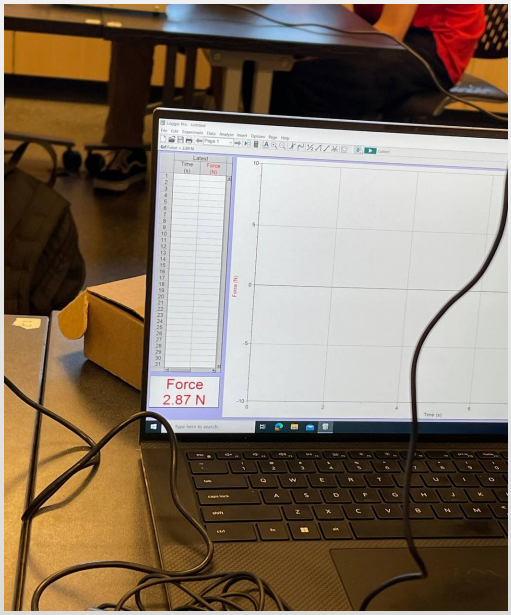
Force Test on the Apparatus: Data



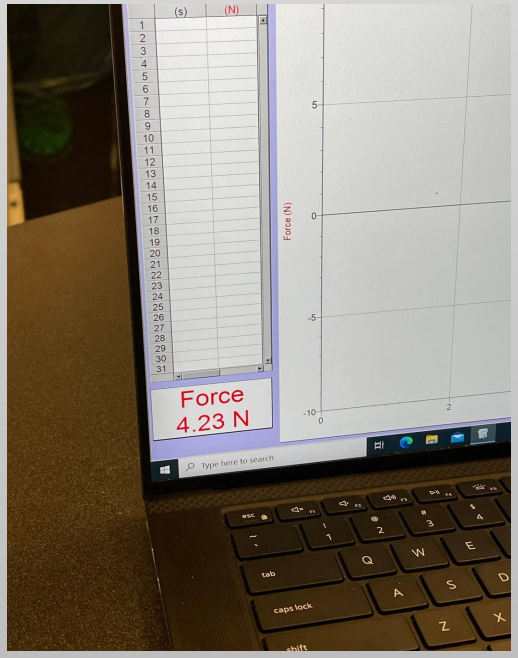
Used 4 different force testers in order to test the force on each corner of the apparatus; Used a level to make sure the forecis accurate when balanced



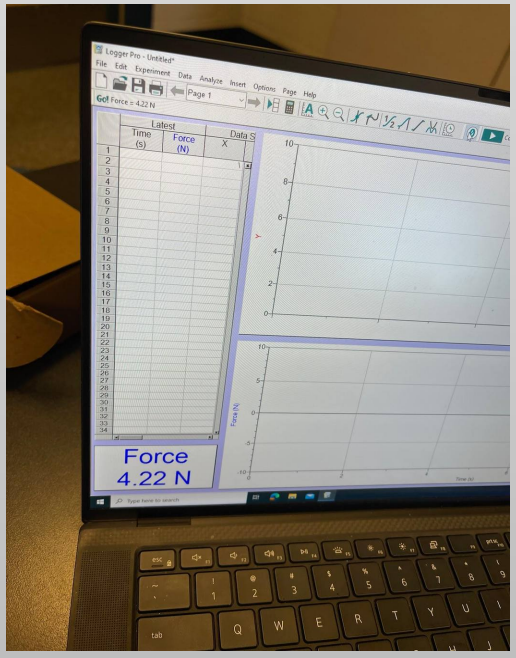
Front left corner



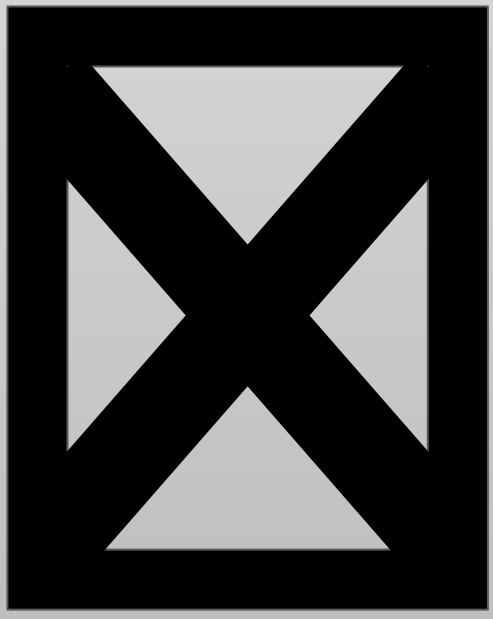
Front Right Corner



Back Left Corner



Back Right Corner



Author:	Date:	Witness:	Date:
Vivian Nguyen	5/12	Elijah LaMothe	5/12

Weekly Logs

Week 31: May. 12 - May. 16

Goals for This Week

- Prepare sheet metal final base
- Find a way to make the sonar detection more consistent
- Have final modification onto final base for the end product

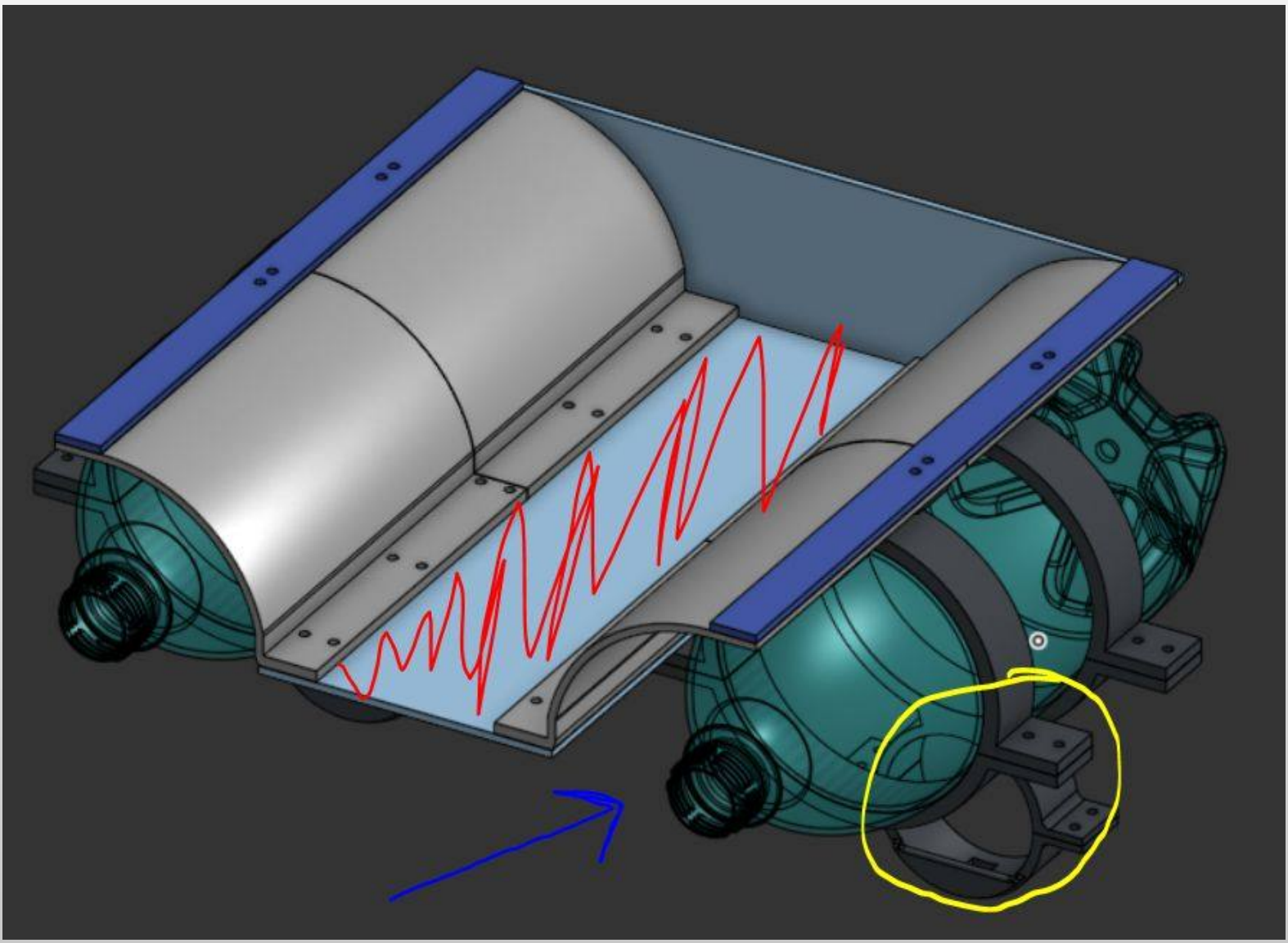
Tasks	Reflection
Update Notebook	- Add the force test on all four sides of the apparatus - Add video of the pre-made propeller performance -
Build	- We got two liter bottles and attached it into the bottle holders; the bottles did not fit perfectly in between the holders so we stuck memory phone underneath the bottle to fill in the extra space causing the holder to hug the bottle more secure.
CAD	Organize the onshape CAD files and look for ways the new design could be optimized/any discrepancies with the design in general

Goals for Next Week

- Build and assemble the entirety of the model, make sure the acrylic parts are cut too
- Find any weak points/discrepancies in the design and issue fixes
- Test out the new model in water with sensors and propellers

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/12	Elijah LaMothe	5/12

New Base Design and Reasoning



Centered and raised compartment for electronics allows us to keep the important parts of the robot dry and protected from the water

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/15	Elijah LaMothe	5/15

New Base Design and Reasoning Continued



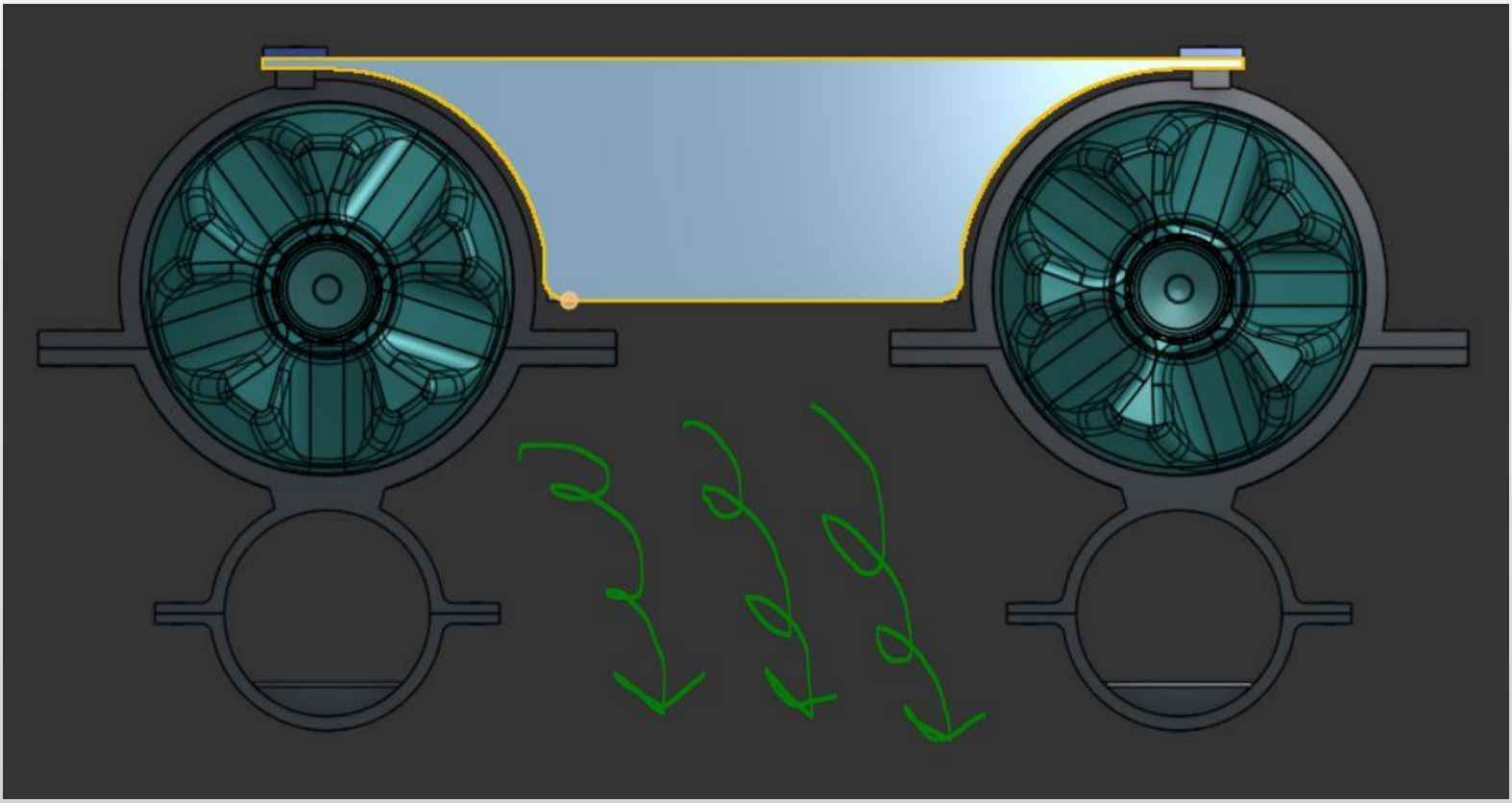
Refer to image on the last page

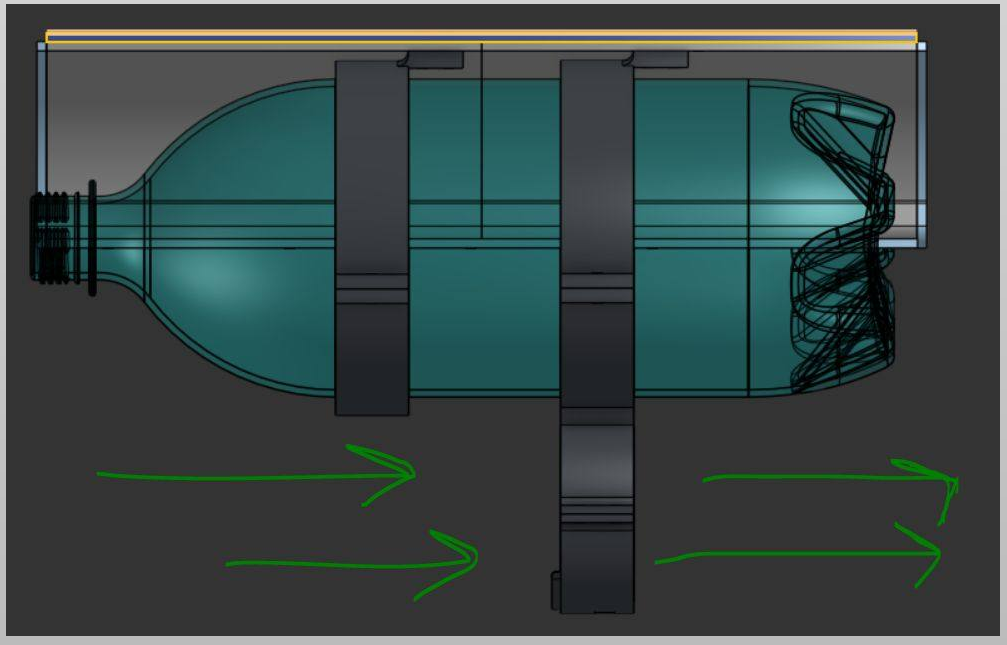
2 Liter bottle design ensures that we will be able to float under nearly any amount of weight that we would need to complete this project. These also allow us to be much more consistent when it comes to surface skimming as the bottles are much more buoyant than any design we had before.

To secure the 2 Liter bottles, we have made adapters that wrap around the circumference of the bottle, and even attach our bilge pumps. These allow us to continue to use the same bilge pump mount design as before and effectively use the space the 2 Liter bottles give us.

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/15	Elijah LaMothe	5/15

New Base Design and Reasoning





Clean passthrough design ensures that little to no water is colliding with obstacles under the surface so our pathing is as consistent as possible and our speed is maintained.

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/15	Elijah LaMothe	5/15

Weekly Logs

Week 32: May. 19 - May. 23

Goals for This Week

- Assemble new design; complete modification
- Print out all parts to new design
- Find 2 liter water bottles for buoyancy
- Modify any parts needed

Tasks	Reflection
Robot Apparatus Build	The assembly of the robot went smoothly. The mounts for the bottles and motors are made well and adapt nicely to the geometry of the objects. The acrylic pieces also work very nicely and perfectly align with the rest of the robot build.
Possible Problems with the build	There are many folds and edges in the robot which means an increased risk of water leaking through. There are many ways we could go about fixing this. Some of the solutions we proposed are using the marine epoxy to seal the edges and covering it with duct tape.

Goals for Next Week

Run the robot in the pool and see if the creases in the robot cause a significant leak. Also test the sonar redirection algorithm in the pool and see how well it works in the school pool.

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/19	Elijah LaMothe	5/19

Underwater sonar sensor we ordered



Did not deliver in time

Why do we want an underwater sonar?

The current sonar sensor has to be in the perfect position above the water to be able to consistently detect distance. With an underwater sonar sensor we don't have to worry about any water damaging the electronics. The underwater sonar sensor would also allow us to easily and consistently detect walls from underneath the robot.

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/19	Elijah LaMothe	5/19

Weekly Logs

Week 33: May. 26 - May. 30

Goals for This Week

- Take the robot to the pool to see how the new base performs in the water
- Make any adjustments to the base of the robot (poxy the base, add tape, adjust the height, etc.

Tasks	Reflection
Robot in pool	The robot maneuvered well in the pool, however the sonar redirection algorithm did not work well at all. It has inconsistencies detecting certain parts of the wall. Water also leaked to the interior of the base causing some of the electronics to get wet. To mitigate this, we added marina poxy to the perpendicular points of the base so that no water would leak through the cracks.
Sonar Redirection Algorithm	The sonar redirection algorithm really did not go as planned. We've noticed that the sonar is inaccurate when it comes to navigating the pool and it is inconsistent with different types of walls. For example, the wall we have in this pool compared to the wall that typical home-built pools will have. Also, the base of the robot is too low in the pool so the sonar sensor is at significant risk of getting submerged in the water. To solve this, we adjusted the base so that it rested a little higher on the surface of the pool with some spacers.

Goals for Next Week

- Ultimately test out the final results of the apparatus
- Work on maintenance on the physical device, so the filter basket, water bottle holders, and leakage
- Work on servo with basket cover mechanism
- Finalize everything in the notebook

Author:	Date:	Witness:	Date:
Vivian Nguyen	5/26	Elijah LaMothe	5/26

Limit Switch V.S. Sonar

Winner







The limit switch sensor would decrease complication in our system as we only have to deal with two states of being (1 and 0). The limit switch is also capable of being adjusted on the z axis if it is too high or low compared to the pool surface

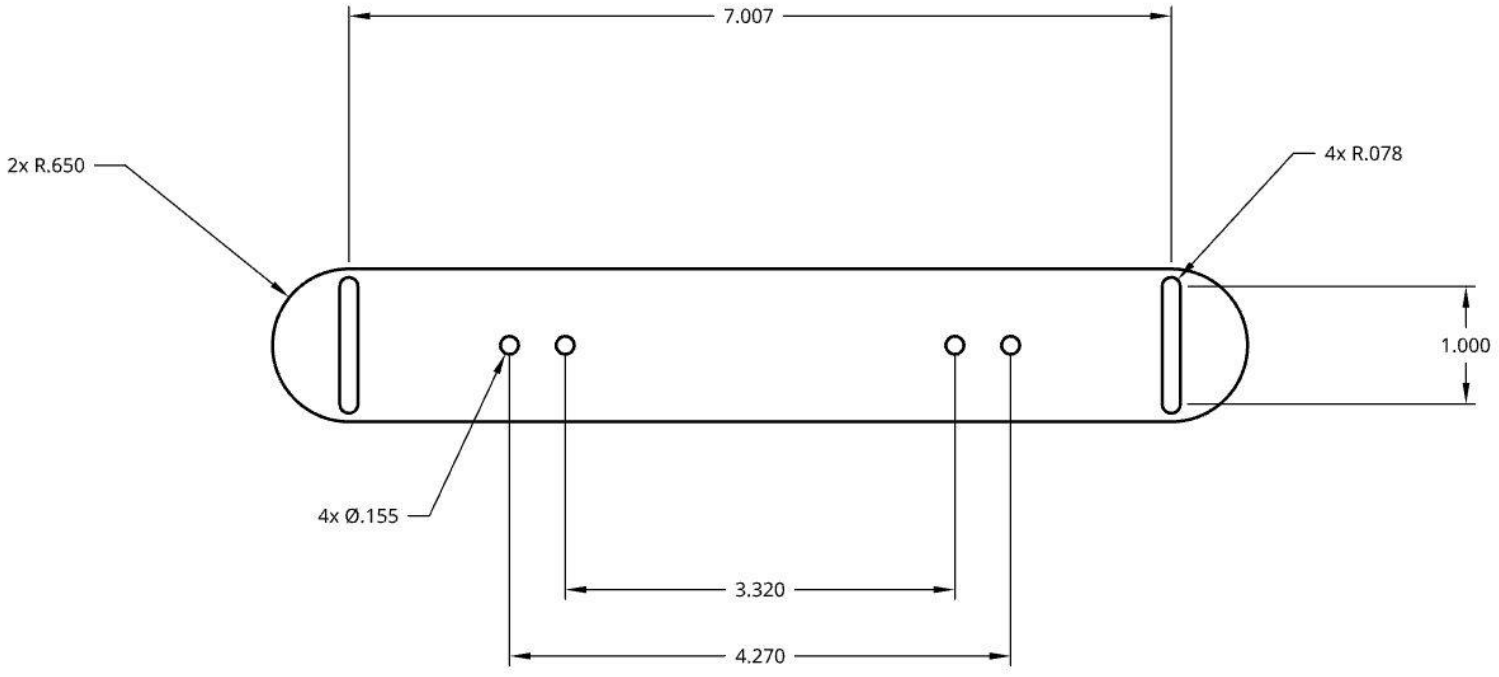
The sonar sensor is inconsistent with different types of walls. Also, if any water comes in contact with the sensor, its performance and ability to function decreases significantly compared to the limit switch.

Author:	Date:	Witness:	Date:
Vivian Nguyen	6/2	Elijah LaMothe	6/2

Button Sensor Redirection Concept

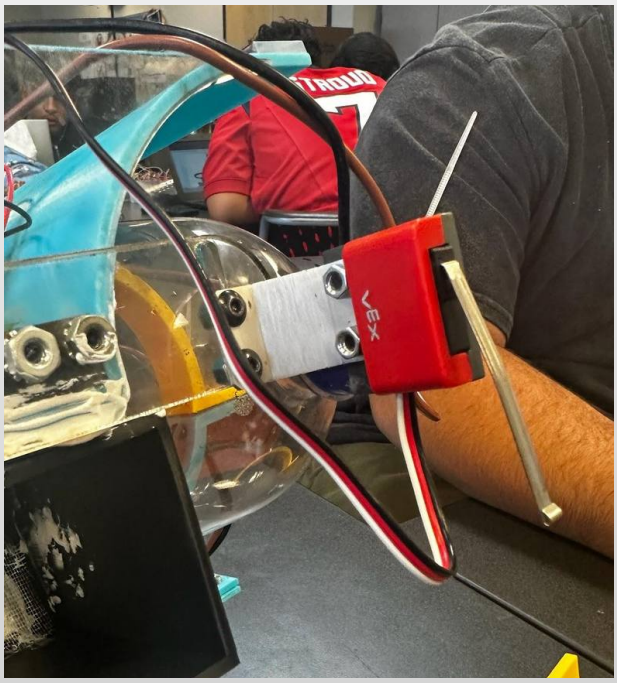


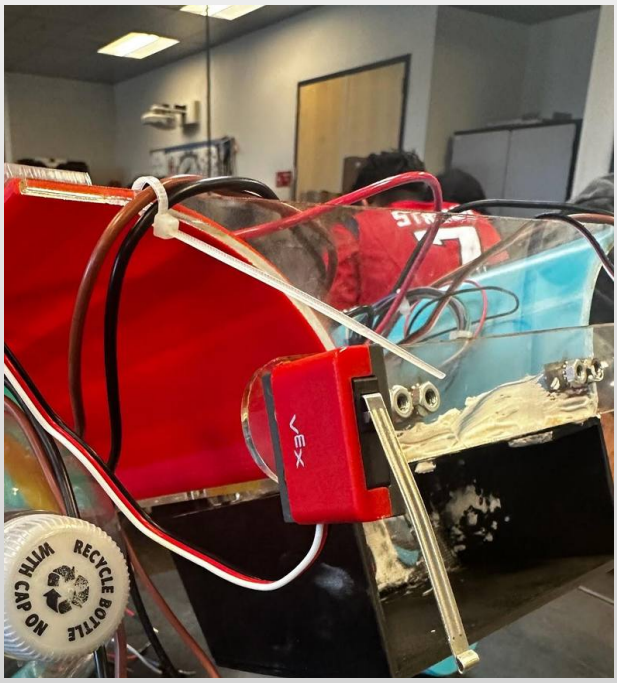
This entire drawing/concept will basically hold two limit switch buttons; one for the left side and on for the right. This will attach directly to the front of the robot apparatus. The slotted design for the buttons allow us to adjust them on the z axis in case we notice they need to be higher/lower in respect to the surface of the water.

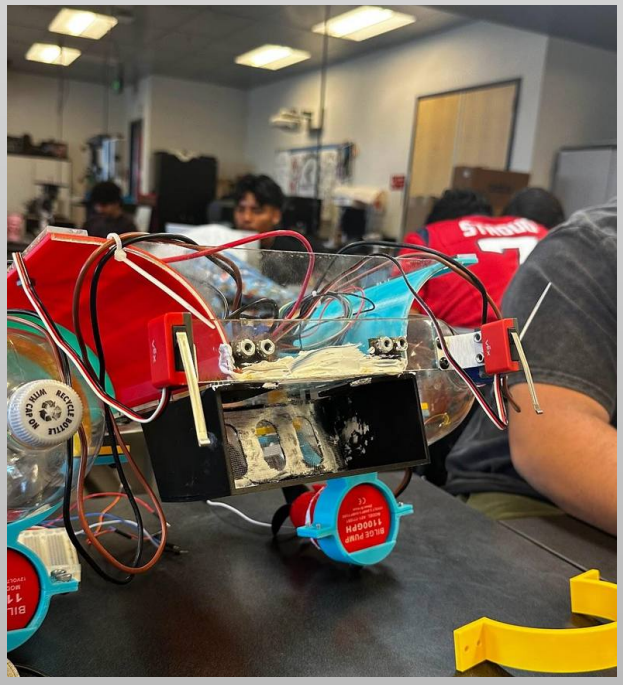


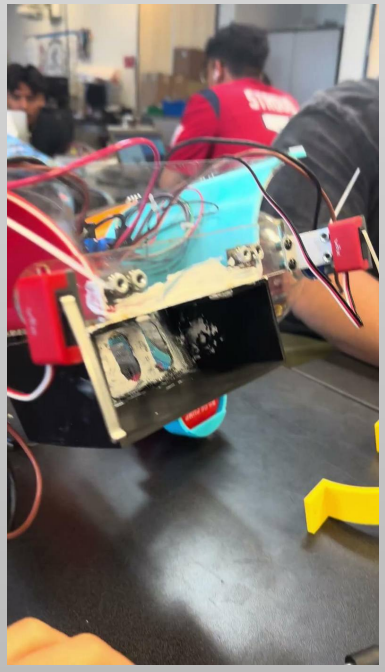
Author:	Date:	Witness:	Date:
Vivian Nguyen	6/2	Elijah LaMothe	6/2

Button Sensor Redirection Implementation





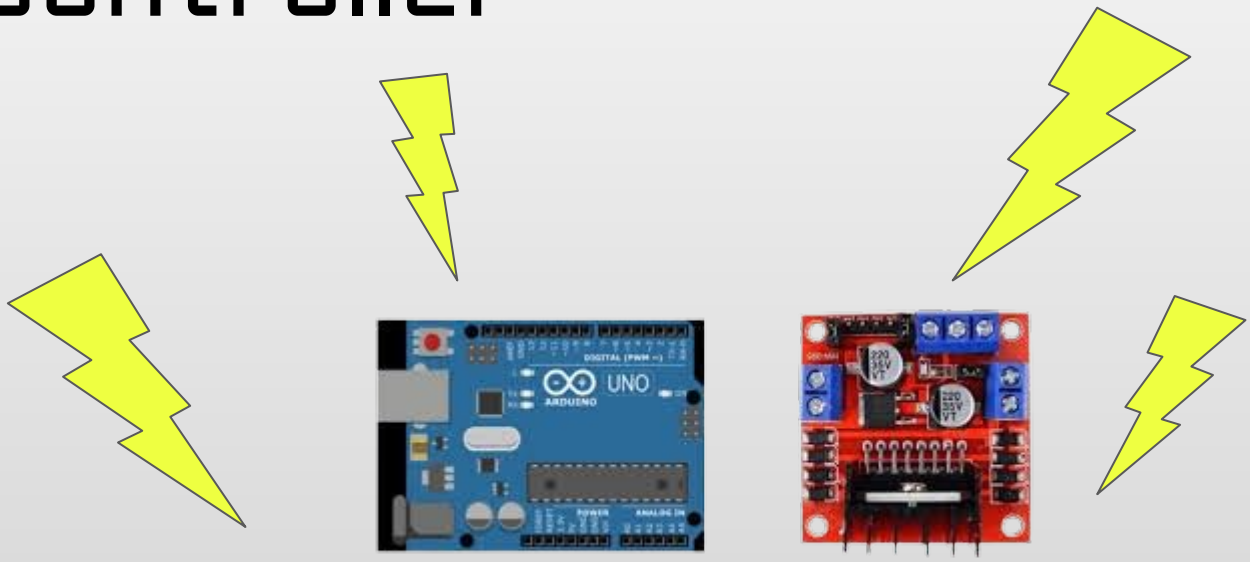




Author:	Date:	Witness:	Date:
Vivian Nguyen	6/2	Elijah LaMothe	6/2

“Broken” arduino and motor controller





At the start of the week we found out that the bilge pumps were not getting power. First, we thought it was because the button sensor implementation had logic issues, but we soon found out it was due to the arduino and motor controller. In our previous test a little water came in contact with the electronics which I believe caused them to short circuit/not function properly. We switched out the arduino and motor controller with completely new ones which fixed the issue.

Author:	Date:	Witness:	Date:
Vivian Nguyen	6/2	Elijah LaMothe	6/2

Weekly Logs

Week 34: June. 2 - June. 4

Goals for This Week

- Ultimately test out the final results of the apparatus
- Work on maintenance on the physical device, so the filter basket, water bottle holders, and leakage
- Work on servo
- Finalize everything in the notebook

Tasks	Reflection
Build Updates	Filter Basket - Modify the original filter basket to lie flat on the bottom of the apparatus - Take out two flaps on the servo to be an open close mechanism for the debris coming in and out - Figure out a way to attach the filter basket to the bottom
Notebook	Update various parts: - Update the last two weeks of the notebook - Include photo update of the build, pictures of new holders, filter basket, wiring, physical updates
Testing	- The robot drove straight for a few seconds before spinning in a circle. We are attributing this issue to a faulty bilge pump which will be tested later in the week. Likely on Thursday/Friday.

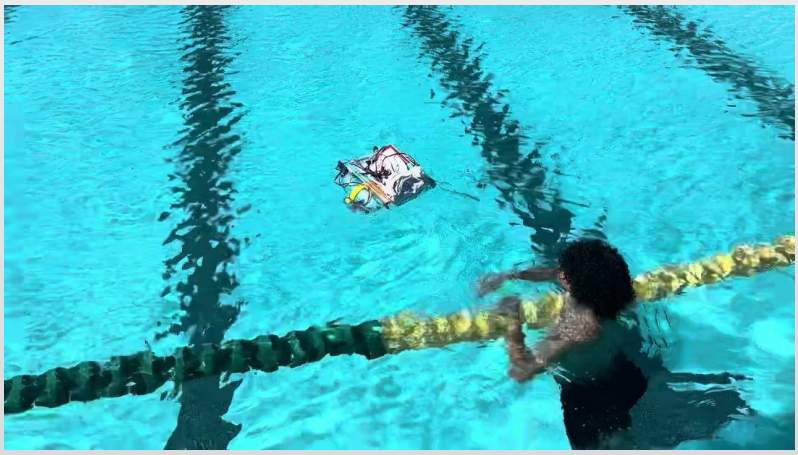
Goals for Next Week

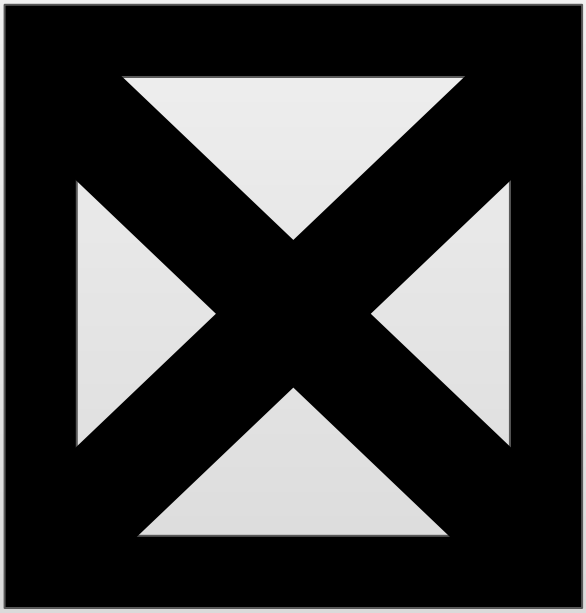
Pool Surface Cleaner Project Complete

Author:	Date:	Witness:	Date:
Vivian Nguyen	6/2	Elijah LaMothe	6/2

Robot Pool Test with button sensors







During this test we noticed that the robot would drive forward straight for a few seconds, then start spinning around in a circle. We concluded that this was not a logic error because we ran code that just span the motors for the bilge pumps and the same issue occurred. Our second guess is that one of the bilge pumps might be defective so we will try replacing it with our spare. (To be tested/documented on the next page)

Author:	Date:	Witness:	Date:
Vivian Nguyen	6/5	Elijah LaMothe	6/5